



AHEAD OF THE CURVE® SERIES

The Cable(s) Guy: Initiating On Datacenter Connectivity

September 30, 2025

Connectivity is a fundamental aspect of datacenter and AI that merits its own category of coverage, but with an effort that remains unified across networking and compute.

We initiate on datacenter connectivity as a distinct category within our broader semis and AI coverage, informed by the respective roadmaps across accelerators and hyperscalers.

Our investment framework for connectivity asserts that low-level tech differentiation and project diversification are key.

Initiating on ALAB (Hold), MTSI (Buy), and SMTC (Buy) while assuming coverage of CRDO (Buy). Further key implications for AMD, AVGO, MRVL, and NVDA.

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Please see addendum of the report
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INITIATING COVERAGE

The Cable(s) Guy: Initiating On Datacenter Connectivity - Ahead Of The Curve

THE TD COWEN INSIGHT

We are initiating on Datacenter Connectivity and Networking Infrastructure based on our view that connectivity is a fundamental aspect of datacenter & AI that merits its own category of coverage in the context of a unified effort across networking and compute. We develop a bottom-up, port-level connectivity model illustrating a potential **>\$75B networking silicon TAM by 2030**.

Our Thesis: Connectivity Worthy Of Its Own Focus

As AI workloads have scaled to clusters now exceeding 100,000 accelerators, the speed, reach, and power draw of the connections between these accelerators is becoming nearly as important as the accelerator itself. To enable the next generation of AI scaling, investors see optical links and co-packaged optics as being required. However, in our view both **copper cables (active and passive) and pluggable optics will grow strongly** well into the next decade, and **investors are underappreciating the opportunity scale up networking represents**. Our conclusions are underpinned by our proprietary, bottom-up forecast that aligns our views on accelerator units with networking speeds and connectivity media, and has important implications for key connectivity stocks such as **Broadcom, Marvell, Credo, Astera Labs, MACOM, and Semtech**.

Amdahl's Law observes that **connectivity has the potential to bottleneck entire AI workloads** and is thus integral to full-stack optimization of both training and inference. Further, a fundamental property of all-to-all connected networks (such as a backend AI network) is that **the number of edges (connections) grows exponentially for a given linear increase in the number of nodes (accelerators)**.

The era of Generative AI all-to-all scaling, constrained by the realities of Amdahl's Law, is driving the most significant acceleration of demand for networking technology since the 1990s. We are initiating on Datacenter Connectivity and Networking Infrastructure due to our belief that this **demand is durable, and connectivity in the age of AI is most effectively covered with a unified effort across networking and compute**. As an outgrowth of our datacenter semis coverage, we therefore initiate on **Astera Labs (ALAB, Hold), MACOM Technology (MTSI, Buy), and Semtech (SMTC, Buy)**. *With this report, we also transfer lead coverage of Credo Technology (CRDO) to Sean O'Loughlin from Joshua Buchalter, maintaining a Buy rating.*

What Is Proprietary?

To analyze the puts and takes across what we identify as the key debates for networking and connectivity, we created a proprietary model that **aligns our existing assumptions around AI accelerator growth with the associated networking infrastructure** such a buildout would require. Against these unit volumes and speed estimates, we then **estimate relative share of physical media (copper vs. optics) at each speed**, then map market share assumptions to **specific company revenue builds**. *We make this fully-dynamic industry model available to clients.*

Financial & Industry Model Implications: A Material TAM Developing

Our model illustrates the potential for a **~\$90B networking infrastructure TAM** to develop by 2030 across physical layer connectivity and switching, **>\$75B of which is silicon (rather than systems)**. We believe **this is much larger than investors contemplate**. Consider, Gartner's forecast for C2029 Wired Communications semiconductors is just \$48B—our comparable estimate for C2028 is **>\$68B from datacenter-only networking silicon** (a figure that does not include systems-level revenue that some such as NVIDIA capture or Broadband wired communications market). Further, we note that networking silicon by 2030 will likely rival

the size of the entire automotive semis market, and **exceed the size of the entire server CPU market by 2026**.

This has clear implications for SMID cap companies that are the focus of this report, but is also impactful for not only those traditional "networking" companies such as Marvell & Broadcom but traditional "compute" companies such as NVIDIA and AMD. Three things that stand out to us from a model standpoint: (1) the growth is significant, even after the explosive past two years; (2) non-NVIDIA scale up networking specifically is a **brand new >\$8B TAM** with largely **no incumbent**; and (3) optical DSPs remain alive and well...though the incumbent Marvell appears under threat.

What To Watch: Scale Up Domain Size and Overall Cluster Size Key Variables for Model; Transceiver Power Draw Key for Co-Packaging

We make many assumptions in constructing our model (and make no claims to being the arbiter of truth on any of them), but perhaps most impactful are our assumptions regarding the **number of scale up switches required per XPU** and **overall cluster size**, which we expect will continue to trend upward (and therefore increase the connector/XPU attach rate).

More qualitatively, we believe the dual optimization points of power draw and signal integrity for pluggable optics will be the determining factor in CPO adoption, rather than any technical specs regarding co-packaged solutions themselves. If some combination of linear optics (fully linear or receive-only) and lower-power full DSPs can deliver 1.6T with an acceptable power envelope, that is likely to materially slow the pace of CPO deployment.

Additionally, **progression in (and adoption of) NVIDIA's networking roadmap** will be key to monitor, and is one of the reasons we believe coverage of networking and compute as a unified entity is critical. NVIDIA's significant influence in the supply chain has the capacity to sway the market in one direction or another in a material almost all-or-nothing type way. Especially impactful in our view would be if NVIDIA were to adopt CPO for scale up—an effectively captive design element that would see rapid adoption effectively **by fiat**.

Stock Conclusions:

Aligned against our two-prong investment framework and the key and emerging debates, we view **Credo (Buy)** as well positioned given its role as a key innovator in both copper (AECs) and pluggable optics (LRO). **Semtech (Buy)** and **MACOM (Buy)** are likewise well positioned to benefit from ACCs and LPO, though their diversified business models dampens the acute impact to the P&L. **Astera Labs (Hold)** has the best business model in our view in terms of customer and platform diversity and should see strong near-term growth as its PCIe switches ramp to volume, but we believe could face headwinds if Scale Up Ethernet becomes the protocol of choice.

Beyond the four stocks specifically highlighted as "datacenter connectivity," we see material headwinds for **Marvell (Hold, Buchalter, separate downgrade [here](#))**. While we are less concerned regarding the adoption of CPO than we believe are many investors, our confidence in the sustainability of the pluggable ecosystem is based on non-Marvell factors and we model >10% share loss to LRO and LPO solutions for the company over the next 5 years. When we couple this view with our work earlier in the year highlighting the challenges facing custom silicon from a structural standpoint ([here](#)), we feel compelled to lower our rating to Hold.

Finally, conversely we highlight the potential opportunity for **Broadcom (Buy, Buchalter)** to expand its networking TAM if Scale Up Ethernet (SUE) is adopted. Specifically, while we do not (yet) incorporate this directly into our model, **SUE-based switches could generate >\$8B by 2030**. This is likely subsumed by the size of the ASIC franchise at that point, but it is incremental upside we do not believe is likely embedded in investor expectations at this point.

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Figure 1 The One Page Version of This Report

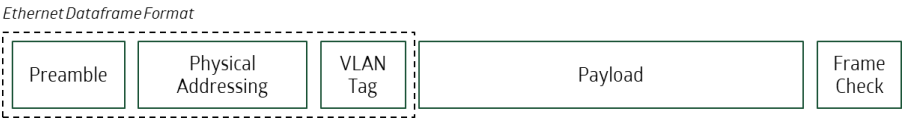
INVESTMENT FRAMEWORK			KEY DEBATES			EMERGING DEBATES	
	Differentiate with IP in the Physical Layer	Achieve Customer & Project Diversification	Where Are We in the AI Cycle?	When Will Optics Displace Copper in the Rack?	When Will Co-Packaging Displace Pluggable Ecosystem?	Whose Open Scale Up Protocol Will Prevail?	How Will Emerging Photonics Tech Disrupt Incumbents?
TD Cowen View	Long-run returns likely accrue to those that leverage proprietary tech to differentiate at the lowest level	Those who achieve customer & project diversification should command a higher multiple over time	Still early days in the overall AI investment cycle, with reasoning models & agents driving inference demand	Copper is likely to remain the preferred solution intra-rack through at least 224G SerDes generations	Pluggable ecosystem via innovations such as LPO & LRO is likely to be preferred over co-packaged solutions	UALink has AMD's Infinity Fabric heritage, but impressive latency from TH Ultra makes SUE a serious alternative	Start-ups are getting attention, but established supply chain likely makes SiPho preferred long-term.
 (CRDO, Buy)	Positive Industry-leading SerDes IP underpins broad connectivity portfolio	Neutral-to-Positive Highly concentrated today (AWS) but improving through 2026.	Positive Strong beneficiary of AI investment, especially in scale out.	Positive A key enabler of copper-based solutions with AEC market leadership.	Positive Innovator driving adoption of LRO; greenfield opp in half-retimed TRX.	Positive Does not compete in switching; SUE adoption would open new scale up market for AEC	Neutral Largely on the electrical side of optical modules, uninvolved in OE/laser.
 (SMTC, Buy)	Positive Strong mixed signal IP portfolio in signal integrity & wireless/analog "dark arts."	Neutral-to-Positive Diverse business, but DC exposure can be concentrated.	Neutral-to-Positive Benefits from AI uplift, but muted impact given other businesses.	Positive A key enabler of copper-based solutions with ACC market leadership.	Positive A key enabler of linear pluggable solutions such as LPO/LRO.	Neutral Largely agnostic to scale-up protocol given analog nature of product portfolio.	Neutral Largely on the electrical side of optical modules, uninvolved in OE/laser.
 (MTSI, Buy)	Positive Strong mixed signal heritage and portfolio across niche applications.	Neutral-to-Positive Diverse business, but DC exposure can be concentrated.	Neutral-to-Positive Benefits from AI uplift, but muted impact given other businesses.	Positive A key enabler of copper-based solutions with ACC opportunities.	Positive A key enabler of linear pluggable solutions such as LPO/LRO.	Neutral Largely agnostic to scale-up protocol given analog nature of product portfolio.	Positive Wide variety of exotic process technologies used allows for flexibility.
 (ALAB, Hold)	Cautious SW-defined architecture delivers programmability and observability but invites concern on trade-offs req'd.	Positive Strong relationships with both HSP & NVDA/AMD; Interop Lab ensures smooth deployment.	Positive Strong beneficiary of AI investment, especially in custom accelerators.	Neutral-to-Positive Taurus AECs enable longer reach copper; lower market share <u>mutes</u> impact.	Neutral Not a significant player in optical ecosystem; more server side, on-board content.	Cautious Well-positioned in UALink, but widescale adoption of SUE would be negative	Neutral Not a significant player in optical ecosystem.
 (AVGO, Buy)	Positive Industry-leading SerDes IP coupled with industry-leading switching and routing IP	Neutral-to-Positive Diverse business outside of AI, but TPU/Google remains the major driver of custom silicon	Positive Strong beneficiary of AI investment in networking and custom accelerators	Neutral-to-Positive AECs enable longer reach of copper; lower market share <u>mutes</u> impact	Neutral Would benefit from CPO adoption, but is not required for growth	Positive Incumbent player in Ethernet switching and de-facto leader advancing SUE	Positive Growth opp, though would be small in the context of overall business
 (MRVL, Hold)	Positive Strong portfolio of SerDes, switching fabric, custom HBM and security IP	Positive Diverse business in optics, but custom AI silicon remains inherently concentrated	Neutral-to-Positive Strong beneficiary of AI investment in networking, mixed in custom accelerators	Neutral-to-Positive AECs enable longer reach of copper; lower market share <u>mutes</u> impact	Cautious Does not have meaningful LPO/LRO portfolio, could see \$ share loss in DSPs	Neutral-to-Cautious Lower market share in switching; UALink member, but could switch to SUE	Positive Growth opp, potentially helps offset DSP share loss with other content in transceiver

ALAB (\$198.80), CRDO (\$146.01), MTSI (\$123.29), and SMTC (\$61.98) covered by Sean O'Loughlin, CFA.

AVGO (\$327.90) and MRVL (\$82.39) covered Joshua Buchalter, CFA.

Source: TD Cowen.

HEADER



- Preamble:** Executive Summary
- Physical Address:** Key Stocks in Datacenter Connectivity
- VLAN Tagging:** Model & Implications

Preamble: Executive Summary

"The network is the computer."

-John Gage, Sun Microsystems

Communications networks have existed far longer than computers; even digital communication's early history parallels rather than follows that of early digital computation and has historically been an adjacent-but-separate field of technology. However, AI, and especially Generative AI, is a full-stack problem and is governed by the constraints of Amdahl's Law. This is a theme we have touched on frequently ([here](#), [here](#), [here](#)). In short, **connectivity has the potential to bottleneck an entire AI workload.**

Further, it is a mathematical fundamental property that in an all-to-all connected network, **the number of edges grows exponentially for a given linear increase in the number of nodes.** That is, for an XPU cluster (which is inherently all-to-all), the number of *connections* in that cluster (of all types) will grow faster than the number of accelerators themselves.

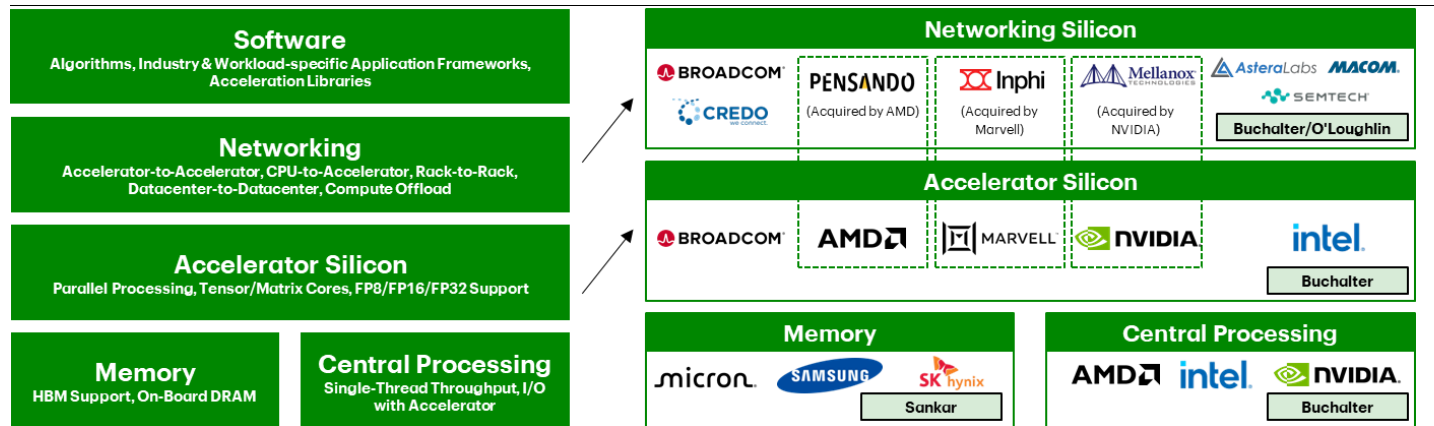
The era of Generative AI scaling of all-to-all accelerator clusters, constrained by the realities of Amdahl's Law, is driving the most significant growth of demand for networking technology since the Dawn of the Internet. We are initiating on Datacenter Connectivity and Networking Infrastructure as a distinct category not only based on our recognition of this demand, but also our confidence in the *durability* of this demand.

We take the view that **connectivity is most effectively covered as a unified effort across networking and compute**, informed by our coverage of such key semis companies as NVIDIA, AMD, and Broadcom. Alongside their hyperscale customers and such foundational model builders as OpenAI and Anthropic, accelerator companies play a key role in pushing the cadence of network and connectivity upgrades.

Further, in pursuit of a more-optimized full-stack solution, many of these companies are aggressively moving into areas that would have been more traditionally understood as pure "networking," such as NVIDIA's Spectrum Ethernet switching platform and AMD's Pensando acquisition and roadmap.

Understanding these compute roadmaps and product offerings at a granular level will be key to informing the broader networking and connectivity space.

Figure 2 AI Is a Full Stack Problem That Warrants an Integrated Full Stack Coverage Approach

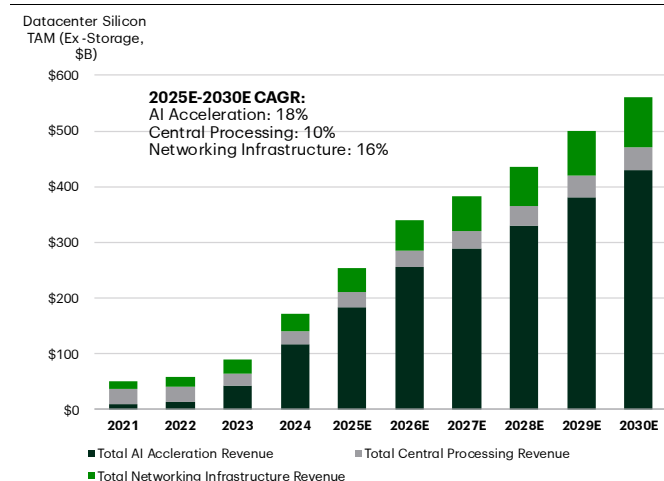


Source: TD Cowen

Adding Granularity to Our Networking Forecasts

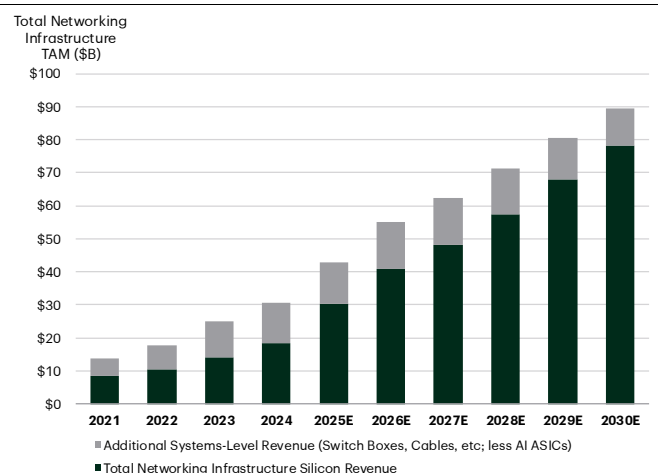
In Spring 2024 we initially published our Datacenter Silicon model ([here](#)) and asserted that while AI Accelerators would be the largest bucket of silicon in dollar terms, Networking Infrastructure would grow the fastest, based on the two principles of Amdahl's Law and the properties of an all-to-all network. Today, we greatly expand upon that Networking Infrastructure model and incorporate it into our broader Datacenter Silicon model, reiterating our view that Networking silicon is likely to grow the fastest of the three major categories of Datacenter compute. Specifically, we model a **~\$90B networking infrastructure TAM by 2030** for our coverage companies. Notably, we estimate that **>\$75B of this TAM is from silicon** specifically (as opposed to systems or other passthrough revenue), and this TAM is growing at **>20% CAGR** from 2025E, compared to a 18% CAGR for accelerators and 10% for central processors.

Figure 3 Networking Infrastructure a Key Category in Datacenter Silicon...



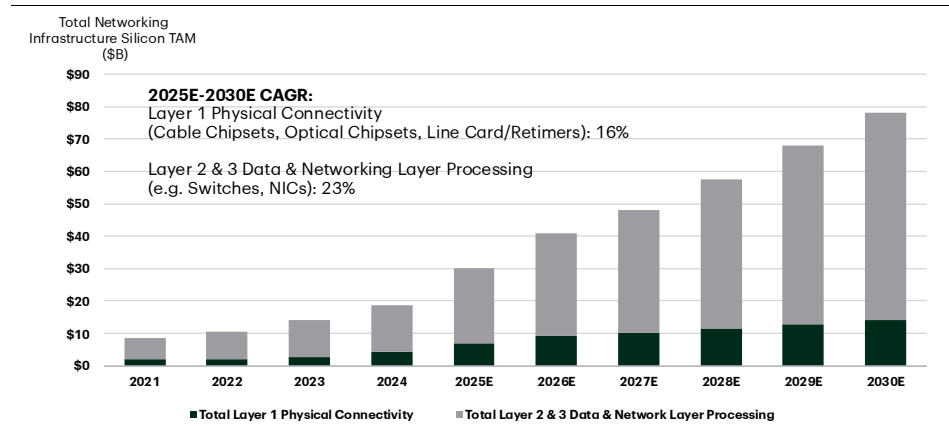
Source: TD Cowen

Figure 4 ...With Networking Silicon Driving the Lion's Share of Growth (>20% CAGR '25 to '30)



Source: TD Cowen

Figure 5 Healthy Growth Across Networking Silicon, Larger \$ Opportunity in Layer 2/3



Source: TD Cowen

The magnitude of this growth is likely underappreciated by many investors who have (appropriately) primarily focused on the accelerator segment of datacenter silicon as it represents the largest in terms of nominal dollars. At a macro level, scale up networking likely represents upside to expectations on both switching and connectivity. In fact, it is likely that even our model underestimates the impact of scale up networks on cables and connectors given we base our estimates on Ethernet ports only.

Identifying a true "consensus" estimate for networking expectations out to 2030 is an imprecise exercise at best, but we believe our >\$75B silicon footprint is likely higher than consensus. Consider that as recently as June 2025, Marvell identified a ~\$94B TAM for their datacenter business by C2028 of which ~\$55B was attributed to compute and only ~\$39B to networking. Likewise, many investors are currently anchoring off Broadcom's "\$60B to \$90B SAM in 2027" commentary from late 2024 ([here](#)), of which ~20% was attributed to networking (~\$18B of networking across its three ASIC customers at the time). By contrast, our apples to apples estimated TAM for Marvell in 2028 is ~\$58B (vs. their \$39B) and for Broadcom is ~\$47B (vs. their \$18B, although we recognize our figure is full-market and theirs referred to just three customers).

For the large and mega cap semis supplying the datacenter market such as AMD, Broadcom, Marvell, and NVIDIA, each's respective accelerator franchise is likely to largely subsume their connectivity businesses both in terms of dollars and in those of stock narrative. However, for those smaller companies such as **Astera Labs**, **Credo**, **MACOM**, and **Semtech**, the next five years could be (and are expected by many to be) transformational to each's P&L. Of these four, we see most upside to consensus estimates for **Semtech** with Datacenter alone exceeding \$1B by C2029 (from ~\$150M in C2024). For **Credo** we see upside to both AEC deployments and Optical DSPs out to 2030, especially from 1.6T LRO deployments, though we recognize at >20x F2027 consensus Sales this is better known. For **MACOM**, we see an opportunity to drive above-consensus growth in Datacenter from not only LPO (likely contemplated by street estimates) but also CW lasers (potentially not yet in estimates, nor in ours). Finally, we believe **the risks in scale up switching posed by Broadcom are too great to ignore for Astera** and model its Scorpio-related revenue growth plateauing as we approach 2030.

Returns in Datacenter Connectivity Likely to be Defined by IP & Diversification

Our foundational investment framework for semis has long been that technological differentiation drives pricing and gross margin leverage, which in turn drives stock performance over the long run.

We continue to view our coverage through this lens and apply it to our expanded coverage of Datacenter Connectivity. Specifically for Connectivity and Networking Infrastructure, we identify two corollaries to this broad investment framework. Over the long run, we believe the companies that are most likely to outperform in Connectivity & Networking are those who:

1. **Differentiate with Proprietary IP in the Physical Layer;** and
2. **Achieve Customer and Project Diversification**

Differentiating in the Physical Layer

We believe long-run returns are likely to accrue to those connectivity companies who can leverage proprietary intellectual property to differentiate at the lowest level, by either going **faster**, going **more accurately**, using **less power**, or some combination of all three.

We believe the datacenter (and computing overall) is amidst two simultaneous paradigm shifts: the advent of Generative AI, and the trend away from the monolithic, CPU-heavy software-defined compute infrastructure towards a heterogeneous, workload-specific, hardware-defined infrastructure. Both paradigm shifts are impacting connectivity. These paradigm shifts combine to **accelerate the pace of innovation** in networking, as well as **increase the demands on the network itself**.

Further, the largest investors in AI infrastructure (and therefore the most significant growth drivers for network infrastructure spend) have **demonstrated a willingness to "do it themselves"** where advantageous to their overall compute footprint. The most salient example of this is the many projects developing custom accelerators today, but consider also the white box network hardware market. If it becomes important enough and can be achieved with merchant/commodity silicon or merchant IP, we believe it will be **difficult for those suppliers to maintain a healthy gross margin profile**.

Achieve Customer and Project Diversification

Customer concentration is, to a certain extent, a fact of life for those serving hyperscale datacenters. The market segment is itself customer concentrated, much like those who supply wide-body jet aircraft would be inherently customer concentrated. However, when accounting for this fact, we prefer those companies who can achieve not only customer diversification (recognizing the inherent concentration) but also *project* diversification within those customers.

Recognizing and building confidence in project diversification is difficult—it is not a reported metric, of course, as is customer diversification. In fact, project concentration is often only recognized in hindsight (Credo at Microsoft and Semtech at Meta are two prominent examples). But in our view, all else equal, a more diverse portfolio across platforms and product categories is likely to both perform better fundamentally and also command a higher multiple from those investors who prefer the lower volatility of earnings.

Taking Sides in the Key Connectivity Debates: The Cycle, Copper, and Co-Packaging

Zooming into a more near-term time horizon, we see three key debates ongoing, and two debates emerging. The three ongoing debates are:

1. **Where Are We in the AI Cycle?** Given our coverage is on average up ~40% YTD and >200% over the past two years, how much upside remains?
2. **When Will Optics Displace Copper for Intra-Rack Connectivity?** As speeds increase to feed the AI beast, at what point will copper's physical limitations prompt the full adoption of optical links inside the datacenter?
3. **When Will Co-Packaging Displace the Pluggable Ecosystem?** With power as the primary constraint for datacenter deployments, every watt counts. At what point will the theoretical benefits of co-packaged optics overcome the operational drawbacks in vendor lock and serviceability of CPO?

In the "emerging debates" category, we would add:

4. **What Will Be the Non-NVLink Scale Up Protocol, and Who Will Benefit?** As AI systems move from server-sized scale up domains to rack-scale and beyond, which "open" scale up protocol will prevail: AMD-led UALink (nee Infinity Fabric), Broadcom-led Scale Up Ethernet, or a combination of the two?
5. **How Will Emerging Photonics Tech Disrupt (or Not) Incumbent Players?** Is silicon photonics and other novel optical technology a threat or opportunity for companies such as Marvell and MACOM in optical transceivers?

Rapid fire, **our take on each** is that:

1. While growth since 2023 especially has been admittedly eye-popping, we believe we are **still early days in the overall AI investment cycle**. As our recent work demonstrated ([here](#)), **AI inference is profitable today**. Further, each hyperscaler (and perhaps increasingly each sovereign nation) faces a decision matrix characterized by Prisoner's Dilemma-like dynamics. **Underinvestment in AI presents an existential threat to the largest businesses in the world**—and the risks to underinvestment vastly outweigh those to overinvestment. More on this debate [here](#). This is likely to benefit **many if not all compute and networking companies**. Finally, we believe the scale of networking revenue specifically within the broader AI theme is underappreciated by investors.
2. By incorporating active components where necessary and continuing to push the envelope on what was thought possible in SerDes design, we believe **copper is likely to remain the primary intra-rack medium** through at least 2030. In fact, with the introduction of 7m AECs from Credo, Broadcom, and Marvell, we believe there is a potential for the trend from copper to optics to **reverse**, with some inter-rack optical links being displaced in back-end networking especially. Blithely, we believe **the next line rate generation will be copper's last—and always will be**. More on this debate [here](#). This is likely to benefit **Credo** and **Semtech**, and to a lesser extent **Astera**, **Marvell**, and **MACOM**.
3. Similarly, we believe the **pluggable ecosystem for optics is likely to persist** far longer than many expect, thanks to such innovations as Linear Pluggable Optics (LPO), Linear Receive Optics (LRO), and the ongoing innovation to lower DSP power draw at 3nm and below. More on this debate [here](#). In our view this benefits **Credo**, **MACOM**, and **Semtech**, and presents a potential headwind for **Marvell**.

On the emerging debates:

4. In our view, the market is most likely **to rely upon Ethernet's physical layer and hardware**, regardless of the transport layer protocol that is implemented. This is effectively the schema AMD is implementing in its forthcoming Helios rack—while the datalink and transport layers are UALink, the physical layer relies on Ethernet (Broadcom's Tomahawk switches). While AMD (accurately) calls this UALink (nee Infinity Fabric), from a hardware vendor perspective it is Ethernet—and we struggle to identify any who have outperformed Broadcom in Ethernet switching hardware (other than *perhaps*...NVIDIA). More on this debate [here](#). This represents an opportunity for **Broadcom**, a potential new market for **Marvell**, and given what we believe are too-high investor expectations, an emerging threat for **Astera**.

5. We view the trend toward greater integration of the optical engine in silicon as a **growth vector** for Marvell & Broadcom and a **key enabler of LPO and CPO**. We are also monitoring innovative 3D photonic fabrics such as those from Lightmatter and Celestial AI, but do not forecast a meaningful disruption to the current optical interconnect paradigm in our time horizon (out to 2030). More on this debate [here](#). We believe this benefits **Marvell** and **Broadcom**, and we see potential for **MACOM's** laser portfolio to benefit as well.

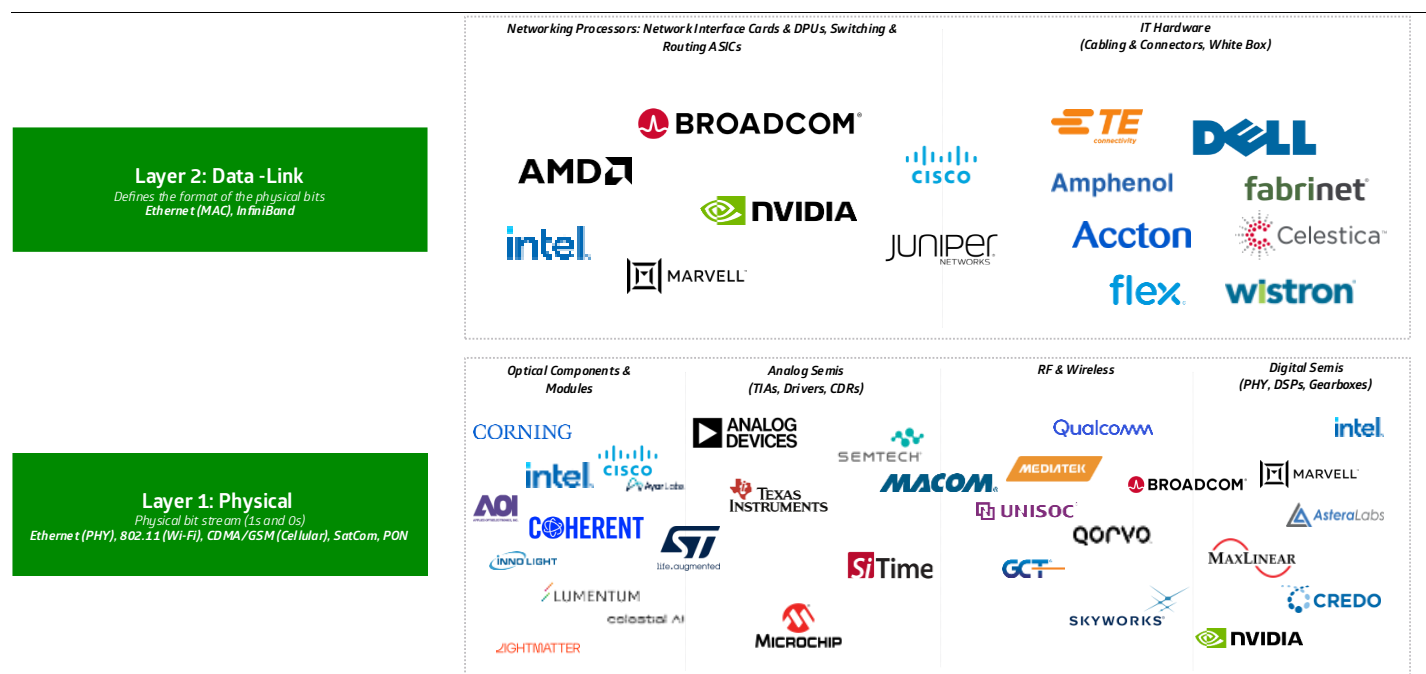
What Is in The Rest of This Report:

An Ethernet dataframe is standardized and divided into sections. With tongue firmly in cheek, we have aligned the sections of our report to align with these sections of an Ethernet message. After the Preamble (this Executive Summary), we identify the key stocks in the datacenter in the Physical Address field. We use the VLAN tagging field to describe our model and its implications before expanding on these points in the Payload field. Finally, we close with our own Frame Check Sequence—an appendix that dives deep into the technical underpinnings of this report.

Physical Address: Key Stocks in Datacenter Connectivity

We highlight 20+ stocks below based on relative positioning within each layer of the datacenter networking infrastructure stack (alphabetically in each section, though different aspects of each company's portfolio may be highlighted depending on the section). In our view, a rising tide is likely to lift many if not all of these boats, but true outperformance will be defined by the investment framework and debates we describe above and expand upon below. Given we are primarily focused on datacenter connectivity, we focus on those companies with material datacenter exposure. We do include some in the visual below who are key players in the broader internetwork infrastructure but have relatively minimal exposure to datacenter environments, such as Qualcomm and other wireless infrastructure companies.

Figure 6 The Network Infrastructure Ecosystem, Layers 1 & 2



Source: TD Cowen, Company data. Juniper Networks is now part of HP Enterprise

Layer 1: Key Low-Level Component and Processing Companies

Analog Devices (ADI, Buchalter): Analog Devices does provide a portfolio of I/O and connectivity products for optical and cable networking equipment, which sits within its Communications segment. These include digital and analog I/O, redrivers, clock and data recovery, DSPs, and more. We see the company as a limited beneficiary of datacenter connectivity simply in the context of its broader portfolio and end market exposures, particularly as it services lower lane rate speeds.

Astera Labs (ALAB, O'Loughlin): We view Astera as a key player in the datacenter connectivity ecosystem, with its initial traction in PCIe retiming (Aries) now expanding to PCIe switching for front-end and scale up (Scorpio). Its COSMOS observability software suite adds significant value to customer deployments and adds an installed-base aspect to Astera's sockets. However, we are more concerned on investor expectations regarding scale up switching, a key growth vector for the company that we see as facing stiff headwinds from competition such as Broadcom.

Broadcom (AVGO, Buchalter): With a historical legacy in the optical and wireless markets, we expect Broadcom to continue participating and holding ~20% market share of the optical market. Broadcom would materially benefit from a greater than anticipated adoption of CPO, as it would be very likely that Broadcom would be able to capture at least some of the value that currently accrues to transceiver vendors in the form of higher switch ASIC ASPs. That said, in the context of the company's larger AI franchise, we think optics are becoming less impactful to both financials and narrative.

Credo Technologies (CRDO, O'Loughlin): Credo remains the market leader in AECs, and we see Credo's optical DSP franchise as likely to grow strongly in the 1.6T generation (and even gain share in 800G and below, though more slowly). Credo is a key innovator driving adoption of linear receive optics (LRO), looking at a greenfield opportunity in half-retimed transceivers. We think successful diversification of its sales from AEC will be a critical driver for the stock, with the magnitude of the opportunity not fully appreciated by investors.

Intel (INTC, Buchalter): Intel had previously designed and manufactured silicon photonics transceivers before selling the business line to Jabil. Though Intel exited the *transceiver* business in 2023, it continues to invest in the PIC side of silicon photonics and has shipped more than 8 million PICs since 2016. That said, we see this as less impactful to Intel's overall franchise, both from a financial and narrative perspective.

MACOM Technology Holdings (MTSI, O'Loughlin): We see MACOM as well-aligned with our investing framework and well-positioned in the key debates. MACOM's lasers and high-performance connectivity solutions play a key role in important new categories such as active copper cables (ACC) and linear pluggable optics (LPO), with its CW laser portfolio well positioned to gain share as silicon photonics proliferates. While Datacenter is just one of three segments the company serves, we also see attractive secular growth drivers in its two other businesses and believe MACOM is well situated to deliver steady, diversified growth over time.

Marvell Technology (MRVL, Buchalter): We are cautious on Marvell's DSP and optics franchise, despite its leadership position. As the industry moves towards faster data rates, we see newer technology such as linear pluggable optics (LPO) and linear receive optics (LRO) and increased competition in DSPs that will challenge Marvell's incumbent market share, which could result in material content loss for the company. Silicon photonics could present a growth opportunity that offsets share loss with content elsewhere in the transceiver, but we think this debate is in its early stages. Further, while we expect AECs to be a strong market opportunity that enables the longer reach of copper, we think Marvell's lower market share mutes this impact comparatively.

Microchip (MCHP, Buchalter): Microchip provides a wide range of products for datacenter interconnect networks; for example, Microchip's META-DX family of devices consists of multi-purpose Ethernet MACs/PHYs that support rates from 1–800 GbE with up to 2.4T capacity. Microchip also supports retimers, gearboxes, and flexible I/O to enable connectivity to a variety of optical modules, DACs, and Ethernet switches. However, in the context of its broader end market exposures and product portfolio, we see Microchip as a more limited beneficiary of the uplift in datacenter connectivity.

NVIDIA (NVDA, Buchalter): At GTC 2025, NVIDIA had announced its initial CPO-enabled switch platforms, starting with Infiniband switches with Ethernet versions to follow. We anticipate these initial CPO deployments will be relatively low volume and, despite NVIDIA's growing presence in the scale out and scale up markets, a smaller driver of the company's overall financial performance and stock narrative in the near-medium term.

Semtech Corp. (SMTC, O'Loughlin): We see Semtech as well-positioned in datacenter connectivity and poised to further lean into this attractive growth story. Specifically, we believe Semtech's Signal Integrity products across FiberEdge and CopperEdge are industry-leading. Semtech's TIAs and linear equalizers put it in pole position to capture share in important new categories such as active copper cables (ACC) and linear pluggable optics (LPO), but its opportunities are not limited to these form factors as analog content remains key in both retimed and linear optical transceivers. We believe Semtech has the potential to generate >\$1B from Datacenter alone by 2029.

STMicroelectronics (STM, Buchalter): STMicroelectronics has positioned its silicon photonics business as a key enabler for next-generation optical modules, providing photonic integrated circuits (PICs) developed in collaboration with AWS that support 200 Gbps per lane, enabling future 800 Gbps and 1.6 Tbps pluggable optical solutions. We see this as an early, but emerging opportunity for STMicroelectronics, as AWS will be working with ST and a leading optical module provider to use the PIC100 in its datacenters; however, we think it is unlikely to be a needle-mover for the company in the near-medium term given the size of its other businesses.

Other Key Components & Processing Companies Not Under Coverage

Lasers, Transceivers, & Pluggable Optical Module Manufacturers: In addition to the companies above, the datacenter connectivity ecosystem is enabled by a number of other component suppliers, manufacturing lasers (such as Lumentum and Coherent), glass (such as Corning), and other optical transceiver content and packages providers (such as Amphenol, Fabrinet, Applied Optoelectric, Hisense, Source Photonics, Innolight, and many others). We discuss the interconnects specifically [here](#).

Photonics-Focused Startups & Pure Plays: Photonics especially is an area of start-up investment and innovation, with interesting ideas coming out of the labs and beginning to come to market. These companies are typically smaller cap or even private, but we are monitoring the developments in this area as a potential source of disruption for the incumbents we focus on elsewhere. A non-comprehensive list of examples includes Avicena, Ayar Labs, CelestialAI, Lightmatter, and POET Technologies.

Layer 2: Data-Link Layer Processing & Switching

Advanced Micro Devices (AMD, Buchalter): AMD expanded its portfolio into networking with its acquisition of Pensando ([here](#)), bringing SmartNICs and DPUs into its roadmap to provide a more optimized, full-stack solution. We see AMD as less impacted by the switch market given switches are not currently part of AMD's core product portfolio and is thus not a direct participant of the scale up market for either UALink or SUE. That said, AMD is a founding member of the UALink consortium and has donated its own Infinity Fabric to the organization. Longer-term, we see AMD continuing to advance and use UALink at the data-link and transport layer, with physical layer optionality.

Broadcom (AVGO, Buchalter): Broadcom is a leader in datacenter routing and switching applications, we think for both scale out and scale up. As scale up becomes a more significant market, we think Broadcom will emerge as the leading open-standard Ethernet (and PCIe) switch player given its early time-to-market with Scale Up Ethernet. With our expectations that Broadcom will become the second largest market share player (behind NVIDIA), we see a significant opportunity for Broadcom in the long term.

Cisco (CSCO, Not Covered): Cisco is a leading provider of switching and routing hardware, and the Internet is, to a large extent, delivered to the world on the back of Cisco systems. Specific to silicon, Cisco historically had leveraged its own switching ASICs and routing fabrics, selling its switches as a bundle. Broadcom's entrance into the merchant switching silicon market effectively enabled the creation of the "white box" switching market that was much closer in nature to the server market (in which a CPU from AMD/Intel is assembled into a server built by Dell/HPE/Supermicro). In 2019, Cisco announced its Silicon One merchant silicon offering and participates in the datacenter connectivity market primarily in scale out Ethernet and optics.

Intel (INTC, Buchalter): We see Intel as a limited beneficiary in the growth of the data-link layer processing and switching markets. Though the company had gained exposure through its 2019 acquisition of Barefoot Networks, which produced programmable network Ethernet switch silicon, Intel has since halted production on its networking chips in 2023. The company has a small portfolio of NICs and IPUs, but this remains a smaller portion of the company's business in comparison to its CPU franchise.

Juniper Networks (Now Part of HPE, Not Covered): Similar to Cisco, Juniper has historically developed and used its own silicon inside its switching and routing equipment. While the combined HP Enterprise/Juniper strategy remains early days, Juniper largely does not participate in the datacenter connectivity market in a meaningful way and is much more service provider and enterprise-focused.

Marvell (MRVL, Buchalter): We come away from our work neutral on Marvell, as we see the company as a limited participant in the scale up market, with minority share in UALink, PCIe, and Ethernet switches (~10-20%). That said, we still expect revenue to grow at a healthy clip (off limited contributions) through the end of the decade to >\$700M, primarily as a result of participation in the Ethernet market. While we estimate this as ~20% of Marvell's networking infrastructure silicon revenue, we note this would only represent ~5% of total datacenter revenue in 2030E, limiting its impact on Marvell's overall revenue and earnings power.

NVIDIA (NVDA, Buchalter): With its early mover advantage, NVIDIA has created a strong position in scale out with its Infiniband and Ethernet offerings and is the market incumbent with its NVLink offering. In particular, we see the proliferation of NVIDIA-based systems as a wide moat for the company in the scale up switch market. Net, even as other offerings come to market, we see NVIDIA maintaining its market leadership at ~70% share through the end of the decade, with scale up switching revenue growing to almost \$16B (~30% CAGR 2025E-2030E).

Non-Silicon Data-Link Layer Hardware & Contract Manufacturers

In addition to silicon vendors, the link-layer is enabled by a significant number of hardware vendors and contract manufacturers that leverage the processors of the companies above and incorporate them into systems. These include such contract manufacturers as Celestica, Flex, Wistron, Accton, and more.

VLAN Tagging: Model & Implications

"All models are approximations. Essentially, all models are wrong, but some are useful."

-George Box

Our full working Excel model is available on request and is reproduced in full at the end of this report, [here](#).

While the theoretical first principles of Amdahl's Law and the mathematics of all-to-all networks give us confidence in the long-term growth drivers in networking overall, to speak with confidence regarding any specific outcome requires a more granular exploration of the potential outcomes. To that end, we have developed a bottom-up, ports-based model that attempts to **unify ports, switches, cables, and most importantly revenue** across the datacenter connectivity space.

We integrate discussion of specific point estimates throughout the rest of the report where appropriate, but from a high-level we note four key highlights that emerge from the work: (1) the magnitude of datacenter networking infrastructure is striking, with the revenues approaching the **size of the whole automotive semis end market**; (2) an **entirely new TAM in scale up switching** is scaling to multiple tens of billions of dollars by 2030; (3) despite the clamor surrounding LPO and CPO, **the optical DSP is far from dead**; and (4) likewise despite the intense focus on 200G SerDes, **100G lane rates will retain the lion's share of connections** through at least 2028.

Model Takeaway #1: Datacenter Networking Infrastructure Growth Is Significant

In an era during which eye-popping growth numbers for AI-related projections have become almost routine, we are struck by the sheer magnitude of the market, even when systems-level revenue uplift is removed and we focus exclusively on silicon. Specifically, at ~\$75B in 2030, this silicon-only TAM is effectively the size of the current automotive semis end market. This is admittedly a self-fulfilling prophecy given the existence of this report, but we believe our model demonstrates a strong justification for this TAM based on its connectivity to our broader Datacenter Silicon Model.

Model Takeaway #2: Scale Up Switching a Multi-Billion Dollar Opportunity

Scale up switching is one of the few segments of the model that is truly greenfield for 2025+. As an identifiable revenue category, it was reported into being when NVIDIA made the decision to segment its fifth generation NVLink Switch into its Networking revenue rather than Compute (reflective of the switch silicon now commanding its own tray rather than being housed on the base board). Forecast to grow at a ~38% CAGR to 2030, scale up switching is **one of the fastest growing buckets in our model** and one in which there is **no true incumbent** outside of NVIDIA. We take the view that Scale Up Ethernet is likely to see broad adoption, but we also categorize this as an "emerging" debate and we have relatively lower conviction levels on this conclusion—and if we're wrong, there is potentially a **>\$8B market up for grabs** for non-NVLink switching.

Model Takeaway #3: The Debate is Over—Scale Out is Ethernet

In contrast to scale up, which remains an open debate, we see scale out as having largely coalesced around Ethernet rather than NVIDIA's proprietary Infiniband. Specifically, we forecast that Ethernet is likely to outgrow Infiniband in 2025 for the first time since the "ChatGPT moment," and believe this paradigm shift is likely to persist to at least 2030. We believe that NVIDIA itself has embraced this transition, having recently highlighted the momentum around its Ethernet portfolio ([here](#)), and we believe NVIDIA will remain a key competitor in the Ethernet connectivity market on both the NIC and the switch side.

Model Takeaway #4: Optical DSPs Remain Alive and Well

Seemingly a steady drumbeat since OFC 2023 has been the imminent Death of the DSP in optical links. However, when you attach numbers to the discussion, this appears to be far from the case. In fact, when you include linear receive solutions (a half-DSP if you will), we anticipate PAM4 DSPs to grow at a healthy ~15% CAGR to 2030—hardly an ex-growth market.

Model Takeaway #5: 100G Per Lane SerDes Still Dominant Through 2028

Finally, we note that while investors are eagerly tracking the arrival of 200G per lane SerDes on both switch and transceiver platforms (likely to arrive in volume by 2026), we view 100G SerDes as likely to remain the predominant lane speed through at least 2028. Incidentally this aligns with commentary from such management teams as Credo's, who have in the past noted they viewed their opportunity on 100G and even 50G to be just as large near-term as those on 200G. One might say that our model... lends *credence* to this view.

Model Methodology: Port Growth Tied to Accelerator Forecast

The foundation of our model is a by-speed port-level forecast, informed by our accelerator forecast and paired with a by-SerDes lane switch ASIC forecast. For each port speed, we then estimate the respective share of physical media across copper (active and passive) and fiber (retimed, linear, and co-packaged). Finally, we estimate market share for each major category of revenue and then tie these market share estimates to reported or estimated networking/datacenter revenues for NVIDIA, Broadcom, Marvell, Credo, Astera Labs, Semtech, and MACOM.

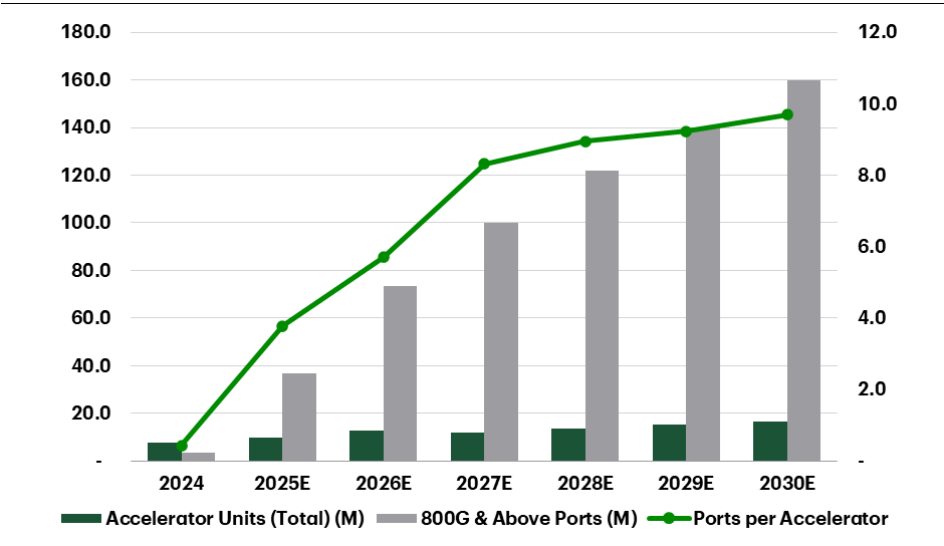
We leverage third-party data providers 650 Group, Omdia, and Lightcounting for historical data and some forecasts, but where our view differs from what we believe is that data provider's view, we adjust as required.

One of the most important aspects of our model is illustrated in rows 61 to 70 (Fig. 7, below). We explicitly tie our forecast for 800G and above ports to our accelerator unit forecast, noting an implied Ports per Accelerator metric based on current inputs. Importantly, we believe **Ports per Accelerator should increase over time as cluster sizes grow**. This is an important tenet of all-to-all connected networks and is something we have discussed in the past ([here](#)), but we feel it bears highlighting as it is a material driver of our above-consensus high-speed port forecast.

The specific number of ports required for a given cluster size is not a straightforward calculation, as it depends on the specific network architecture as well as the radix of switches being used. Leveraging discussion from Enfabrica, we estimate that a 32K XPU cluster would require ~2.5x ports to XPUs, while a ~100K XPU cluster would require >9x ports for each XPU. To achieve a 1M XPU cluster (which a number of industry participants have described as a goal for the C2027 timeframe, [here](#)), the number of ports required per XPU likely approaches 20x. Clearly, not every accelerator shipped in 2027 will be to a 1M XPU cluster, but we incorporate a growing "attach rate" for ports to accelerators in our modeling.

The growing ratio of ports to accelerators is **the key assumption** in our model, as it directly informs our view of port growth for 800G and above. This in turn flows through to our assumptions on the size and growth rates of key connectivity markets such as AEC, ACC, and LPO/LRO, and ultimately company revenue.

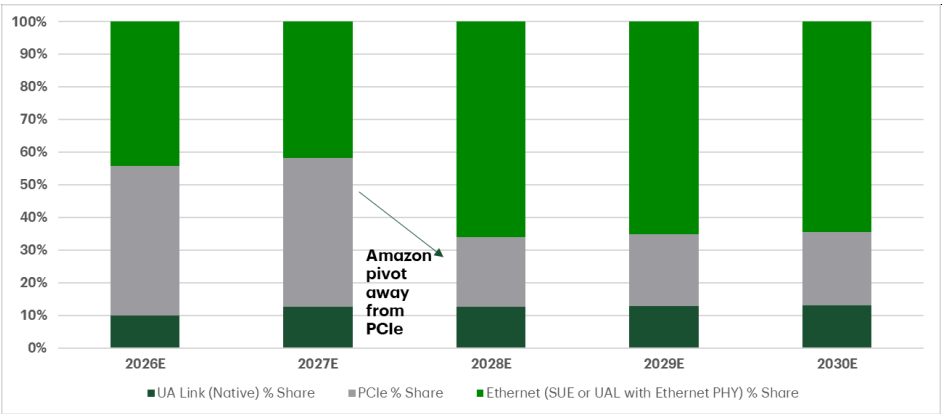
Figure 7 Our Port-Based Connectivity Model Contemplates Growing Cluster Sizes



Source: TD Cowen estimates

Similarly, our assumptions regarding scale up switch protocol choices drive large shifts in revenue between buckets and by proxy between companies. Specifically, we think it is likely that Amazon will move off PCIe and opt for either Ethernet or UALink at the physical layer. In our model, we assume they opt for Ethernet (either SUE or UALink-over-Ethernet as AMD is currently implementing). This drives a 900 bps increase in scale up switching share for **Broadcom** and results in Astera's 2028 scale up switching revenue to decline year-over-year. Clearly this is a high level of precision for a forecast three years away, and we do not claim clairvoyance but rather highlight this modeling artifact to emphasize this was a deliberate choice we made (and to engage in debate on this topic).

Figure 8 Our Scale Up Estimates Contemplate a Rapid Share Shift Away from PCIe Toward Ethernet-based Switching Driven by Amazon



Source: TD Cowen

Mechanically, the model is divided into four high-level functional sections:

- **Section A: Ports & Plugs** establishes the baseline for the rest of the model by estimating specific numbers of ports shipped (annually) on both switch and NIC side. These ports are then allocated physical media across passive copper (DAC); ACC, AEC, and AOC integrated cabling solutions; optical transceivers (fully retimed, half-retimed, and linear); and co-packaged optics. Doing this *by-speed* enables us to specifically allocate share of the various physical media to its appropriate use case. For example, the share of passive copper rapidly declines as line rates increase as passive cables become reach-limited beyond 100G lane rates (so 800G+ line rates).
- **Section B: Layer 1 Cables & Link Layer Processors** takes the aggregated estimates from Section A and applies ASP and revenue share assumptions to the major categories of connection. Further, we estimate line cards and on-board retimers/redrivers (admittedly with much lower precision).
- **Section C: Layer 2 & 3 Switches and NICs** estimates scale out and scale up revenues attributable to the network layer processors and fabrics. We estimate Ethernet switch ASIC revenue based on SerDes lanes shipped—and importantly, we harmonize the total bandwidth shipped with our estimates in Section A.
- **Section D: By-Company Revenue Builds** takes the market-level forecasts (and the revenue shares we have estimated for each major category) and aligns these figures to the reported or estimated Datacenter/Networking revenue each company has described. This requires reconciliation from outside the model at times (for example, to include Marvell's storage controller franchise), but we believe represents a reasonable estimate for the composition of each's respective datacenter connectivity franchise.

Overall, our resulting model contemplates a networking TAM of ~\$78B in silicon only by 2030, growing from an estimated ~\$18.5B in 2024 and \$30.1B in 2025E, representing a 21% CAGR (2025 to 2030). This is a **silicon-only** estimate, as we attempt to segment out systems-level revenue. Specifically, we attempt to estimate the value of the ASIC inside NVIDIA Spectrum-X switches and DPUs, though they are sold as full systems. This allows us to compare NVIDIA's Ethernet franchise on an apples-to-apples basis with Broadcom's, for example. Likewise in the AEC market, we estimate the ASP of just the chipset itself, recognizing Credo goes to market with a full cable solution while Astera and Marvell recognize the DSP/module only.

Figure 9 Our Expanded Datacenter Networking Infrastructure Model

Total Networking Infrastructure Silicon	2021	2022	2023	2024	2025E	2026E	2027E	2028E	2029E	2030E	CAGR (25-30)
Section B: Layer 1 Physical Connectivity											
Cable Chipsets (ACC, AEC, AOC)	\$ 129	\$ 236	\$ 225	\$ 378	\$ 1,235	\$ 1,792	\$ 1,993	\$ 2,124	\$ 2,219	\$ 2,376	14.0%
Optical Chipsets (DSP + Analog incl LRO, LPO, CPO)	\$ 635	\$ 785	\$ 935	\$ 1,680	\$ 2,828	\$ 3,974	\$ 4,570	\$ 5,010	\$ 5,828	\$ 6,713	18.9%
Line Cards & Retimers (Ethernet and PCIe)	\$ 1,109	\$ 1,131	\$ 1,489	\$ 2,186	\$ 2,766	\$ 3,454	\$ 3,770	\$ 4,227	\$ 4,700	\$ 5,129	13.1%
Total Layer 1 Physical Connectivity	\$ 1,873	\$ 2,151	\$ 2,648	\$ 4,244	\$ 6,828	\$ 9,220	\$ 10,332	\$ 11,361	\$ 12,747	\$ 14,217	15.8%
Section C: Layer 2 & 3 Data & Network Layer Processing											
Ethernet Switch ASICs & NICs	\$ 6,445	\$ 7,865	\$ 9,276	\$ 10,715	\$ 15,421	\$ 21,410	\$ 24,761	\$ 29,296	\$ 33,307	\$ 37,185	19.2%
Infiniband Switch ASICs & NICs	\$ 314	\$ 538	\$ 2,159	\$ 3,198	\$ 3,624	\$ 3,306	\$ 3,204	\$ 3,009	\$ 2,820	\$ 2,671	-5.9%
Scale-Up Switch ASICs	\$ -	\$ -	\$ -	\$ 385	\$ 4,278	\$ 7,080	\$ 9,828	\$ 13,868	\$ 19,103	\$ 24,046	41.2%
Total Layer 2 & 3 Data & Network Layer Processing	\$ 6,759	\$ 8,404	\$ 11,435	\$ 14,299	\$ 23,323	\$ 31,796	\$ 37,793	\$ 46,173	\$ 55,231	\$ 63,901	22.3%
Total Networking Infrastructure Silicon Revenue	\$ 8,633	\$ 10,555	\$ 14,083	\$ 18,543	\$ 30,151	\$ 41,017	\$ 48,125	\$ 57,534	\$ 67,978	\$ 78,119	21.0%
Additional Systems-Level Revenue (Switch Boxes, Cables, etc.)	\$ 6,587.19	\$ 9,002.55	\$ 14,168.82	\$ 21,978.28	\$ 29,171.83	\$ 45,752.29	\$ 54,958.34	\$ 67,711.84	\$ 81,839.34	\$ 96,122.40	26.9%
Less: AI ASICs (Broadcom & Marvell)	\$ (1,500.00)	\$ (1,893.59)	\$ (3,144.28)	\$ (10,042.52)	\$ (16,295.30)	\$ (31,796.05)	\$ (40,847.13)	\$ (53,909.70)	\$ (69,096.98)	\$ (84,790.85)	39.1%
Total Networking Infrastructure Revenue (\$M)	\$ 13,720	\$ 17,664	\$ 25,107	\$ 30,479	\$ 43,028	\$ 54,973	\$ 62,236	\$ 71,336	\$ 80,721	\$ 89,450	15.8%
y/y Growth (%)		29%	42%	21%	41%	28%	13%	15%	13%	11%	

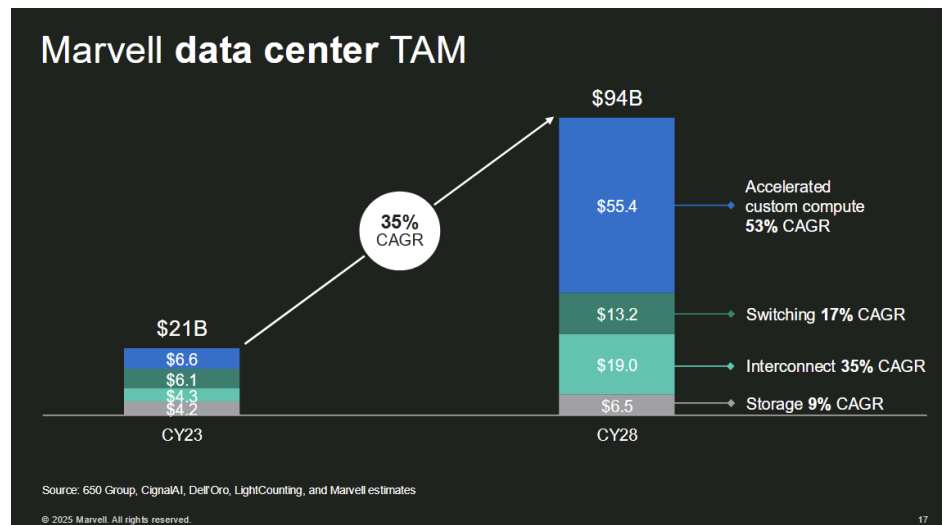
Source: Lightcounting, 650 Group, TD Cowen estimates

Forecast Comparisons: Where We Differ

Overall, we believe our model is likely above investor expectations for networking infrastructure, though we acknowledge the wide range of estimates that exist for total market size and the rapidity with which those estimates are being revised. We believe three specific sub-segments of our model are likely materially different than the broad market consensus: (1) we believe our scale up switch expectations are higher (and skew more towards Ethernet) than many contemplate; (2) we believe our explicit modeling of Linear Receive Optics (LRO) adoption is unique and the contemplated pace is likely more rapid than many expect; and (3) conversely our expectations for CPO penetration are likely below market consensus, to the extent that investors were modeling such an adoption.

Regarding the total market, the most recent assertion regarding the size of what we would call the Networking Infrastructure segment of datacenter came from Marvell during its recent Custom Silicon Webinar ([here](#)). During the event, management described a ~\$94B TAM for the company across custom compute, switching, interconnect, and storage. From our perspective, we believe the Switching and Interconnect buckets largely align with our model, and we do not model the storage controller market beyond our estimates for its contribution to Broadcom and Marvell's respective revenues.

Figure 10 Our Forecasts are Materially Higher than Marvell's TAM Expectations, Though We Recognize Potential Overlap Between Custom Compute & Scale Up



Source: Marvell

Our **\$57.5B** modeled TAM is clearly higher than the ~\$32B in the Switching and Interconnect buckets described by Marvell. However, during the event Marvell described an additional ~\$15B of "Custom XPU Attach" that potentially overlaps with other parts of our model such as scale up switches and network interface cards. The resulting ~\$47B TAM remains ~20% smaller than our estimate. Some third-party estimates are more optimistic, such as those from 650 Group (whose data we leverage in constructing our "actuals") while others are more pessimistic. We think Marvell's TAM likely represents somewhere in the middle.

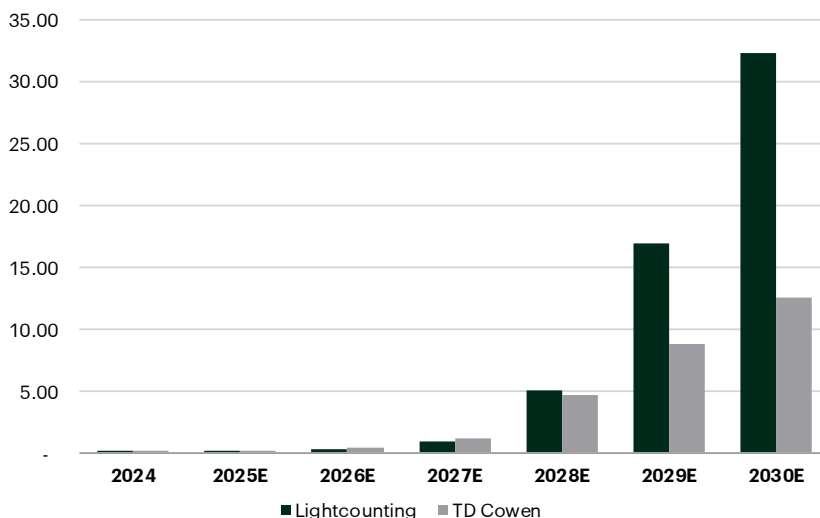
When we consider the specific areas where we believe we are materially different than market expectations, there is an even greater paucity of "consensus." Scale up networking is perhaps the least well-defined sub-segment as it has only recently come to the forefront of investor conversations. In fact, we believe the first use of the term "scale up networking" that appears in a transcript from a company *not* named NVIDIA was during Broadcom's March 2024 AI Infrastructure Day—and the scale up protocol being discussed was PCIe.

During a recent UALink investor event, Astera Labs estimated the scale up specific Scorpio-X switch had a TAM of ~\$2.5B by 2028 (i.e., scale up switching ex-NVLink). This TAM would likely include all accelerator scale up environments other than NVIDIA's (which uses NVLink) and Google's TPU (which uses a unique optical circuit switch topology for scale up). We model non-NVIDIA, non-Google scale up switching of >\$4B by 2028 based on our expectations for ~6.5M accelerator units at that time (with the discrepancy likely due to differences in ASP assumptions). Conversely, on the high side, Broadcom has commented that its scale up opportunity could be 5x to 10x its scale out opportunity. Perhaps over time that may prove to be the case, but we think Broadcom should give themselves more credit—in our model we estimate an Ethernet switch ASIC market still >2x larger than scale up by 2030.

On Linear Receive Optics, we have not encountered any specific market forecasts for that market (again unsurprising given it is a single vendor with a product only a year into its launch). We believe LRO will become increasingly relevant as the 1.6T transceiver market matures.

Finally, on CPO, we believe Lightcounting's forecasts are well regarded by investors (and we certainly do not claim any intellectual authority or superiority with respect to their work which we admire greatly). However, we remain skeptical regarding the adoption of CPO at scale, despite its technical benefits. Thus, by 2030 despite otherwise similar overall unit estimates, our CPO penetration is approximately half that of Lightcounting's.

Figure 11 Our CPO Port Forecast is ~50% Lower Than Lightcounting's



Source: Lightcounting, TD Cowen

Company-Specific Implications

Figure 12 Implications of Our Industry Model at the Company Level

	2021	2022	2023	2024	2025E	2026E	2027E	2028E	2029E	2030E	CAGR (25-30)
Broadcom											
Total Broadcom Networking Revenue (Reported/Est.)	\$ 7,382.28	\$ 9,300.00	\$ 11,825.50	\$ 17,101.59	\$ 25,158.94	\$ 45,166.01	\$ 55,301.16	\$ 71,360.66	\$ 88,546.78	\$ 104,653.74	33.0%
Marvell											
Total Marvell Datacenter Revenue (Reported/Est.)	\$ 1,785	\$ 2,409	\$ 2,217	\$ 4,151	\$ 6,009	\$ 6,853	\$ 8,185	\$ 9,497	\$ 11,144	\$ 13,754	18.0%
NVIDIA											
Total NVIDIA Networking Revenue	\$ 1,566	\$ 2,718	\$ 8,575	\$ 12,990	\$ 19,673	\$ 21,781	\$ 25,299	\$ 28,655	\$ 32,620	\$ 37,778	13.9%
Credo											
Total Credo Datacenter Revenue	\$ 57	\$ 143	\$ 128	\$ 288	\$ 839	\$ 1,263	\$ 1,577	\$ 1,873	\$ 2,147	\$ 2,394	23.3%
Astera											
Total Astera Datacenter Revenue	\$ 81	\$ 117	\$ 396	\$ 775	\$ 1,287	\$ 1,472	\$ 1,487	\$ 1,806	\$ 2,012	\$ 2,012	21.0%
Semtech											
Total Potential Semtech Datacenter Revenue	\$ 44.94	\$ 58.70	\$ 74.77	\$ 142.13	\$ 278.63	\$ 443.28	\$ 590.44	\$ 735.81	\$ 869.97	\$ 1,017.18	29.6%
MACOM											
Total MACOM Datacenter Revenue	\$ 133.70	\$ 149.00	\$ 157.06	\$ 213.60	\$ 313.15	\$ 436.92	\$ 529.45	\$ 605.93	\$ 705.00	\$ 811.03	21.0%

Source: TD Cowen

Astera Labs (ALAB, O'Loughlin): We model Astera Labs as putting together a ~21% CAGR to 2030 and exceeding ~\$2B in revenue by that time. This growth is likely primarily driven by its Aries retimers initially, though scale up-focused Scorpio is a major driver in 2026 and 2027. Towards the end of the decade, we anticipate a potential deceleration in Scorpio growth as the scale up market coalesces around the Ethernet physical layer and competition from Broadcom and Marvell intensifies.

Broadcom (AVGO, Buchalter): We model Broadcom's "Networking" revenue to exceed \$55B by 2030, but importantly at least \$30B of this figure is based on our expectations for the company's custom silicon programs, especially TPU. We leverage Broadcom's old segment reporting structure, which is why this accelerator revenue ends up in our "Networking" estimate. Specific to datacenter infrastructure silicon, we model strong growth for Broadcom, with a CAGR of 26% expanding its connectivity franchise to >\$22B as its switch ASICs continue to dominate the merchant market (that now includes scale up in addition to scale out). Specifically, we believe by 2030 Broadcom could generate ~\$9B from scale out and front-end switching, \$3B in server connectivity (NICs), and \$5B in scale up switching. Frankly, all of this will be irrelevant to the stock if CEO Hock Tan's expectations for AI revenue come to fruition, but paradoxically it remains that Broadcom is the leader in datacenter networking infrastructure...while accelerators will be what matters most to AVGO.

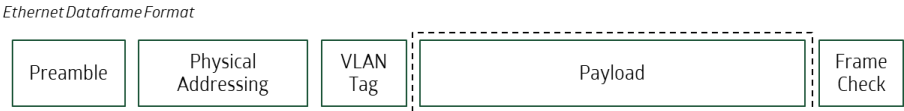
Credo (CRDO, O'Loughlin): We see Credo's revenue as likely to grow at a ~23% CAGR to ~\$2.4B by 2030, though we would anticipate this is likely a conservative estimate given what we see as the strong arguments in favor of AEC and LRO adoption. We believe that investors underappreciate the extent to which Credo will have an opportunity to participate in the optical DSP market, and in fact we model that part of Credo's business as growing >40% annualized (off a smaller base, of course).

MACOM (MTSI, O'Loughlin): We model MACOM's Datacenter segment as expanding to >\$800M by 2030, more than doubling with a 21% CAGR. Importantly, this \$800M does **not** include material contribution from the company's CW laser portfolio given our lack of visibility into that market (for now). As we describe below ([here](#)), the increasing adoption of silicon photonics and therefore CW lasers could drive a multi-hundred million dollar TAM for MACOM by 2030. This could represent important upside to our modeled growth rate.

Marvell (MRVL, Buchalter): We model Marvell's Datacenter segment (which now far exceeds 70% of the overall business) as likely to grow at a ~17% CAGR to 2030—the lowest growth rate in our model. Even more importantly, much of this growth is, like Broadcom, from custom silicon. Driven by headwinds in its optics business only mildly offset by potential participation in scale up switching, we see Marvell's *non*-ASIC Datacenter business as growing just 6.5% annually to 2030. This could prove conservative (especially in the form of increasing focus on pluggable coherent optics for long-haul fiber and datacenter interconnect), but in a model that has many elements of such conservatism, the delta between market and Marvell's growth is stark. We discuss these dynamics more in our separate downgrade of the company to Hold.

Semtech (SMTC, O'Loughlin): We see datacenter (currently reported as a part of Semtech's Infrastructure end market) as a strong growth driver for the company, approaching nearly 30% CAGR and exceeding \$1B by the end of the decade. This is largely due to our modeled growth for optical transceivers, with an additional tailwind from active copper. On-board linear equalizers (such as those potentially being used by NVIDIA in future rack-scale architectures) would represent upside to our model given the difficulty associated with granularly estimating the size and penetration of such redriver solutions.

PAYLOAD



IP & Diversification: Our Investment Framework for Datacenter Connectivity

The Key Debates: Cycle, Copper, and Co-Packaging

The Emerging Debates: Scale Up and Photonics

IP & Diversification—Our Investment Framework for Datacenter Connectivity

Our coverage of datacenter connectivity is an organic outgrowth of our datacenter compute coverage, and our investment framework is informed by this origin. We take the view that (1) technological differentiation drives gross margins which drives returns, and (2) customer diversification is difficult but valuable.

Differentiate in the Physical Layer

Semiconductor companies serving datacenter applications have always competed in the three dimensions of price, performance, and power. Datacenter operators prefer the lowest cost, best performance, and lowest power (often having to optimize for two out of the three). We view datacenter connectivity as being defined by these same characteristics: seeking the fastest data rates with the lowest error, lowest power, and lowest cost.

Datacenter compute in general is amidst two simultaneous paradigm shifts: the adoption of Generative AI throughout the economy, and the increasing heterogeneity of hardware being deployed to accelerate workloads that were formerly software-defined and run on the CPU. This is a topic we discuss at length in our datacenter architecture deep dive, *The Next Datacenter Decade* ([here](#)). Datacenter connectivity is likewise experiencing and is in some ways enabling these paradigm shifts.

The fundamental principle of delivering the lowest power, highest performance solution—supercharged by the advent of GenAI and heterogeneous computing—leads us to conclude that those companies that can deliver differentiation at the most fundamental level are most likely to gain share at an attractive margin.

Further, the largest investors in AI infrastructure (and therefore the most significant growth drivers for network infrastructure spend) have **demonstrated a willingness to "do it themselves"** where advantageous to their overall compute footprint. The most salient example of this is the many projects developing custom accelerators today, but consider also the white box network hardware market. If it becomes important enough and can be achieved with merchant/commodity silicon or merchant IP, we believe it will be **difficult for those suppliers to maintain a healthy gross margin profile**.

When viewed through this lens, we see Credo, MACOM, and Semtech as being well positioned given their respective portfolios of SerDes IP, laser and RF tech, and analog/mixed signal expertise. Conversely, we see Astera's software-defined architecture as providing other benefits (such as fast time to market and observability) but we believe this may limit the company's ability to compete at the very bleeding edge of data rate speeds.

Credo: SerDes IP Enables Multiple Market Opportunities

TD Cowen View on Impact/Positioning: **Positive**

Credo was founded in 2008 by a team of mixed signal connectivity designers from Marvell and has developed a portfolio of high performance serializer/deserializer IP solutions that underpin the company's entire product portfolio. In fact, Credo began as an IP provider and pivoted into developing products more recently. The company owns 78 patents with 21 pending, primarily related to the Ethernet standard, cable technology, and SerDes cores.

The company's mixed signal-heavy approach to SerDes and DSP development has allowed it to deliver high-performance connectivity solutions at either lower power or higher process geometries than competitors. In fact, until recently Credo had pursued an "n-1" process tech strategy, optimizing for cost by developing on a process geometry one node behind its peers (thus delivering a cheaper solution).

Today, Credo has moved off its n-1 strategy, not because it has needed to progress to the leading edge node to match its competitors, but rather because it sees an opportunity to *beat* its competitors on power and performance at these smaller nodes. Credo's potential at these lower process geometries was recently demonstrated at OFC ([here](#)), where the company showed a 3nm, 244G-based 1.6T DSP that was pulling <12W (compared to >15W from Marvell's solution).

Semtech: Historical Strength in Signal Integrity A Hidden Gem

*TD Cowen View on Impact/Positioning: **Positive***

Semtech was founded in the 1960s with a specific focus on military and aerospace applications. The company has cultivated a niche in mixed-signal and high-performance analog ICs, marrying silicon-on-insulator (SOI) processes with custom analog front-ends that span timing, power management, and signal-conditioning functions. This heritage of building application-specific building blocks has positioned Semtech as a supplier of "silicon primitives" that are later assembled into larger systems for telecommunications, industrial IoT, and datacenter connectivity.

Specific to datacenter connectivity, Semtech's differentiation rests largely on its portfolio of transimpedance amplifiers (TIAs). The company's TIAs are considered best-in-class, and Semtech is one of only a few companies offering eight-channel PAM4 53GBd linear TIAs (i.e., can enable 1.6T at 100G/lane). It owns at least 30 patents related to TIA and other electro-optic circuit design innovation, emblematic of the company's focus on the "dark arts" of mixed-signal/analog circuitry.

Especially notable is Semtech's deep domain expertise in exotic semiconductor processes and materials such as silicon germanium (SiGe) technologies from foundry partners such as Tower Semi. Having worked in SiGe for decades, Semtech's organizational knowledge regarding exotic analog design is deep and is maintained by much of the senior engineering talent having been with the company for more than thirty years (for example, CTO Mike Wilson joined Semtech in 1995).

MACOM: Strong Heritage & Broad Portfolio of Mixed Signal Solutions

*TD Cowen View on Impact/Positioning: **Positive***

MACOM traces its roots to the early 1950s, when its early founders leveraged their experience at the MIT Radiation Lab during WWII to consult for and supply the US Army Signal Corps with magnetrons. Over the ensuing seven decades the firm has built a catalog of mixed-signal components—ranging from GaAs and InP power amplifiers to silicon-photonics modulators—that serve aerospace, defense, telecom and datacenter markets. Its corporate tagline, "From RF to Light," captures the breadth of this heritage, as well as the unifying philosophy of the company's technology portfolio.

At the physical layer, MACOM's competitive moat is anchored in a portfolio of proprietary process-specific intellectual property. The company's GaAs and InP processes embed patented epitaxial structures and device geometries that enable low-noise figure amplifiers, high-linearity mixers and ultra-wideband photonic drivers with performance margins that are difficult for pure-silicon rivals to match. In silicon photonics, MACOM has recently demonstrated its unique combination of process and packaging expertise with the launch of its co-packaged 200G/lane photodetector/

amplifier IC in which the photodetectors are stacked on the TIA, offering space and power savings for module designers.

Because the company's patents (of which there are over 700) are deeply embedded in the device architecture rather than in peripheral circuit topologies, MACOM can differentiate even at the lowest physical-layer level where alternative vendors typically rely on off-the-shelf silicon foundries. While MACOM's markets are deliberately niche, its IP-centric strategy reinforced by a long lineage of custom semiconductor processes provides strong and stable foundation to deliver innovative differentiation in what the company describes as the "highest power, highest data rate, and highest frequency" signal applications.

Astera Labs: Software-Defined Architecture Gives Us Pause

TD Cowen View on Impact/Positioning: **Cautious**

Astera Labs was founded in 2017 with a vision to re-define connectivity solutions for AI. The distinguishing feature of Astera's entire product portfolio is its software-defined, platform-centric nature. The company describes its solutions as a "matrix of microcontrollers & sensors" that enables expansive programmability and observability functions for Astera's retimers and switches. Notably, Astera's IP is rooted in this software-definition rather than physical circuit design. Thus, its differentiation is primarily in the control plane and integration into its major customers' respective platform software stacks.

This software-defined approach comes with many benefits. The flexibility afforded by the greater levels of programmability allow Astera to develop "Platform Specific Standard Products," customizing in software while retaining the economies of scale of a standard product. Further, the use of licensed IP (as well as the software-defined architecture) allows Astera to often be first to market with solutions (such as new generations of PCIe, for example). Finally, its matrix of sensors gives hyperscale customers unprecedented visibility into link status and bit errors, all integrated into their existing monitoring stacks through Astera's COSMOS software suite. Because the Astera ICs expose fine-grained metrics (such as per-lane eye-diagram statistics) through a standardized API, operators can embed continuous health-monitoring loops into orchestration software and automatically trigger remedial actions without manual intervention. This level of observability supports the "self-healing" fabric concepts promoted by many hyperscale clouds.

We acknowledge that there are many different points of optimization that hyperscalers solve for—and to date, Astera has delivered its solutions at effectively the same power envelope and line rate speeds as the competition. Further, we note the very prominent example of Arista Networks, whose primary differentiation is in its Extensible Operating System rather than in the design of a proprietary switch ASIC.

Nevertheless, the reliance on externally licensed SerDes and physical layer IP means that Astera's competitive edge is contingent on its ability to sustain superior firmware and analytics capabilities, and perhaps more importantly, for those observability and programmability features to subsume concerns regarding power consumption and speed. In the broader context of trends in datacenter connectivity moving towards *more* analog solutions and *less* digital solutions (e.g., LPO), an inherently software-based approach gives us pause. Accordingly, while the company's software-defined architecture is demonstrating early success, we believe a more cautious assessment is warranted given Astera's limited proprietary ownership over the data plane's fundamental physical layer components.

Broadcom: Industry-Leading SerDes and Switching/Routing IP (Buchalter)

TD Cowen View on Impact/Positioning: **Positive**

Broadcom's industry-leading SerDes IP dates back to the early 2000s when the company was still Avago Technologies. Leveraging deep expertise in high-speed analog design, Broadcom was able to extend its lead from the first introduction of its 10 Gb/s optical module to the industry-first 100 Gb/s PAM-4 SerDes block in 2017 that powered the Tomahawk and Jericho families. Broadcom has continuously refined its SerDes IP for power-efficiency and signal-integrity, maintaining its leadership position and gaining a seat on the Optical Internetworking Forum (OIF) and IEEE 802.3 working groups, where many of its techniques have become reference implementations for datacenter, carrier-grade, and hyperscale networking equipment.

The company is also an industry leader in switching and routing IP, beginning with its acquisition of LSI's networking assets in 2014. The acquisition kicked off the company's StrataXGS switch family with a programmable, fully-meshed fabric. Combining these ASICs with a rich SDK, Broadcom has become the leading silicon supplier for the world's largest datacenter switches (e.g., Cisco, Arista) and carrier routers (e.g., Juniper, Nokia).

This legacy of co-evolving high-speed SerDes and flexible switching fabric IP, along with an emphasis on open standards, has hardened Broadcom's status as the backbone of modern Ethernet and programmable networking ecosystems, in our view.

Marvell: Broad IP Portfolio Enables Diverse Datacenter Optics And XPU/XPU Attach Business (Buchalter)

TD Cowen View on Impact/Positioning: **Positive**

Founded in 1995, Marvell's portfolio today is powered by its broad IP and over 10,000 patents worldwide. The company has a strong SerDes portfolio that delivers industry-leading data-rate density, power efficiency, and signal integrity across a wide range of protocols. The company's SerDes IP is highly configurable, supporting multi-lane aggregation, adaptive equalization, and on-chip clock-data-recovery, which enables OEMs to consolidate multiple physical-layer standards into a single silicon die. This flexibility has made Marvell a favored supplier for hyperscale cloud operators and telecom carriers that demand ultra-low latency and deterministic performance in optical transceivers, ASICs and system-in-package (SiP) solutions.

Beyond Marvell's SerDes offering, Marvell holds a wide portfolio of IP including switching fabric IP, high-bandwidth memory (HBM) controller IP, and security IP. The programmable switching fabric IP supports multi-tenant datacenter switches, high-performance NICs and emerging AI accelerator interconnects. Marvell's HBM controller IP tightly couples compute engines to HBM stacks, delivering up to several terabytes per second of sustained throughput. Finally, Marvell's security IP suite provides end-to-end protection for silicon at the hardware level without sacrificing performance.

Taken collectively, we see Marvell's IP portfolio, both internally developed and acquired over time, positioning the company to be a differentiated solutions provider. We note this is apparent not only in its ability to serve the datacenter optics business, but also in its early success in developing custom XPU and XPU attach silicon for hyperscale customers.

Diversification: The Double-Edged Sword of Serving Hyperscale

Supplying the hyperscale datacenter market is a paradox of scale and concentration. On the one hand, the sheer volume of compute, storage, and networking capacity being procured by these companies is part of what makes datacenter the most attractive end market is semis. Companies are faced with the reality that just a handful of customers account for the majority of annual unit shipments, and a single design win or loss can be material to the P&L, even for those with a multi-billion-dollar revenue base.

There is therefore a persistent tension between the diversified customer base that is typically pursued as an ideal for most semis companies and the transformational potential of serving high-speed connectivity for hyperscale datacenters. In our view, there is a certain extent to which this is simply a fact of life for most. As we often like to point out, \$4T market cap NVIDIA is customer concentrated as well. However, we believe that those companies who can achieve not only customer-level diversification (with at least the five North American hyperscalers) but also *project-level* diversification within each of those hyperscalers' footprints are likely to command, over time, a higher multiple due to the lower volatility of its financials.

When viewed through this lens, we see Astera as well positioned given its strong relationships with both accelerator partners and hyperscale customers. We note Credo as being relatively less-well positioned but see its diversification improving in the near-term as additional customers ramp. Attacking the problem from the complete opposite direction, we view MACOM's deliberate emphasis on its *non-datacenter* opportunities as likely to help mitigate the volatility possible in its Datacenter segment. Finally, we see relative balance for Semtech in this framework as the business itself is quite diversified, but that is set against a datacenter business that is likely very project-specific and concentrated. As we have observed in the past (with companies such as Monolithic Power and even Semtech itself), a broadly diversified business outside of datacenter can matter less to stock performance if the *narrative* becomes about AI.

Credo: Scars of Past Concentration Issues Fading as Growth Heals All Wounds

*TD Cowen View on Impact/Positioning: **Neutral (But Improving)***

Customer concentration has been the defining characteristic of Credo's four-year history as a public company. Its first material ramp with AECs was to support a pilot project at Microsoft in which the company was exploring using two switches at the top of the rack to improve availability for the servers in that rack. When the explosion in ChatGPT's popularity caused Microsoft to pivot to an all-hands-on-deck NVIDIA deployment (based on Infiniband and Mellanox connectivity), Credo was left holding the colloquial bag (of cables).

More recently, we sensed investors were wary of echoes of the past as Amazon-derived revenue exceeded 85% in F3Q25 (JanQ). Despite reporting a 12% beat in the print and 17% upside in the guide, the stock fell -14% T+1 as investors fixated on the extreme customer concentration. Concentration has come down from those extreme levels (35% as of the JulQ), but as a key supplier for Amazon's back-end scale out networking (we believe associated with Trainium build-outs), we see the potential for Amazon revenue to surge again through the Trainium 3 ramp.

Interestingly, Credo's customer concentration is a bit more apparent than others' as the company sells full cable solutions rather than just the chipset. This has the effect of end customers appearing directly in the company's financials. This is in contrast to others who are more typically shipping to a contract manufacturer or ODM (who are often dual-sourced for the same customer). Thus, all else equal, Credo would *appear* more diversified if they were simply shipping into the supply chain rather than to their end customer. However, we believe control over their supply chain has other benefits that far offset this purely cosmetic appearance of diversification. This includes greater flexibility and quicker turnaround time in the case of architecture adjustments, as well as a likely greater level of visibility into their ultimate customer's inventory and deployments.

That said, we do see Credo's diversification improving. Specifically, the company has spoken to having "three to four" >10% customers in the next few fiscal quarters as additional hyperscalers ramp and non-traditional players such as xAI continue to increase the scope of their direct datacenter investments. Further we also see Credo's strong optical DSP opportunity as likely to assist with the diversification from a project point of view.

Semtech: Skating to Where the Puck is Going, But You Have to Have Your Head Up Coming Across the Ice

*TD Cowen View on Impact/Positioning: **Neutral***

At a high level, Semtech's business is perhaps the most diversified in our coverage, given its highly diverse LoRa portfolio as well as the IoT Systems business acquired with the Sierra Wireless transaction. However, new CEO Hong Hou has, among other things, been deliberately emphasizing Semtech's opportunity in the datacenter, especially to support such applications as ACCs and LPO that have high analog content, benefitting Semtech's amplifier and equalizer product portfolio.

This is prudent, in our view, given the opportunity that we illustrate in the subsequent pages of this report and in our separate initiation. However, as we have seen, this can introduce volatility into the narrative as the early scale ramps for some of these more novel technologies will inherently be highly concentrated. This was what played out for the company in February 2025, as anticipated revenue from ACCs to support NVL36 x 2 deployment evaporated as the end customer decided against scaling that form factor. Consider that this had zero to do with anything in Semtech's control and was made for reasons exogenous to the ACCs themselves...and shares fell 31% in a day.

We believe Semtech's strategy is the right one, and in skating to where the puck is going it will be well positioned to benefit from the strong secular trends we describe in this report. But when coming across the ice looking for the outlet pass, if you don't keep your head up you run the risk of getting body checked to the ice (losing the puck in the process, at times).

MACOM: Deliberate and Consistent Messaging on Opportunities Outside Datacenter Has Kept Investor Focus Grounded

*TD Cowen View on Impact/Positioning: **Positive***

Like Semtech, MACOM's diversity is aided by its two large non-datacenter focused segments, specifically Telco and Industrial & Defense. However, in contrast to Semtech's deliberate emphasis of its datacenter exposure in recent quarters, MACOM management takes pains to remind investors that Datacenter is just one of three diverse end markets the company serves, and its growth prospects are healthy in each.

Industrial & Defense has been an especially valuable anchor during the past five years that have seen significant volatility (in both directions) for the company's Telco and Datacenter segments. Defense especially within I&D has been stalwart, with its sticky sockets and long product lifecycles. In fact, management has estimated that its I&D segment is now ~65% Defense vs in the past when it has been majority Industrial. This is reflective of the strong growth seen in such defense applications as EW and radar through what has been a difficult period for most semis companies serving the Industrial market.

We believe MACOM's diversification and its deliberate messaging is likely to dampen its volatility, which has the potential to afford it a higher long-term earnings multiple over time.

Astera Labs: The Interoperability King

*TD Cowen View on Impact/Positioning: **Positive***

For a datacenter connectivity pure play, we believe Astera's revenue base is highly diversified. The company boasts engagements with each of the major hyperscalers and ships directly to ODMs and contract manufacturers supporting NVIDIA and AMD as well. To support its product development and ensure seamless deployment of its solutions, Astera has established an Interoperability Lab to rigorously test functionality and compliance with end points from across the compute industry.

This hyperfocus on interoperability enables Astera's customers to deploy Astera solutions with relative "plug and play" ease. We believe this has been key in driving Astera's success with multiple customers—because Astera's initial ramp was serving sockets that have an established market (PCIe retiming) and its interoperability meant it could easily serve as a second source, the barriers to adopting and qualifying Astera products are likely materially lower than peers.

Going forward, Astera faces the same customer concentration "victims of their own success" risks as others such as Credo, especially as the company's Scorpio-X PCIe switches ramp (we believe likely to be relatively concentrated at Amazon). However, the company's early success and traction across hyperscalers *and* with the accelerator providers position Astera well to scale beyond its early adopters.

Broadcom: Diverse Business Outside of AI, But TPU Remains The Main Driver For Now (Buchalter)

*TD Cowen View on Impact/Positioning: **Neutral-to-Positive***

Broadcom's switching and routing business serves an exceptionally diverse customer base that spans the full spectrum of the networking ecosystem. The company's portfolio serves hyperscale cloud operators, large telecom carriers, and both large and mid-market enterprise customers. The company has a global customer reach with both direct sales to OEMs/ODMs and channel sales.

That said, while it does have multiple customers/projects (including Meta's MTIA custom chip), the vast majority of Broadcom's custom business is currently tied to Google's TPU custom accelerator. We see this franchise as an outsized driver of the company's AI business currently (even with the incumbent switching and routing position). However, we do note that management expects this business to diversify meaningfully following a 4th customer's (widely believed to be OpenAI) recent conversion to production order, set to ramp to \$10B in rack-level revenue in Broadcom's F2H26. This would represent meaningful diversification away from TPU-concentrated revenue, though the ramp itself will still need to be proven.

Marvell: Diverse Business in Optics, Though XPU/XPU Attach Is Inherently Concentrated (Buchalter)

*TD Cowen View on Impact/Positioning: **Positive***

We believe Marvell's datacenter business (outside of its custom XPU franchise) is well diversified, with management talking to both cloud, enterprise, and carrier customers. The company has noted that its optical solutions and Ethernet PHY portfolios are used by more than 150 customers across the three markets. Marvell sells its optical solutions to a broad ecosystem including OEMs, ODMs, and module makers (such as InnoLight, Acacia Communications, and Lumentum) worldwide that integrate them into devices.

On the other hand, the custom XPU and XPU attach businesses are inherently customer concentrated; though there is high interest in creating a custom part (as noted by the company's >50 XPU and XPU opportunities across >10 customers), wins today are counted in far fewer numbers, and even less for those in production. That said, we think customer concentration concerns will ease if and when Marvell's XPU and XPU attach projects ramp in volume.

The Key Debates—Cycle, Copper, and Co-Packaging

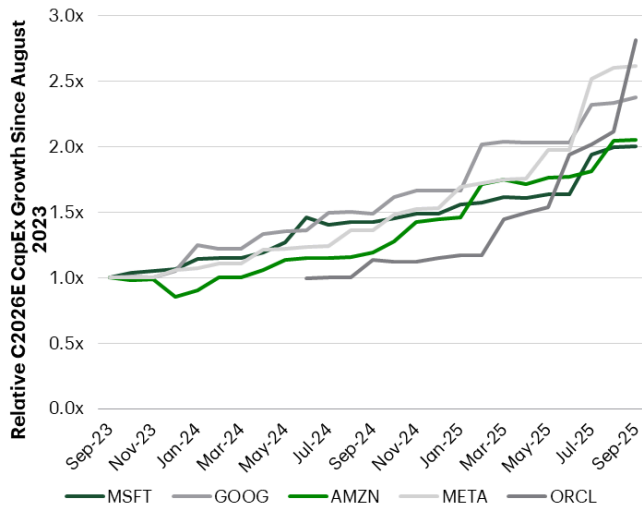
In the near term, stock performance for most datacenter connectivity companies is a function of specific project ramps and design wins, typically associated with a specific hyperscale customer. This is why the diversification leg of our framework is important. But over a longer time horizon, we see three key debates as likely to materially impact each company's datacenter franchise in the next two to three years. We collectively refer to these debates as the Three C's: Cycle, Copper, and Co-Packaging. Overall, we believe (1) we are early in the AI investment cycle; (2) active components are likely to extend the reach of copper for longer; and (3) innovations in analog are likely to obviate the need for co-packaging optics until at least 3.2T.

Cycle Just Starting: AI Infra Stronger for Longer

As we have discussed ad nauseam in our Ahead of the Curve reports [The Datacenter's Next Decade Ahead, Driven By Acceleration & Generative AI](#) (2024) and [A Rose By Any Other Name: Reframing & Dissecting the GPU vs. ASIC Debate](#) (2025), we believe that computing is undergoing two fundamental paradigm shifts at the same time: the rise of workload-specific hardware acceleration and the advent of generative AI as the most important workload in the datacenter. These paradigm shifts have and continue to drive and enable fundamental changes in datacenter compute architecture, especially at scale. Rather than the software-defined, central processing-heavy infrastructure that defined the first decade of cloud, the next datacenter decade (and beyond) will feature hardware for AI, underpinned by compute and networking semiconductors.

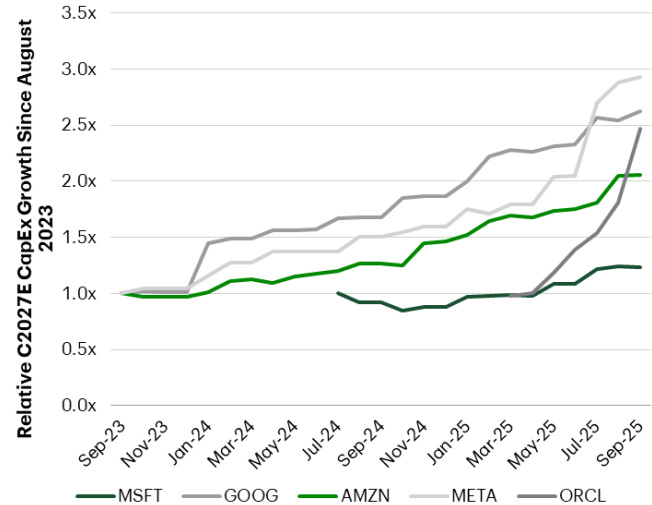
The rise of GenAI especially has driven an explosion in demand for AI acceleration hardware, both for training ever larger models and running at-scale, production-level inference of large language models. Despite persistent concerns from investors on hyperscale CapEx seemingly reaching new high-water marks every year, demand appears to continue unabated, outstripping the pace at which cloud providers can build datacenters and deploy servers. Over the past two years, we have seen reiterated and raised CapEx from the largest hyperscalers, now bolstered by a new wave of sovereign investment announcements that will drive AI hardware demand.

Figure 13 Consensus C2026 CapEx Estimates For Hyperscalers Have Been Revised Up >2x Over The Past 2 Years...



Source: TD Cowen, FactSet

Figure 14 ... As Have Consensus C2027 CapEx Estimates



Further, Generative AI is already a profit-making endeavor for key hyperscalers, as we discussed in our accelerator deep dive report, *A Rose by Any Other Name* ([here](#)). In our unit economics build-up and project-level profitability analysis, we analyzed various inputs such as costs, utilizations, and pricing, including training costs. Our findings show that **Gen AI inference hosting already has an attractive ~high-40% to low-50% gross margin**, indicating investment will continue as long as demand is there. Further, we note that the growing deployment of reasoning models increases the total amount of compute (in the form of tokens) consumed to achieve bleeding edge levels of intelligence. Though new use cases are still being defined, the profitable nature of GenAI applications are demanding huge investment in datacenters across every component.

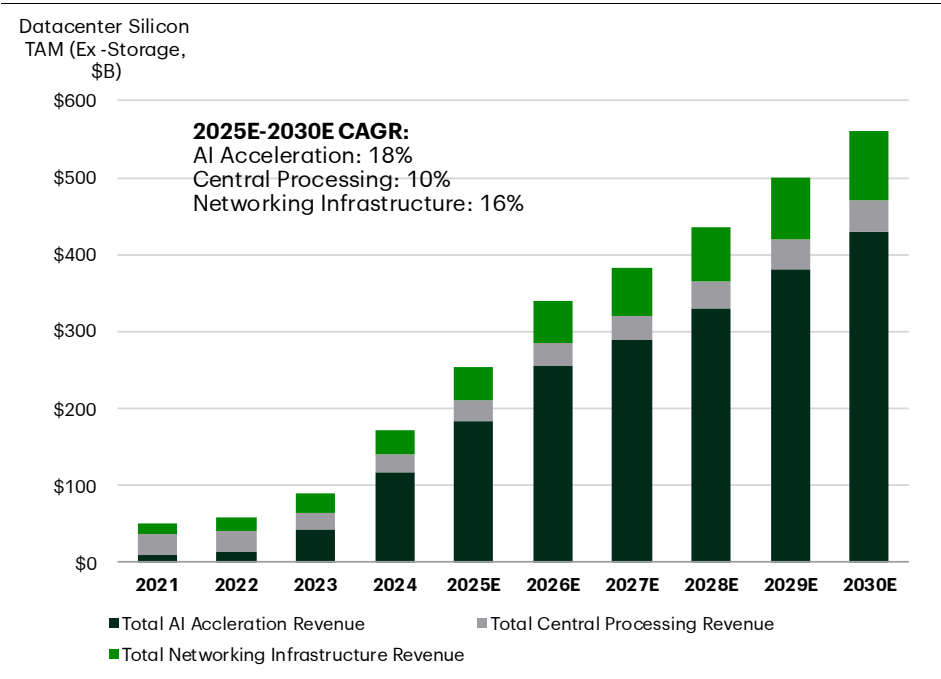
The steady flow of investment and compute capacity constraints suggests that any potential digestion could be pushed beyond 2026 at the least, with emerging buildouts and deployments being scheduled for 2027 given extended lead-times for infrastructure (e.g., Oracle-OpenAI, see work from our colleagues Derrick Wood and Mike Elias [here](#)). We also observe more capital flowing into AI investment from non-traditional sources, i.e. outside hyperscaler's reported CapEx. In our own datacenter TAM model, we have (very) conservatively incorporated a digestion year in C2027... a digestion year that has now been pushed out twice – we originally modeled it for C2024!

Thus, we believe the industry is still in the early innings in the AI investment cycle. While long-term TAM estimates remain difficult to prove or disprove given the rapid pace of innovation and buildout driving strong secular growth, our datacenter silicon model (excluding storage) projects a TAM of \$561B by 2030E, with a 17% CAGR from 2025 across accelerators, central processing, and networking.

Within our TAM, we anticipate the lion's share of value to flow toward AI acceleration of the next decade; however, we model **Networking Infrastructure** (including systems) to be one of the fastest growing categories within our TAM, growing at a ~16% CAGR from 2024 to ~\$90B, or 20% of the total market in 2030E and the second largest category by revenue. Given the constraints of Amdahl's Law and the electrical and photonic properties of moving data at ever increasing speeds, innovation is needed to prevent physical networking infrastructure from becoming a bottleneck.

Active electrical cables, active copper cables, active optical cables, linear pluggable and linear receive optics, and silicon photonics are all likely to play a role in solving for these constraints.

Figure 15 TD Cowen Forecasts Networking Infrastructure as 20% of the Datacenter Silicon (Ex-Storage) TAM in 2030E



Source: TD Cowen

We recognize the stock performance for many networking companies over the last two years has demonstrated that AI as a growth vector is no secret. However, we continue to believe our estimates for networking specifically exceed those of many investors, and for especially the smaller cap companies that are a focus for this report, we see significant upside potential still if the next 5 years play out as we are modeling.

Copper is Dead, Long Live Copper: Active Components Extend Life

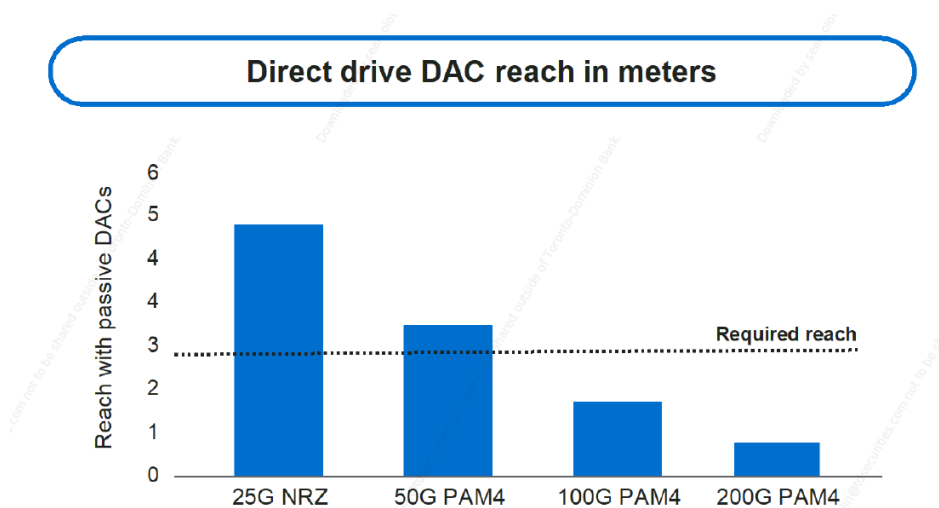
A perpetual topic of debate among investors is how quickly optical links will displace copper connections inside the datacenter. This trend has been a strong growth vector for companies such as Inphi (subsequently acquired by Marvell), and the prevailing view seems to be that copper is approaching its physical limits in the datacenter.

However, we take the view that by incorporating active components where necessary and continuing to push the envelope on what was thought possible in SerDes design, we believe **copper is likely to remain the primary intra-rack medium** through at least 2030, despite concerns that intra-rack would go optical. In fact, with the introduction of 7m AECs from Credo, Broadcom, and Marvell, we believe there is a potential for the trend from copper to optics to **reverse**, with some inter-rack optical links being displaced in back-end networking especially. Blithely, we believe **the next line rate generation will be copper's last—and always will be**. This is likely to benefit **Credo** and **Semtech**, and to a lesser extent **Astera**, **Marvell**, and **MACOM**.

The adage reminds us, "copper when you can, optics when you must." This guiding philosophy for interconnect speaks to the fundamental attractiveness of a copper solution as compared to fiber. Copper is cheaper, easier to deploy, and more reliable than optical links. However, copper also suffers from significantly higher **attenuation** than fiber cables (see Appendix below).

The rough rule of thumb states that, all else equal, the reach of a copper cable halves with each doubling of speed. Specifically, a Direct Attach Copper (DAC) cable can reach ~7m at 10G, but only 5m at 25G and just 2-3m at 100G per lane. Given that a server rack is ~2m high and typically the first hop from a server is to a Top of Rack (ToR) switch, a reach of just 1 or 1.5m at 200G is not suitable for many applications.

Figure 16 Passive Copper is Running Out of Juice

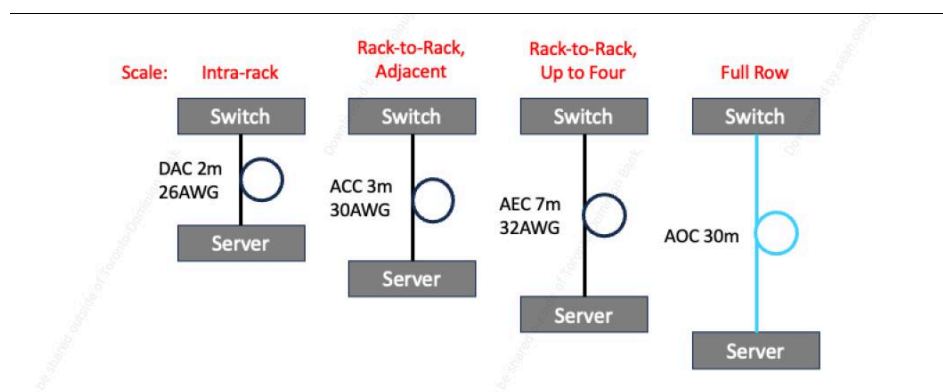


Source: Marvell courtesy of LightCounting

In contrast to copper, fiber cables experience little attenuation, with fiber attenuation more relevant in distances of *kilometers* rather than meters. However, optics are expensive—not only in terms of cost, but also in terms of power. An optical signal requires conversion from electrical to optical (and back). This process (which occurs in the optical module) requires power—a key motivator of solutions like CPO and LPO discussed below ([here](#)).

Introducing active components into the cable while *maintaining a full electrical link* is an attractive stopgap between pure passive copper (DAC) and implementing optics (in this case likely an AOC). To that end, two new categories of cable have emerged in the last few years: the Active Electrical Cable (AEC) and the Active Copper Cable (ACC). ACCs and AECs extend the reach of copper links at 100G and beyond, allowing high-reliability, low-cost copper to remain the backplane interconnect medium for high-speed AI networks.

Figure 17 Active Components Extend Reach of Copper at 100G



Source: LightCounting

Brief Aside: Re-Driving vs. Re-Timing

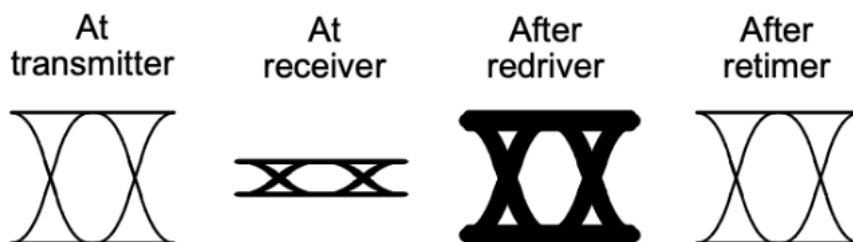
The fundamental difference between an ACC and an AEC is that an AEC includes digital processing whereas an ACC is a pure analog solution. More broadly, this distinction is described as "re-driving" vs. "re-timing." An analog solution is called a **redriver**, while a digital solution is a **retimer**.

A **redriver** primarily compensates for attenuation in a channel, essentially amplifying the received signal to extend the distance the signal can travel. A redriver is attractive for its very low power and its sub-nanosecond latency characteristics. Because the redriver takes a pure analog approach, the amplification is taking place at the speed of electricity and is not bound by a digital processor's clock speed. This makes redrivers attractive in applications where latency is important such as scale up domains. The downside of a redriver is that no "cleaning" of the signal takes place—signal *and* noise is amplified, which potentially limits the scope of redriving applications.

A **retimer** uses a digital signal processor to recover a signal and fully re-transmit it. A retimer leverages SerDes and a physical coding sublayer (PCS) to digitally recover the original signal's clock and use complex methods such as decision feedback equalization (DFE) to eliminate inter-symbol interference (ISI), reducing the noise that is eventually re-transmitted to the destination. Clearly this is a much more intensive operation, and it comes with higher power and higher latency. But because the signal is being digitally processed, retiming can also be implemented with additional features. For example, "protocol-aware" retimers can implement forward error correction (a Layer 2 signal integrity application) or provide advanced telemetry or even packet inspection.

A more detailed discussion of the physics of signal transmission can be found in the appendix, [here](#).

Figure 18 Redrivers Replenish Amplitude while Retimers Re-Send the Signal



Source: semiengineering.com

Redrivers and retimers are general categories, and the terms can apply to either specific products, or to the functionality of a given link. For example, one could describe an optical link as either "retimed" i.e. contains a DSP, or "redriven" i.e. is a linear solution. The key difference between a redriver and a retimer is the presence of digital logic for signal processing.

Active Copper Cables Extends Low-Latency Copper to 5m (And Beyond?)

ACCs are a redriver-based solution, essentially placing a linear equalizer at the end of a DAC to boost the signal through the cable. **Semtech** and **MACOM** are currently the two key suppliers of ACC-enabling chipsets.

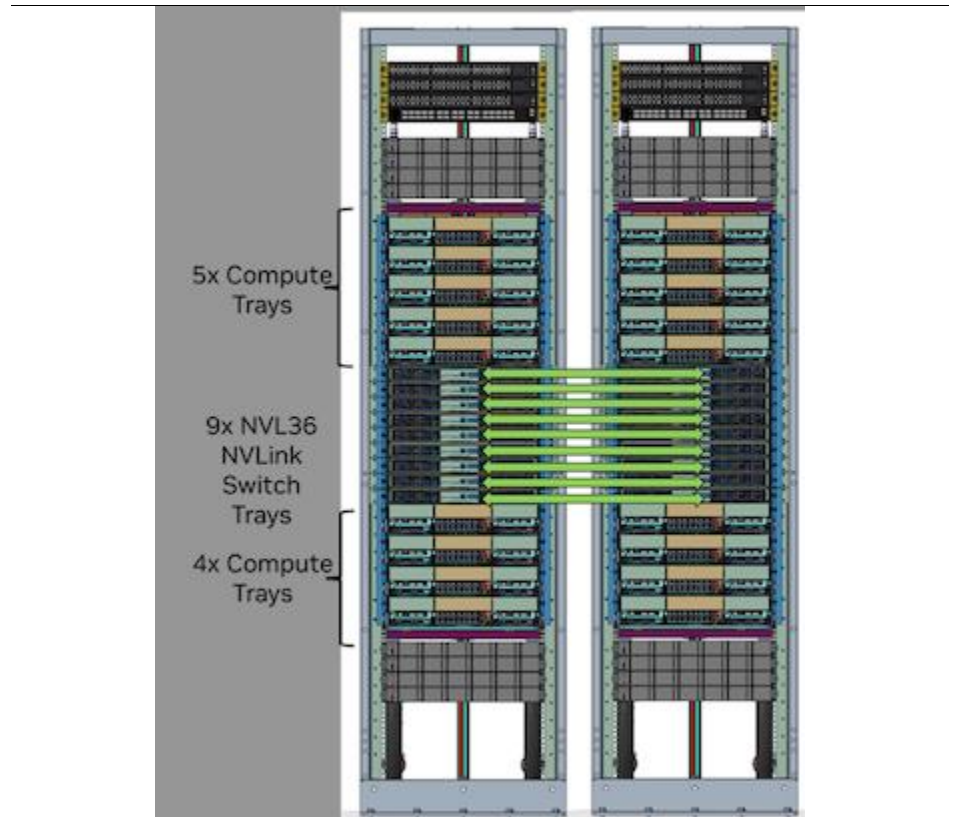
Semtech's CopperEdge portfolio extends copper's reach to up to 5m at 112G SerDes speeds or to 3m at 224G, with less than 100 picoseconds of latency (0.1 nanosecond) and less than 1.1pJ/bit at 1.6T (~1.7W). Likewise, MACOM's corresponding product family (MAEQ-40904) boasts sub-200 picosecond latency and can compensate for ~16 dB of insertion loss at 112G SerDes. Other suppliers of high-speed ACC chipsets include Parade Technologies & Nubis Communications, while companies such as Analog Devices, Texas Instruments, and Microchip offer redrivers at lower lane rate speeds (often for industrial applications).

Early Adoption of ACCs Has Come in Fits & Starts, But Next-Gen Scale Up Domains & Racks at NVIDIA Likely Drive Demand in ~2027+

The first high-profile application of ACCs in the datacenter has a checkered history, as they were originally going to be used to provide connectivity in the NVL36x2, dual-rack application of NVIDIA's NVL72 rack-level solution. As shown in Fig. 19, the NVL36x2 required ACCs to bridge the two sets of NVLink switches in the two racks, maintaining the 72-GPU scale up domain while reducing the per-rack power footprint and weight (the green arrows in the figure below).

We believe Semtech and MACOM had dual-sourced content in these ACCs for an NVL36x2 deployment at Meta, who would have been the first to implement this configuration. However, in February 2025, both companies announced that their anchor end customer (i.e., NVIDIA/Meta) had elected to not implement the dual-rack solution. This resulted in ~\$50M of lost revenue for Semtech (MACOM did not quantify the impact), and materially impacted Semtech's share price in the short term due to the reversal of investor expectations.

Figure 19 NVIDIA's Blackwell NVL36x2 Application Would Have Leveraged ACCs

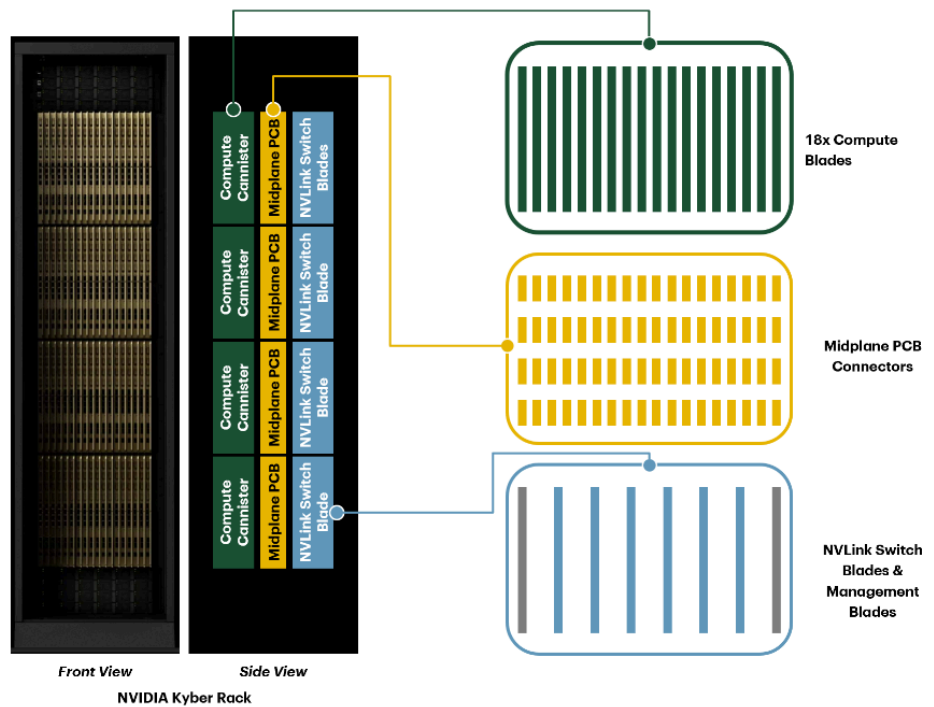


Source: NVIDIA

While the initial ramp of ACCs was volatile, we strongly believe this is not an indicator of the long-term potential for ACCs and other redriver applications in scale up (though it is illustrative of one of our core tenets of our investment framework regarding customer diversification). We view ACCs as especially relevant for scale up applications due to the low latency and anticipate redrivers will be needed in at least some of the copper links on NVIDIA's forthcoming Kyber rack.

Announced at GTC 2025, NVIDIA's Kyber rack is the successor to the current rack-scale solution that supports NVL72 configurations for Blackwell (the "Oberon" rack). Expected to ramp in 2027 with Vera Rubin Ultra, Kyber significantly increases the density of the rack by: (1) rotating the compute trays by 90 degrees and (2) moving the NVLink Switch trays *behind* the compute trays (connected by a PCB-based midplane). The rack will consist of four domains of these compute/switch tray pairings, and we believe the domains themselves are likely to be connected via copper cabling.

Figure 20 Kyber Rack Midplane Likely to Use Active Components Given Lane Speeds and Density



Source: NVIDIA, TD Cowen mock-up based on NVIDIA designs, not to scale

Moving to 200G SerDes for NVLink while maintaining copper backplane will likely require active components in the rack

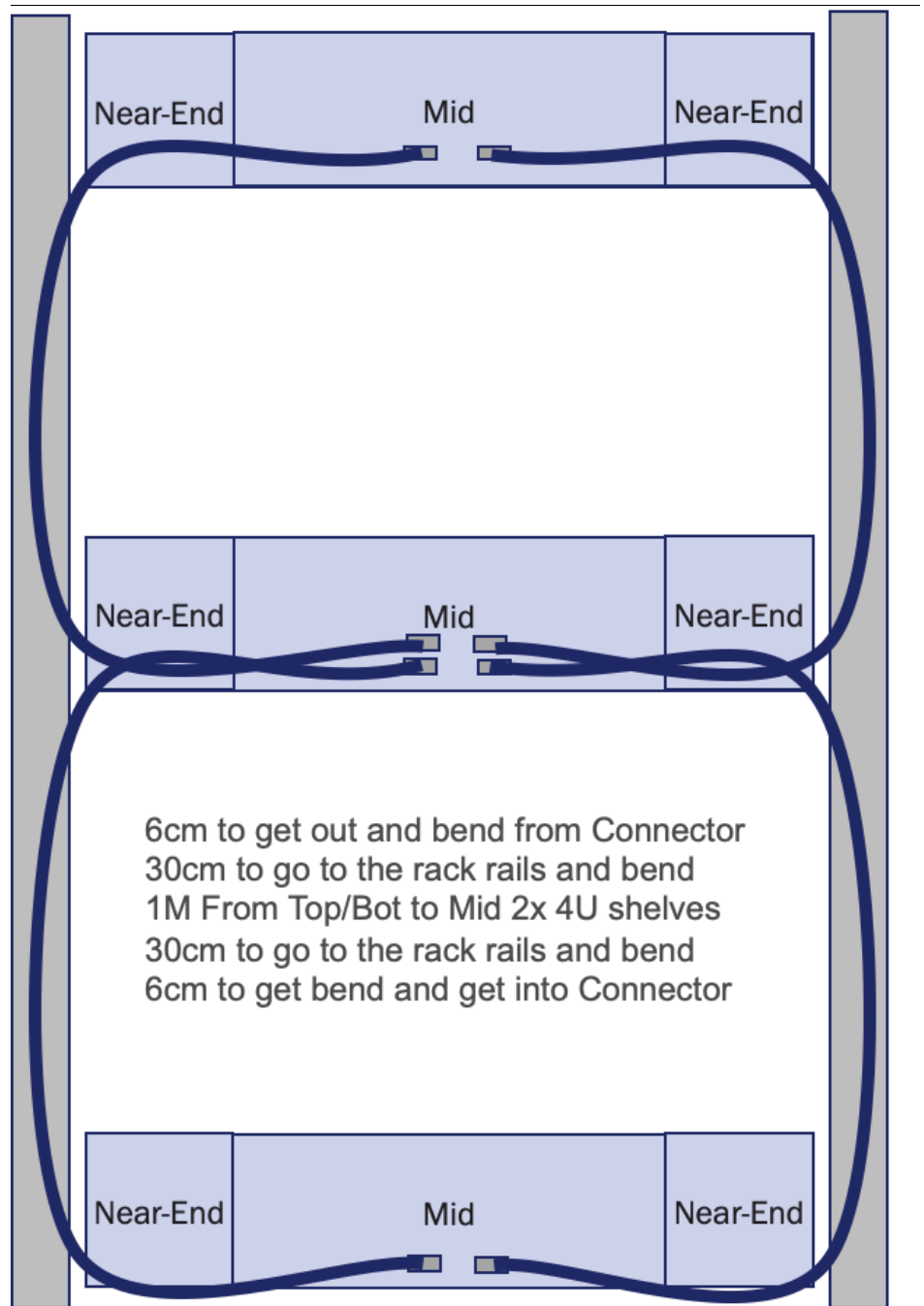
NVIDIA has further indicated that the Kyber rack will coincide with NVLink Gen7, with line rate speeds of 3.6TB/s. Current platforms are using Gen5, 1.8TB/s which is achieved with 18 lanes of 100G SerDes. We believe that either with Gen6 (2026, also 3.6TB/s) or with Gen7 NVIDIA will move to 200G SerDes lane rates. **At 200G, the reach of passive copper will be reduced to less than 1m, and the current NVL72 is >2m high.** To enable copper to connect the four midplanes with the signal integrity and latency required of a high-performance scale up domain, **we believe active copper cables will be required** (or redrivers if some sort of co-packaging is used).

The likelihood that ACCs will be used for many 200G/lane copper links was highlighted recently during Semtech's F2Q26 earnings call, with management commenting that they expect CopperEdge products to ramp in high volume beginning in Q4 and into 2026 alongside 1.6T switches.

Beyond NVIDIA, ACCs & Redrivers Are Ideal for Extending Life of Copper as Backplane Connectivity Medium in Both AI and General-Purpose Applications

The realities of progressing to 200G SerDes is applicable broadly beyond just NVIDIA's ultra-dense scale up racks. At 100G, NVIDIA addressed this directly by moving the scale up switches to middle of the rack, allowing the copper in the Blackwell generation to remain passive. As line rate speeds progress to 1.6T (8x200G), even in a traditional server configuration passive copper will begin to face challenges.

Figure 21 Even From Middle of Rack to Top or Bottom Requires a ~2m Cable



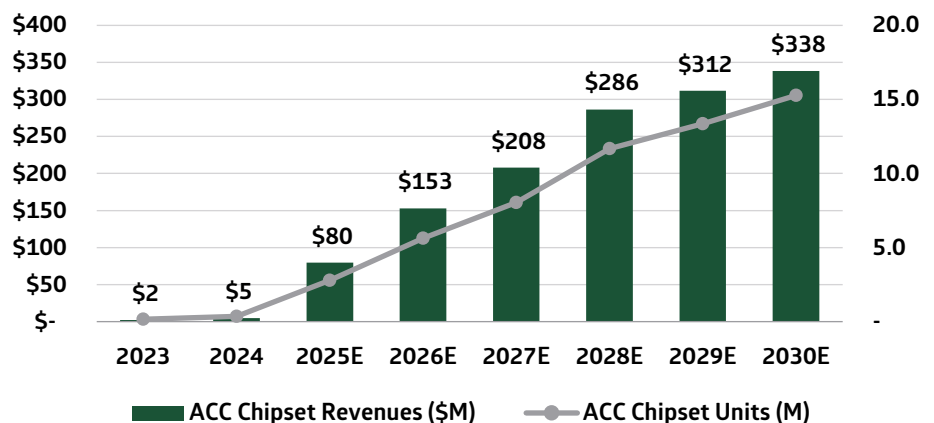
Source: OCP

In our view the key metric to watch in this theme is the adoption of 200G+ SerDes lanes at the switch and NIC, which are beginning to launch now. For example, Broadcom's latest datacenter-focused switch ASIC Tomahawk 6 has two SKUs, one with 128 lanes of 100G and another with 64 lanes of 200G. The 200G share is key for active copper, as it is the lane rate speed that matters rather than aggregate line rate speed (for example achieving 800G via an 8x100G OSFP port has a potential ~2m reach with passive copper, whereas an 800G port achieved with 4x200G QSFP more likely has a reach <1m). We anticipate 200G+ SerDes could command as high as 45% of switch ASIC lanes by 2030. Many of these will be connected with optics or AECs (see below), but for the first hop, NIC to top-of-rack switch configuration, we believe ACCs will become an increasingly attractive option given their cost and power advantages.

We See Potential >\$300M ACC Chipset Market by 2030

We estimate that as many as 10% of 800G and 1.6T ports could implement an ACC-based connectivity solution by 2030. At an ASP of ~\$25 to \$30 (per end), we model ACC driving a \$338M market in five years, while modeling ~5% annual price downs over that time period (typical for mature analog semis). However, we note that pricing is opaque, and our current estimates rely on Lightcounting's 2025 data for ASP.

Figure 22 Active Copper Cable Chipset TAM Through 2030E



Source: Lightcounting, TD Cowen estimates

Company Implications

Based on our model and the share assumptions associated with it (in which we assume Semtech retains majority market share), ACCs are likely to materially benefit Semtech's revenue growth, while having a more muted contribution for MACOM. That said, this is essentially a greenfield market and could be much closer to 50/50 share than we are modeling. Our views on the companies are discussed below, in alphabetical order:

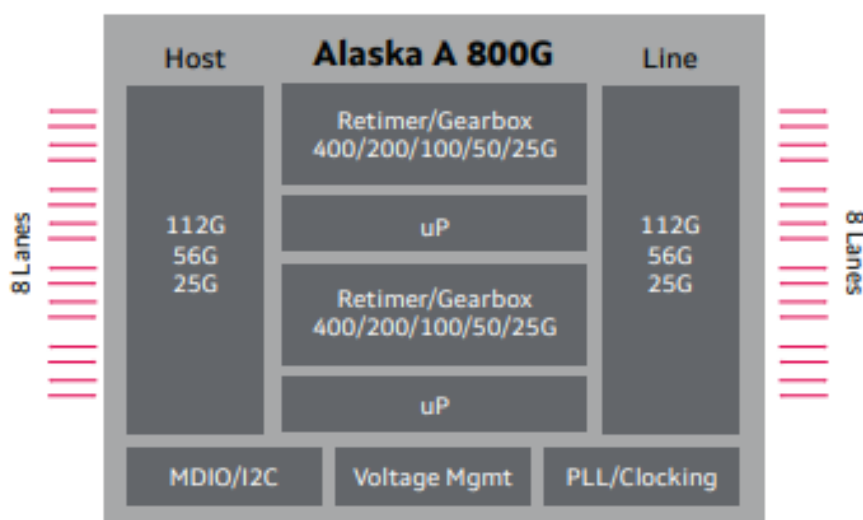
MACOM Technology (MTSI, O'Loughlin): We estimate MACOM's ACC revenue as potentially growing to ~\$50M by 2030, but this is based on our view of Semtech as possessing a first-mover advantage. If the market were closer to a 50/50 split, MACOM could generate >\$100M per year from ACC chipsets.

Semtech Corp (SMTC, O'Loughlin): We estimate Semtech's ACC revenue contribution could be as high as \$270M (not counting linear equalizers used in on-board applications). This would represent ~25% of modeled Semtech Datacenter sales by 2030 and a similar share of its total dollar growth over the next five years.

Making Copper Great Again: Active Electrical Cables Extend Copper Across the Rack

Active electrical cables (AECs) are a retimer-based solution, differentiated from ACCs by the presence of digital processing. As is generally true of retimed solutions, AECs contain DSP elements in the pluggable end, and the result is materially higher insertion loss compensation than comparable ACCs. For example, MACOM's 112G SerDes redriver can compensate for 16 dB of loss, whereas Marvell's Alaska A 800G AEC DSP (shown below in Fig. 23) can drive >40 dB at the Nyquist rate.

Figure 23 Active Electrical Cable DSP Improves Signal Integrity Through Conditioning & Digital Logic (At the Cost of Latency)



Source: Marvell

The digital processing in an AEC link allows for additional control-plane management features as well. Astera Labs' COSMOS software platform and Credo's PILOT suite are two examples of how digital logic enables advanced telemetry, diagnostics, and even security features.

The AEC's increased signal integrity and reach comes at a cost of higher power and higher latency. For example, we estimate Credo's 800G AEC solutions typically draw ~10W of power per end and introduce ~5 ns of latency for each DSP. This compares to <2W per end and <1 ns latency of an ACC solution.

Different Markets for PCIe & Ethernet, as AECs Engage Layer 2 Protocol Directly

AECs engage with the interconnect protocol they support, making them a hybrid Layer 1/Layer 2 device. This is important because it means not all AECs are created equal, and investors run the risk of reaching incomplete conclusions if adequate precision is not applied when discussing the AEC market. Specifically, today there are two distinct "AEC" markets—one for PCIe, and another for Ethernet. These are distinct, and the processors used to support each are not interchangeable.

When we use the term "AEC" we will be referring to Ethernet-based retimed cable solutions rather than PCIe...if for no other reason than because we had to pick one

We find this confusion most often arises when discussing the product portfolios of Astera Labs as compared to Credo—while Astera has strong market presence in PCIe-based AECs (specifically their Aries Smart Cable Modules (SCM)), Credo was a first-mover in the Ethernet-based AEC market and still maintains dominant share in that market.

To be crystal clear regarding what we mean going forward, when we discuss AECs and model the AEC market, we are specifically referring to the **Ethernet connectivity market**. This is largely arbitrary (certainly PCIe retimers embedded in a cable are active, electrical, and cables), but for clarity we choose to plant the flag and declare what we mean when we say AEC. To describe Astera's (and Credo's) PCIe cabling solutions we will be specific, and we model those parts within the PCIe retimer category alongside their board-soldered brethren. As the market matures, we would humbly request that the industry come up with a better term for retimed PCIe cables—goodness knows we could use another acronym.

Scale out Backend Deployments Supercharging Early AEC Growth

AEC's initial deployments were for general purpose applications (see below), but growth has significantly accelerated at the market level with the incorporation of AECs for scale out applications in custom XPU deployments, especially in Amazon's Trainium 2 clusters. While Amazon has not specifically detailed their use of AECs in scale out, we believe strong circumstantial evidence points to Credo's AECs being used to connect Trainium 2 pods (not the least of which is the distinctive purple HiWire AEC cables shown below).

Figure 24 Purple Cables on Trainium 2 Heavily Hint at Credo AECs



Source: Amazon

We believe AECs are an ideal choice for short-reach scale out connectivity due to their strong signal integrity properties, high reliability, and cost advantages over either re-driven copper ACCs or optics-based solutions.

Reliability of AECs a key attribute in addition to lower power draw—AEC links don't "flap"

When considered against ACCs, we believe for scale out applications signal integrity is prioritized over latency as scale out (generally speaking) has less intensive latency requirements than does scale up. Further consider that in a scale out domain, there are potentially one or more additional "hops" for the signal to take prior to its arrival at the destination XPU (perhaps through one or more scale out switches). In our view this should elevate the signal integrity requirements, as any errors in the transmitted signal have the potential compound over multiple links. When considered against optics-based solutions, we believe AECs carry a strong cost and power advantage on a like-for-like basis.

Reliability of Copper Opens Door to Challenge Optics for Inter-Rack Connectivity

In addition to the cost and power advantages AECs typically command over optics, the higher reliability of copper-based links is beginning to come to the forefront as a motivator for customer adoption of AECs. This is especially relevant in back-end networks, in which a link failure has the potential to materially disrupt a distributed AI workload.

AI clusters are unforgiving environments for optics. First, the demanding line rate speeds that connect many back-end clusters lower the tolerance for any kind of physical defect due to dust, heat, or mechanical stress due to the tighter signal integrity margins. Related to and compounding this sensitivity to defects is the fact that accelerators are generating many kW of heat per rack, and the high operating temperatures of both the XPU and the optics has the tendency to increase the failure rate of optical transceivers.

There is significant variability in the available estimates for optical transceiver reliability, but daily link failures appear to range between 0.002% and 0.004% for datacenter-scale deployments (according to Alibaba and Google, respectively). While this appears small, this translates to **40 failures per day on a 100,000 XPU cluster connected with ~1M links**. These failures can be highly disruptive and represent either lost time in a training environment or directly in lost revenue in an inference deployment. Per NVIDIA, the impact of dropped packets due to link failure "on AI workloads and NCCL collectives can be **severe**." While training data is rarely made public, Meta noted in its Llama 3 paper ([here](#)) that it experienced 35 network-related interruptions over the course of 54 days on its 64K GPU cluster.

More impactful than a simple failure, as bad as that is, is a phenomenon known as link "flapping." A link "flaps" when it continuously cycles through an up/down state (like a bird's wing flapping up and down). This is especially disruptive and indeed can paralyze an entire back-end network until the flap is resolved. This is because back-end scale out networks will often leverage the Border Gateway Protocol (BGP) for routing. In its simplest sense, BGP links "learn" the network topology and calculate the shortest link to use for the routed message, accounting for the latest topology. When a link is flapping, the network topology is *continuously changing* (as the link keeps alternating between up and down), thus causing the BGP protocol to be in constant flux—almost like a GPS stuck recalculating a route due to the driver doing donuts in the parking lot.

While specific estimates of link flap occurrence are hard to come by, various sources describe the relative frequency of flapping. Alibaba ([here](#)) noted 50K to 60K flapping links per *day* (though the denominator was unclear), and Credo estimates that a 100K GPU cluster could potentially experience a link flap every 90 minutes ([here](#)). Regardless of the precise Mean Time to Link Flap (MTTL), it appears to be a growing issue that is unlikely to be resolved especially as optics progress to 800G/1.6T and beyond.

As Credo, Broadcom, and Marvell have extended the reach of their copper AECs to 7m+, we believe there is a significant opportunity for AECs to replace some of the shorter optical links in a back-end network. Simply put, AEC-based links do not flap. The all-electrical signal path and laser-free transmission mechanism results in a

significantly higher Mean Time Between Failure (MTBF) rate compared to optical links, and AECs have been field-deployed for *billions* of hours with no instances of flapping.

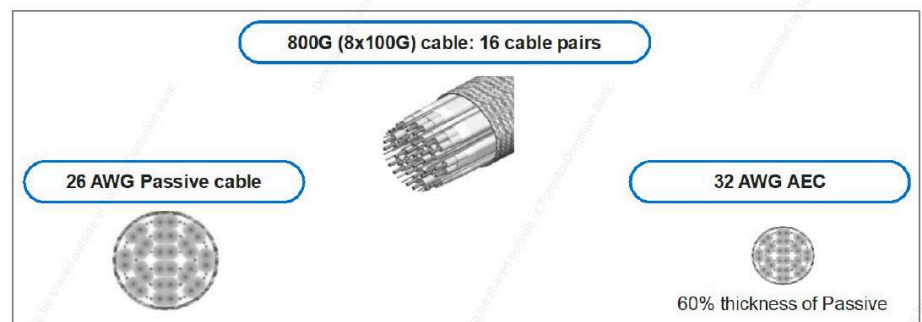
Credo highlighted this use case for AECs last year when it launched its 800G HiWire ZeroFlap AEC ([here](#)), and we believe the 7m AECs have played a key role in scale out connectivity for not only xAI, but Meta as well, as both deploy liquid-cooled GPU clusters to support their AI ambitions. As these deployments become field-tested (and the AEC supplier ecosystem likely extends copper's reach further), we see AECs as becoming the de facto choice for inter-rack, sub-7m scale out connectivity.

Beyond AI, Initial AEC Applications Focused on Intra-Rack, High-Density Applications That Remain Key Use Cases and Long-Tail Growth Opportunities

AECs' first high-profile application was a collaboration between Credo and Microsoft to advance a novel solution for improving reliability in general purpose server racks. Specifically, the project implemented a dual top-of-rack (ToR) switch configuration, effectively creating a "back-up" switch—in the event of a switch failure, the datacenter wouldn't lose access to the entire rack of servers served by that switch.

To implement the dual ToR solution, Credo developed the split-Y AEC, which accomplished two primary end goals: (1) it allowed for true back-up capability by embedding a cross-over switch in the cable itself, described [here](#); and (2) it reduced the gauge (thickness) of the copper cable required, allowing for the higher density that accompanied what was in effect a doubling of the amount of copper towards the top of the rack. The higher cable density enabled by the thinner cables also has applications in core network and telco markets in such configurations as the distributed disaggregation chassis (DDC, [here](#)).

Figure 25 Active Retiming Allows for Reduced Cable Thickness



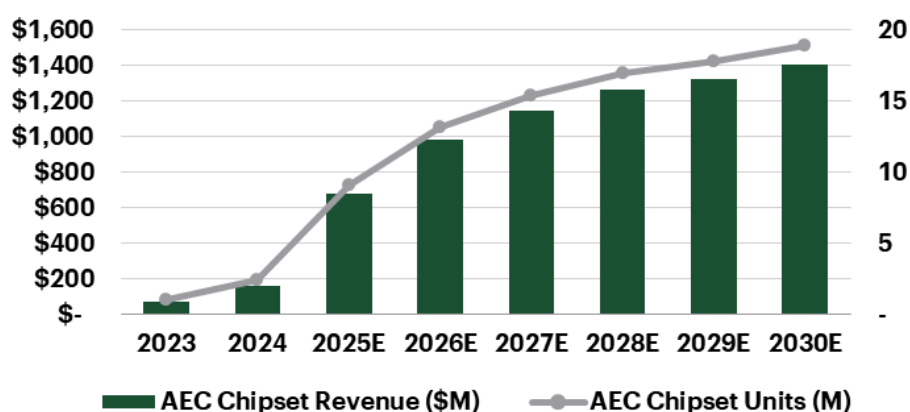
Source: Marvell, courtesy of LightCounting

Investor attention is (appropriately) focused on the opportunity for AECs to support AI deployments, given these ramps are happening now and at significant scale. However, we highlight these "legacy" (four years old) applications for AECs as a durable growth vector that is likely to steadily progress over the next 3+ decades of infrastructure deployment and refresh.

We Model AEC Chipsets As A Potentially \$500M TAM (With System-Level Cables Revenue Upside for Credo)

We believe AECs (and especially 7m AECs) represent an attractive stop gap before implementing more expensive and less reliable optical technologies for inter-rack applications. We model >10% penetration of links above 400G by 2030, and with a materially higher ASP as compared to ACCs (~\$75 per chipset), we view the AEC market as likely to exceed \$1B potentially as soon as 2027 (for just the chipset).

Figure 26 Active Electrical Cable Chipset TAM Through 2030E (\$M)



Source: Lightcounting, TD Cowen est.

Company Implications

Based on our model, we believe the AEC market is likely to remain a strong growth vector for Credo and presents greenfield opportunities for Marvell and Astera Labs. Importantly, while we do model share loss for Credo (from >80% today to ~68% by 2030), we still anticipate >\$1B for the company's AEC business by 2030 and see that as conservative given the spending environment and the company's commentary. Our views on the companies are discussed below, in alphabetical order:

Astera Labs (ALAB, O'Loughlin): We estimate Astera's Taurus Ethernet AEC chipsets and smart cable modules (SCM) were ~\$30M in C2024 and grew to ~\$100M in 2025 as hyperscalers sought to qualify more AEC suppliers. However, we see Astera as likely to retain a smaller portion of the AEC TAM and thus only contemplate ~\$250M in Taurus revenue by 2030.

Credo Technology (CRDO, O'Loughlin): Credo has a first mover advantage and has effectively pioneered the AEC market. Importantly, Credo goes to market with a full cable solution. Rather than selling just a chipset to a cabling vendor (which is the Astera and Marvell business model), Credo has chosen to keep the supply chain monolithic and assemble the cables in-house via their partnerships with Bizlink and Foxlink. This is important because it means Credo has a higher ASP on a like for like AEC (as they sell the whole cable). We estimate a ~50% to 100% uplift in revenue-per-cable for Credo, which drives our >\$1.2B estimate for Credo's AEC-derived revenue (both chipset and cable).

Marvell (MRVL, Buchalter): Marvell faces a dilemma when it comes to the AEC market, as an AEC application that is "replacing" an optical link (or that in the absence of AECs would be an optical link) to a certain extent cannibalizes Marvell's optical DSP business. While Marvell has the technology to aggressively pursue the AEC TAM, we are skeptical they will fully lean into AECs as a solution given the significant stakes for their optical business.

On-board retimers and redrivers are important applications for signal integrity, but much harder to model granularly

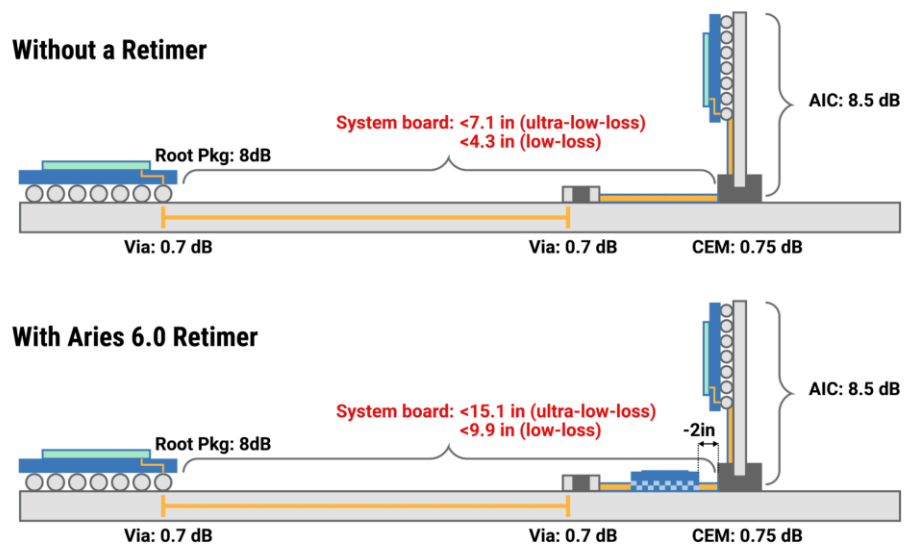
Signal Integrity Beyond Cables: On-Board Redriving & Retiming Take On New Importance in AI Configurations

Beyond cables, the increasing demand for high-speed interconnects in AI servers and switches is increasing the importance of retimers and redrivers to address signal integrity challenges within the system itself. These are opportunities for increased content and ASP uplift that are not captured in our cabling model.

As PCIe evolves from 5.0 (32 GT/s) to 6.0 (64 GT/s), the transition from non-return-to-zero (NRZ) encoding to pulse-amplitude modulation 4-level (PAM4) has introduced new complexities for signal integrity. PAM4 doubles the data rate per lane by encoding two bits per symbol, but it also increases sensitivity to noise, jitter, and inter-symbol interference (ISI). This shift necessitates advanced signal conditioning to maintain reliable communication, particularly in high-density, multi-hop architectures.

An important issue in switches especially is caused by the basic geometry of a single switch ASIC supporting a linear faceplate. The ASIC's downstream lanes "fan out" to 64 or more ports, each with varying trace lengths and impedance mismatches. This causes signal degradation due to attenuation, crosstalk, and timing skew. Retimers and redrivers can be deployed at key points in the signal path to mitigate these effects (for example, on the trace between the outermost ports and the switch ASIC, which have the longest distance to travel).

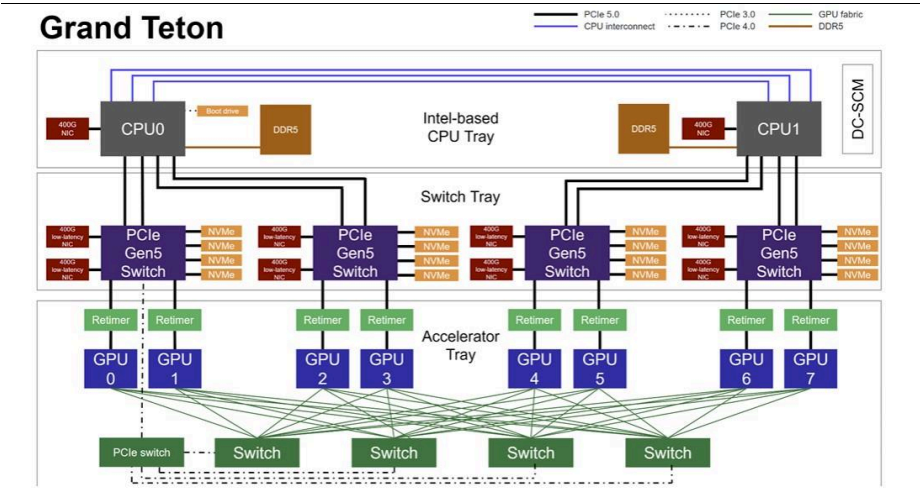
Figure 27 On-Board Retimers Expand Flexibility For Systems Designers



Source: TD Cowen

Within our model, we assume that there are ~4 retimers or redrivers per XPU. This is a rather imprecise SWAG, given the variety of board designs that exist, and the reality could be much higher or much lower than what we have modeled. For example, in the Grand Teton system (an OCP spec), there is one retimer per GPU and one PCIe switch (front-end) for every two GPUs (see below). However, this does not count any retimers that may be required at the scale out or front-end switch due to fan out and is a PCIe Gen 5-based design. With PCIe Gen 6 being deployed, the reach of a copper trace (all else equal) has likely been cut in half.

Figure 28 OCP's Grand Teton Design Uses One Retimer per GPU



Source: Open Compute Project, via NextPlatform

Using this ratio of 4:1 retimer:XPU, we estimate the retimer market could grow to ~\$2.4B by 2030 from ~\$1.5B today. We anticipate Broadcom and Astera will retain the lion's share of revenues in this segment, although Marvell and Credo are likely to compete as well.

Plug & Play: Co-Packaging is No Panacea

Following NVIDIA's high profile co-packaged optics announcements during GTC in Spring 2025, there has been significant investor interest in how fast will CPO penetrate and displace the pluggable optical module ecosystem. CPO promises superior power and signal integrity, but it comes at the expense of interoperability. With power becoming increasingly the primary constraint for datacenter deployments, many investors view CPO as inevitable and imminent.

In contrast with that prevailing view, we believe rather that the **pluggable ecosystem for optics is likely to persist** far longer than many expect, thanks to such innovations as Linear Pluggable Optics (LPO), Linear Receive Optics (LRO), and the ongoing innovation to lower DSP power draw at 3nm and below. In our view this benefits **Credo, MACOM, and Semtech**, and presents a potential headwind for **Marvell**.

Co-packaged optics (CPO) and co-packaged copper (CPC) have recently come to the forefront of industry and investor conversations as line rate speeds transition from 800G to 1.6T at the bleeding edge. By eliminating long, lossy copper backplanes and reducing the number of signal transitions between die, package, and PCB, CPO/CPC architectures can deliver power and signal integrity improvements for scale up and scale out networks. Broadcom has been an early leader in CPO, having launched its concept TH4-Humboldt platform in 2021 and is now sampling its third-gen TH6-Davison CPO switch ASIC. NVIDIA has likewise added CPO for scale out to its roadmap, with Quantum-X CPO (Infiniband) set to sample 2H25 with Ethernet-based Spectrum-X to follow in 2H26. Marvell has demonstrated both CPO and CPC options for its ASIC platforms, most recently at OFC 2025 ([here](#)).

The primary technical benefits of CPO/CPC stem from improved signal integrity and power efficiency. By shortening the signal path from the ASIC to the optical/electrical interface, CPO/CPC reduces cumulative losses from PCB traces, connectors, and stubs—often accounting for 10–15 dB of the total channel budget in traditional systems. This enables higher data rates with lower power consumption, as fewer retimers and redrivers are needed to maintain signal quality. For example, Broadcom's CPO design claims a 60% reduction in power per bit compared to discrete optics, while Marvell's CPC solutions reduce latency by eliminating backplane transitions. These advantages are especially attractive for back-end networking to support AI.

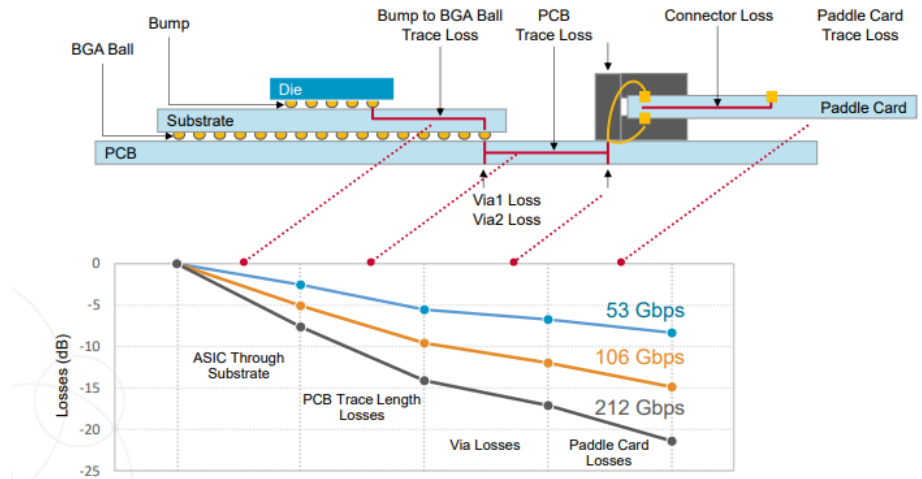
However, we believe CPO/CPC introduces material challenges in terms of serviceability and vendor lock-in. The tight integration of optics and compute hardware complicates field upgrades and repairs, as replacing a faulty transceiver or ASIC often requires swapping entire modules rather than individual components. This contrasts with traditional pluggable optics, which allows for more rapid "hot swapping" of faulty links. Additionally, given the integrated nature of optics and ASICs in a co-packaged solution, there is inherently greater levels of vendor lock-in (for now).

Acknowledging the theoretical performance benefits, we believe these implementation challenges are likely to prompt the industry to prefer pluggable alternatives in the medium-term. In our view, the desire to keep the existing pluggable ecosystem is driving innovative approaches to lowering the power draw of the pluggable module. We believe linear applications such as linear pluggable optics (LPO) and linear receive optics (LRO) are coming of age and will see material adoption at 1.6T. Finally, we see evidence that the industry has not given up on driving the power footprint for full DSP solutions lower, as evidenced by Credo's early progress on 3nm.

Co-Packaging Makes Compelling Power & Signal Integrity Theoretical Arguments

Signal integrity and loss is not exclusively a cabling issue, as the path from die to faceplate contains multiple transitions and copper traces that contribute to loss before the signal even reaches the cable. In high-speed datacenter applications, cumulative losses arise from die-to-package, package-to-board, and board-to-faceplate transitions, each introducing impedance mismatches, stubs, and via inductance, as well as the attenuation in the PCB trace itself. For example, at 212G SerDes speeds, the losses can accumulate to >20 dB between ASIC and faceplate—a 30x increase as compared to 53G SerDes.

Figure 29 At 212G SerDes, Losses Can Exceed 20dB Before Signal Even Exits the Server



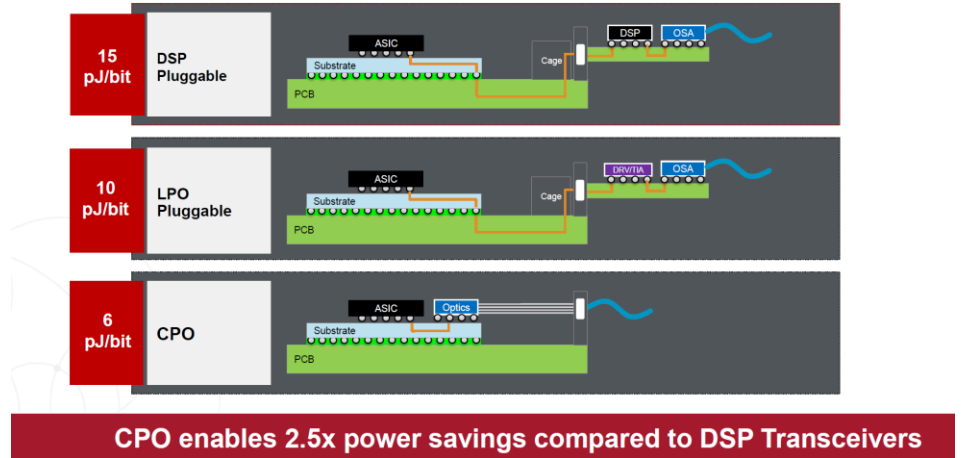
Source: Broadcom

Co-packaging optics and copper interconnects addresses this by minimizing the physical separation between the ASIC and the optical/electrical interface. By integrating optics directly with the ASIC (e.g., silicon photonics or optical engines) or integrating copper cables within the package, the number of transitions is reduced, lowering insertion loss and crosstalk.

These signal integrity advantages have a direct relationship with power consumption. Because the signal has less attenuation due to the shorter time spent in copper and less external noise from ISI and reflections due to fewer transitions, there is less need for active signal conditioning components that increase latency and draw power. Specifically, co-packaging optics removes the need for an additional signal processing chain at the faceplate—the on-ASIC SerDes feeds directly to an optical engine for transmission onto the fiber. Because attenuation in fiber is so low, the additional centimeters of reach into the package itself are negligible, and power can be reduced by ~1/3 as compared to a fully re-timed pluggable solution (as shown in Fig. 30, below).

Figure 30 Co-Packaged Optics Materially Lowers Power Footprint of Signal Path

Power Comparison of Optical Interconnect Technologies



Source: Broadcom

While budgeting at the single-watt level may seem academic in the context of a modern AI accelerator that is drawing well over 1kW at peak power, the volume of optical links in the datacenter add up. NVIDIA estimates that in an "AI Factory" configured datacenter with 100,000 AI servers, optical transceivers might use as much as 40MW of power. If CPO was adopted across the board in that deployment (assuming an all-in power draw of 5W per link at 800G), that would reduce the power consumed by optics to ~11MW. More importantly framed, this would allow the power savings to go towards **~30,000 additional 1kW TDP accelerators** while keeping in the same power budget.

Given that power acquisition and delivery are some of the primary obstacles currently bottlenecking wider-spread deployment of AI datacenters, there is ample *theoretical* backing for the adoption of CPO.

Serviceability & Interoperability Realities of Co-Packaged Solutions Inhibit Widespread Adoption...For Now

Acknowledging the theoretical merits of CPO, we take a more measured stance towards its adoption in live deployments. In our view, the very integration that delivers those performance gains also creates myriad operational drawbacks that are likely to make hyperscalers and others hesitant to adopt the technology at scale.

A primary concern is the loss of granular failure isolation. In a conventional design the optical transceiver is a hot-swappable module; when a link fails the defective unit can be removed and replaced while the switch ASIC remains online. When the optical elements are integrated into the ASIC die, any hard fault in the optical path is inseparable from the silicon that implements the switching logic. Repair therefore requires taking the entire ASIC offline, and in the case of a top-of-rack switch that in turn means losing access to an entire rack of servers. The cascading effect is stark: a single faulty channel can render the whole rack unreachable until the switch is swapped out in a scheduled maintenance window, dramatically inflating the mean-time-to-repair (MTTR) compared with discrete optics.

Additionally, thermal realities exacerbate the failure isolation problem. Optical engines that are normally housed in a controlled ambient environment (often below 40 °C) are, in a CPO architecture, placed directly on the ASIC substrate where temperatures routinely exceed 80 °C. Elevated junction temperature accelerates the degradation mechanisms that dominate silicon-photonics devices—laser efficiency

droop, electro-absorption modulator drift, and polymer outgassing, to name a few (no follow up questions here please). Even high-quality photonic components exhibit hard-failure rates on the order of 100 FIT (one failure per 10^7 hours) when operated within their specified thermal envelope. When those components are forced into the hotter ASIC environment, both the hard-failure and the soft-error rate are likely to increase. The net effect is a higher probability of intermittent outages that are difficult to diagnose because the failure manifests at the ASIC level rather than as a replaceable optical module.

Counterpoint: The Numbers Belie the Serviceability Concerns

Recognizing the serviceability challenges associated with a large-scale CPO deployment, both Broadcom and NVIDIA's respective CPO platforms include mitigation for some of the more extreme potential issues. Perhaps the most impactful design choice both have made has been to use external laser source modules rather than hybrid or integrated lasers. External lasers draw more power but offer two key serviceability advantages: (1) they are away from the hottest part of the system (the ASIC) and thus can be more reliable; and (2) they are readily accessible, making a laser failure (typically the most common type of hard failure) as easy to mitigate as a pluggable transceiver failure.

In addition to external laser sources, NVIDIA has gone one step further and developed detachable optical sub-assemblies (OSA). These OSAs aggregate three optical engines each (with a total of six OSAs on the ASIC) and are replaceable modules that can be swapped out in case of a failure on the assembly itself—further mitigating the risk of a lengthy switch failure incident.

Further, when viewed in light of the actual revenue impacts to a large-scale accelerator deployment, the serviceability concerns fail to make a compelling case. Earlier this year, we estimated a revenue generation potential of ~\$0.04 per second for a 64-XPU cluster with 50% utilization for inference workloads ([here](#)). With a hard-failure rate of ~100 FIT and a soft-error rate 2.5x that, as well as some assumptions around how much time a hard- and soft-error would take to resolve, a CPO deployment would be expected to result in a revenue loss of ~\$57 per cluster, per year—out of a potential revenue generating capacity of nearly \$600K. Even doubling the assumed failure rate fails to amount to much larger than a rounding error in direct revenue impacts.

Figure 31 Revenue Impact to CPO Deployment Lower Than We Expected

Metric	Value	Units	Description
Failures_per_link_per_year	0.001752	failures/link/year	From FIT definition
Hard_failures_per_year	0.224256	failures/year	Total expected hard failures
Soft_failures_per_year	0.56064	failures/year	Soft failures estimate
Revenue_loss_per_hard_failure	324	USD	Lost revenue for one hard-failure event
Revenue_loss_per_soft_failure	72	USD	Lost revenue for one soft-failure event
Annual_loss_hard	72.65894	USD	Total hard-failure revenue impact per year
Annual_loss_soft	40.36608	USD	Total soft-failure revenue impact per year
Total_annual_loss	\$ 113	USD	Combined revenue loss
Gross Profit/yr (GPU v ASIC)	\$575,427		
Less CPO Loss	\$575,314		
% lower	0%		

Source: TD Cowen assumptions

While this conclusion rests on many assumptions, we think the order of magnitude of the impact is likely within the ballpark. A training deployment would have a larger (though less direct) impact given a down link would likely require the training job to reset back to its last checkpoint, but this is no different from a link flap or failure in a pluggable ecosystem (discussed above, [here](#)), though the failure rate in CPO may be higher.

Diversity of Supply and Multi-Vendor Ecosystem the True Barrier to CPO Adoption

In our view, the much greater impediment to widespread CPO adoption is the implications for the multi-vendor ecosystem that the hyperscale datacenter operators currently benefit from. In contrast with the multi-sourced (and frankly lower margin) ecosystem of optical transceiver vendors currently supporting high-speed datacenter connectivity, the co-packaged ecosystem currently consists of effectively three players: Broadcom, NVIDIA, and Marvell. For hyperscalers, who abhor a sole-source and deliberately diversify their supply chain to negotiate better pricing and to hedge against roadmap changes, this lack of interchangeability is a potentially decisive deterrent.

We believe there is likely a financial aspect to hyperscalers' reluctance to adopt CPO as well. Consider that in any given sample of optical vendors, gross margins typically range in the mid-to-high 30s (%) outside of Cisco (whose consolidated margin profile is materially different from that of its optics business). Compared to Broadcom and NVIDIA's 60+% and 70+% semis gross margins, the multi-source pluggable ecosystem has clear price benefits (especially important when considering the volume at which optical links are deployed).

Assuming transceiver ASPs are ~\$300-\$400 each at 800G and above (per LightCounting), this would mean ~\$30,000 worth of 40% gross margin optics are being incorporated into a 64 x 800G switch. If that same cost basis has a 70% gross margin attached to it, it would be reasonable to assume ~\$50,000 in ASP uplift on the switch (which is likely anywhere from \$10,000 to \$30,000). Broadcom specifically has noted as such, having in the past described a ~4x ASP uplift for a co-packaged option ([here](#)).

We discussed hyperscaler behavior in seeking to avoid margin stacking in our comparison of the relative ROI for merchant vs. custom AI accelerators ([here](#)), and we see no reason the market for datacenter connectivity should materially differ.

Over time, we believe a realistic path toward a multi-vendor, interoperable ecosystem for modular co-packaged optics will likely emerge, akin to the hot-swappable optical modules that dominate today's pluggable market. In an ideal world, this would maintain the best of the pluggable ecosystem while also realizing the power and signal integrity benefits of closer integration of optics and ASICs. NVIDIA's CPO platform includes an early nod towards this end goal in its detachable OSAs. Likewise, Coherent has been vocal in calling the industry to action on this topic, publishing open-interface specifications and offering its own modular laser-engine kits that could be licensed to multiple ASIC vendors. However, achieving the breadth of vendor-agnostic parts, common form-factors, and certification processes that make pluggables like QSFP and OSFP so ubiquitous will require coordinated standards work, supply-chain tooling, and proven reliability data, so a fully-fledged, plug-and-play co-packaged ecosystem comparable to today's pluggable world is still likely several years away.

Innovation in Pluggable Ecosystem Obviates Need for CPO at 1.6T

Germane to the cost-benefit analysis for implementing CPO are assumptions surrounding the power draw and signal integrity characteristics of a pluggable module. This is a dynamic rather than a static variable, and we believe innovations such as linear pluggable optics (LPO), linear receive optics (LRO), and frankly just *better* DSPs are likely to continue to lower the power footprint of the pluggable transceiver—keeping the pluggable ecosystem as the preferred solution for at least the 1.6T generation.

Linear pluggable optics made a splash at OFC 2023 when industry luminary Andy Bechtolsheim (co-founder of Arista) made a public call to action to eliminate the DSP within the transceiver. LPO modules rely on analog linear equalization rather than digital signal processing (see Fig. 32). By removing the digital logic-based PMA portion of the Ethernet PHY, LPO solutions can reduce power draw by >50%.

Figure 32 Linear Pluggable Optics Lower Power Footprint of Transceiver by Eliminating Digital Logic

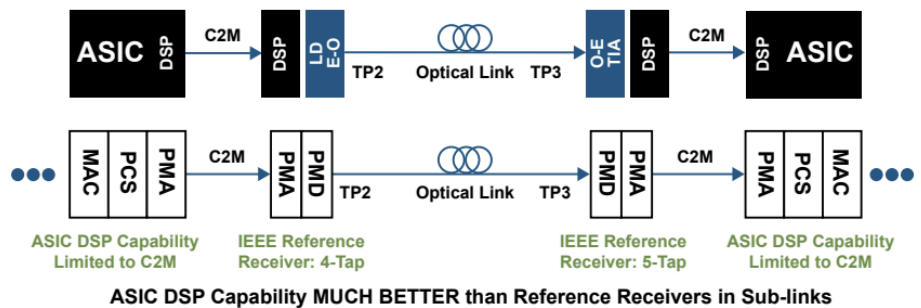


Figure 1a. Retimed Link

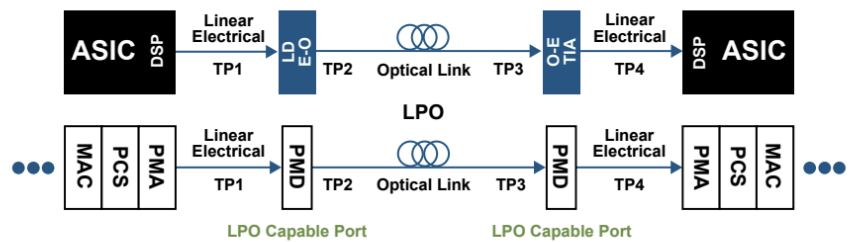
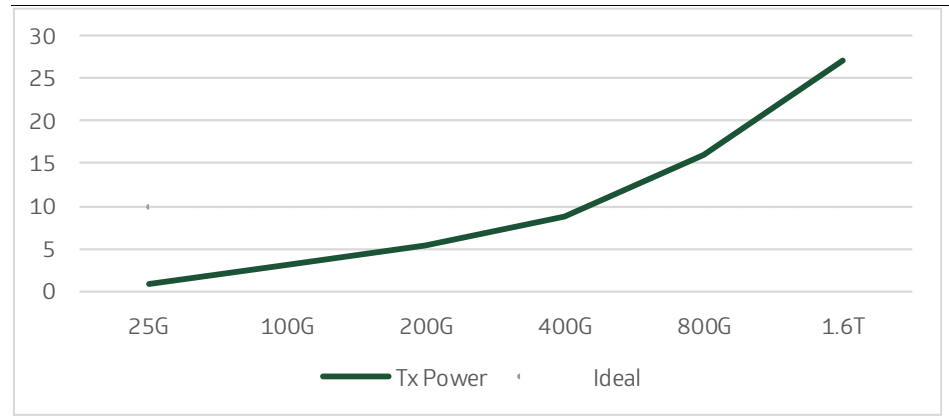


Figure 1b. LPO Link

Source: LPO MSA

This power reduction is significant. We understand that typical target power draw for a transceiver is for the package to come in below 10W, which was relatively easily achieved in the generations prior to 800G. However, at 8x 100G SerDes-based 800G 5nm DSPs, that power draw ballooned to ~15W initially (Marvell's Spica Gen 1 100G SerDes DSP, originally designed for 400G applications). Leveraging an LPO solution allowed for an 800G module to draw ~8W—coming in below the 10W target.

Figure 33 Typical Transceiver Power Draw at 800G Has Blown Through the 10W Ideal



Source: Intel data courtesy of photonics.com, formatted by TD Cowen

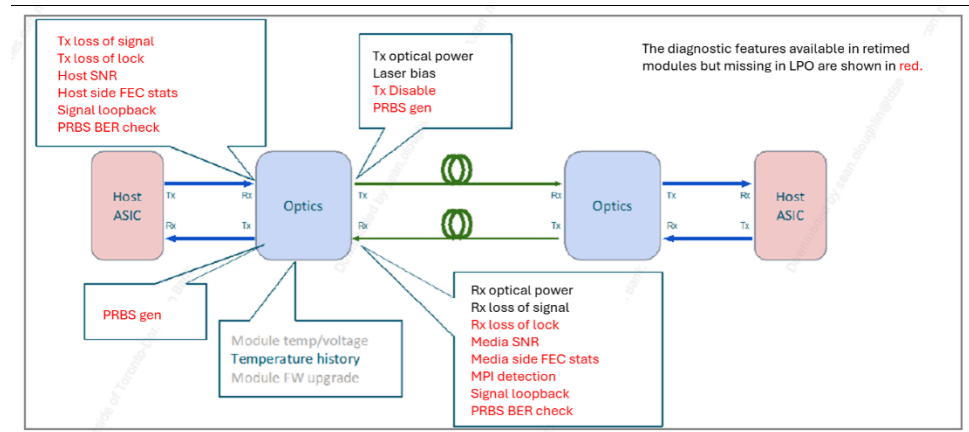
Since the early discussion at OFC 2023, the ecosystem for LPO has evolved materially and culminated most recently with the LPO MSA—a multi-vendor agreement on what are the official specs of an LPO module. In concert with this evolution, the performance of LPO solutions have evolved to the point where it has become legitimately competitive in terms of bit error rates with retimed solutions. The power envelope has continued to decrease as well—at the most recent OFC ([here](#)) Semtech demonstrated an 800G LPO solution that delivered in-spec BER at 5W (implying about ~0.6W per SerDes lane).

LPO is not entirely a free lunch, to be clear. First, it is worth highlighting that the implementation of LPO effectively leverages the SerDes of the ASIC itself to do the heavy lifting. We have heard anecdotally (from multiple sources) that one of the reasons that LPO is feasible today is due to the supreme performance of the 100G SerDes on Broadcom's Tomahawk 5 switch ASIC. Less-performant ASIC SerDes would reduce the overall effectiveness of a linear system.

The second drawback of implementing LPO is that it requires a higher level of alignment between port settings and modules. Early in LPO's development this meant a high degree of manual tuning, reducing the feasibility of a scale deployment. This has been one of the key areas of focus for the LPO MSAs, and we believe much of this manual process has been standardized into pre-calibrated ports.

Finally, the removal of digital logic in the module does come with observability and diagnostics trade-offs. Modern DSPs include blocks that go beyond pure signal processing and engage with the Layer 2 protocol to provide forward error correction as well as telemetry features. For example, as highlighted in Fig. 34, a modern DSP can report to a centralized observability platform if a signal loss or degradation is detected. Because this report is coming from the module itself, it improves failure localization efforts and can improve response time (and therefore utilization and up-time).

Figure 34 A DSP Provides More Than Just Signal Processing

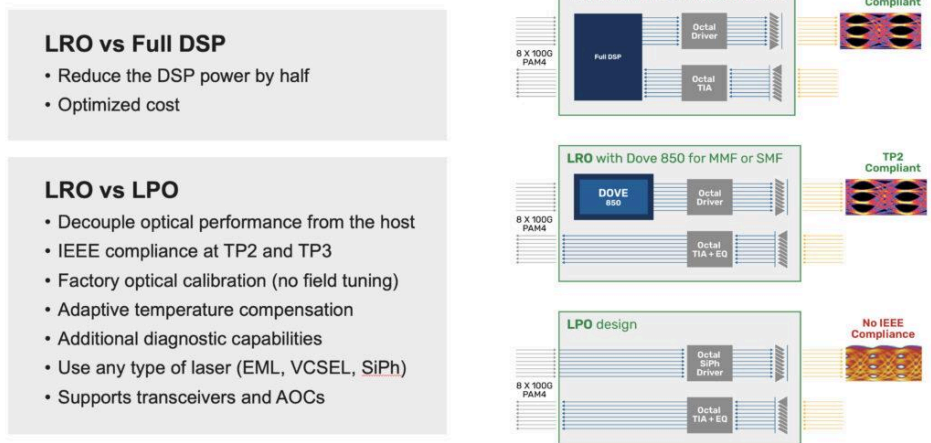


Source: Meta courtesy of LightCounting

Linear Receive Optics-based solution (or Retimed Transmit Linear Receive (RTL) to use OIF's phrasing) takes the core idea of LPO and meets it halfway by adding re-timing only on the transmit path (hence, "linear" on the receive path). Credo has been an innovator in this space, and we believe is the only company to have an LRO solution in the market today (the Dove 850). At OFC 2025, Credo (and Semtech) demonstrated an 800G LRO module that had a power draw of 9.4W at an industry-leading BER ([here](#)). This compares to ~13W for the latest 800G fully-retimed modules.

Figure 35 LRO Removes Half the DSP to Improve Power Specs While Maintaining IEEE Compliance

LRO versus Others at 800G



Source: Credo

Importantly, an LRO-based solution can maintain the diagnostics functionality that an LPO solution lacks—while lowering the power draw to an acceptable level by leveraging the analog signal conditioning in the receive path.

Early days, but LPO and LRO are becoming more than just a demo booth concept as live deployments expand

In the two+ years since OFC 2023, we believe the industry has rapidly moved through the various phases of the classic "hype cycle." In the immediate aftermath of the conference, we observed significant investor attention being paid to LPO and the so-called "death of the DSP," with interest waning over the course of the following year. In fact, in May 2024 we observed discussions during which it was questioned whether LPO would work at all. Meanwhile, we believe those in the industry were doing the blocking and tackling required to make a complex signal integrity solution widely interoperable and scalable. This work resulted in the publication of the LPO MSA initial specification in March 2025.

Today, we believe LPO and LRO deployments have moved beyond test cases and qualifications and are beginning to be implemented for live workloads. On its most recent earnings call, MACOM noted that it had received "high volume **production orders**" for their 100G/lane LPO chipsets to be deployed in a medium reach application. As 800G (both 100G/lane and 200G/lane implementations) becomes a larger share of overall transceiver shipments, we believe LPO and LRO will likewise gain share over time.

Finally, with respect to lowering the power footprint of the pluggable modules, while LPO and LRO eschew full digital retiming, the broader ecosystem still benefits from the rapid advancement of fully-retimed DSPs that are being developed and continually improved. For example, Credo's 3nm-based 200G/lane DSP, demonstrated at OFC 2025 launched recently ([here](#)).

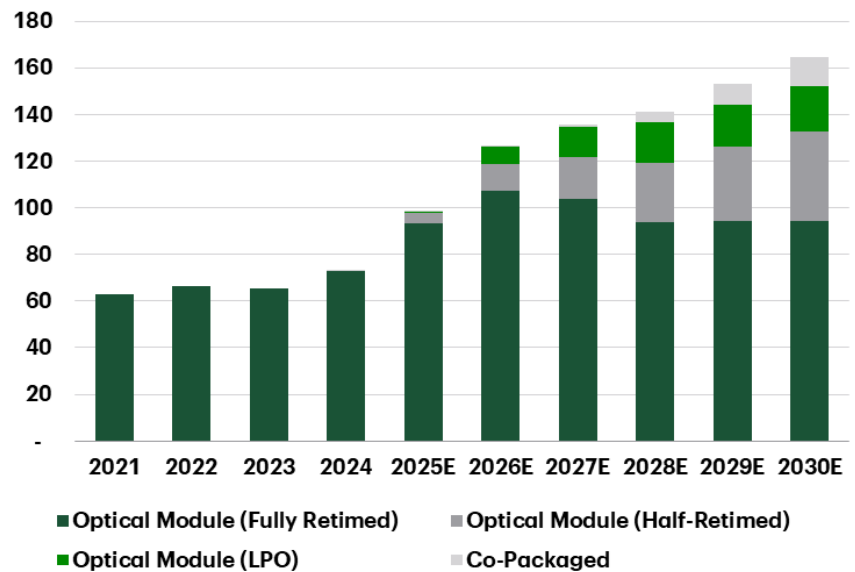
While the precise power draw of the DSP was not publicly disclosed, Credo did note that it expected to enable a 1.6T transceiver to consume "well below 20W". While this remains above the 10W "ideal," it's notable that in the last year, the most power efficient DSP on the market has seen a >5W power reduction (from ~25W with Marvell's initial 5nm offering to "well below 20W" with Credo's 3nm). We believe the industry is actively working to develop next-gen solutions on 2nm that should result in further power savings.

Taken together, we believe the convergence of low-power linear pluggable optics, half-retimed receive modules, and increasingly efficient fully-retimed DSPs eats at the justification for dedicated CPO at 1.6T. The linear ecosystem already delivers the required bandwidth density, latency, and thermal profile using mature, interchangeable form factors, while the incremental power savings from next-generation DSPs further increases the threshold required to make a CPO deployment worth it. 3.2T line speeds may upend this logic chain, but for now we believe the benefits of the pluggable ecosystem outweigh the costs.

Our Model Reflects a Slow Adoption of CPO, and A Growing Share-Loss Problem for Marvell

For all our discussion of copper-based solutions and for all the markets' focus on co-packaging, we see the pluggable optical module ecosystem as one of the primary beneficiaries of GenAI. Given recent stock performance across much of the industry this is not lost on investors, but in our view, what is underappreciated is the extent to which share will shift away from fully retimed, DSP-based pluggables to half-retimed or fully linear solutions. This is starkly illustrated in Fig. 36, below—the fully retimed modules in dark green never regain their peak unit shipments after 2026 due to increased penetration from LRO, LPO, and late in the decade CPO.

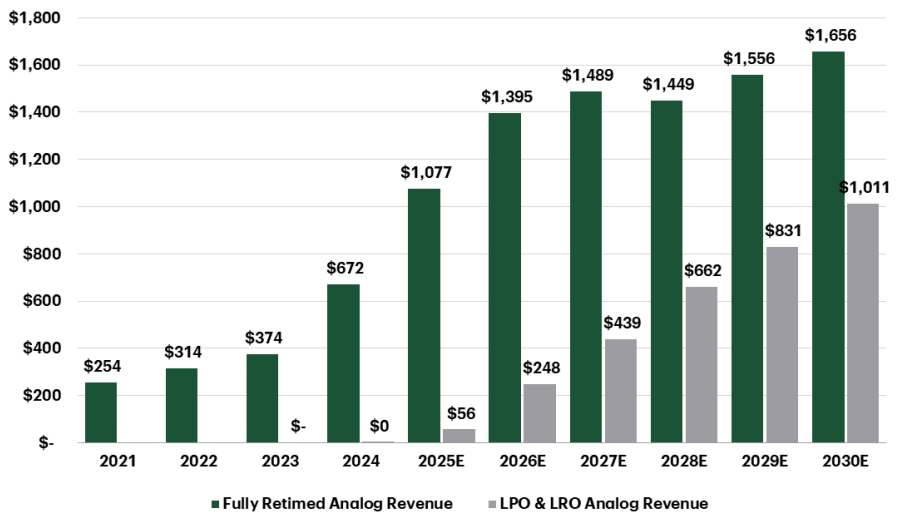
Figure 36 We See Growing Adoption of Analog-Heavy LPO and Half-Retimed Optical Links...



Source: Lightcounting, TD Cowen estimates

This has significant implications for the DSP share leader, Marvell, who has yet to put forward an LRO solution (effectively ceding that market to Credo). Marvell does have capacity to participate in the LPO market through its portfolio of TIAs and drivers, but we believe Semtech and MACOM (and others such as ADI and STMicroelectronics) are better positioned to serve these analog-heavy opportunities given their mixed-signal heritage. To that point, we see LPO and LRO as potentially representing more than half the dollar content for analog semis in optics by 2030—making the ongoing LPO/LRO developments a key trend to monitor for our positive outlook for these companies (Semtech in particular given the outsized focus on the datacenter market from management as compared to MACOM).

Figure 37 ...Driving LPO/LRO Analog Market to >50% of Total \$ Content by 2030 (\$M)



Source: Lightcounting, TD Cowen

Company Implications

Based on our model, we see significant growth for **Credo** in LRO (as well as fully retimed modules), as we believe that will become a very attractive solution and balance between power and performance at 1.6T. Likewise, we see the growing analog content in optical transceivers as benefitting **MACOM** and **Semtech**, and for both the respective revenue bases are small enough that this growth is likely very much needle-moving. **Broadcom** is likely to continue to participate and hold share in what was its historical domain, but in the context of the company's larger AI franchise, optics are becoming less impactful to both financials and narrative. Finally for **Marvell**, as we describe more expansively in our separate downgrade (here), we believe this presents a real threat to what has been a strong growth driver for the company. Our views on the companies are discussed below, in alphabetical order:

Broadcom (AVGO, Buchalter): We see Broadcom's optical revenue as largely holding serve at ~20% market share in specifically pluggable transceivers. Broadcom would *materially* benefit from a greater than anticipated adoption of CPO, as it would be very likely that Broadcom would be able to capture at least some of the value that currently accrues to transceiver vendors in the form of higher switch ASIC ASPs. Given the relatively low penetration of CPO that we contemplate, this is not directly incorporated into our model but would represent upside to Broadcom's Ethernet switching business.

Credo Technology (CRDO, O'Loughlin): We see Credo's optical DSP franchise as likely to grow strongly in the 1.6T generation (and even gain share in 800G and below, though more slowly). Specifically, we see Credo's unit share as going from ~9% in 2024 to >30% in 2030, driven in large part by the increasing penetration of LRO (which we estimate as capturing ~29% of retimed transceivers by 2030). The resulting ~\$1.2B optics business for Credo would at that point exceed the company's AEC sales, by our estimates—a broad revenue diversification that we do not believe is contemplated by even the most bullish of Credo investors.

MACOM Technology (MTSI, O'Loughlin): We model strong growth for MACOM's analog content in optical modules (on the electrical side in this case), contemplating a ~23% CAGR to 2030 and expanding to exceed \$700M over that time. This is a material tailwind for MACOM's revenue growth, and while we see LPO and LRO as a boost to this growth, we highlight that MACOM benefits from broad optics growth overall not just from those linear solutions.

Marvell (MRVL, Buchalter): We see risk to Marvell's DSP franchise from linear solutions and from an increasingly competitive market for PAM4 DSPs. We do **not** model a shocking decline that will hit the model all at once. Rather, we could characterize the threat to Marvell's DSP franchise as a steadily eroding share lead, dampening its datacenter growth compared to peers. Specifically, we see Marvell as growing only ~6% annually in its PAM4 optics business to 2030, under-growing its TAM as its share declines to <40% from >55% today. While Marvell's coherent pluggable optics business may blunt the impact of this somewhat, we believe the datacenter interconnection market is likely to be slower growth than the intra-datacenter connectivity market given the backend networks required to support AI remain largely confined to datacenter reach lengths and below. This optics headwind is one of the key points of our separate downgrade of the stock.

Semtech Corp. (SMTC, O'Loughlin): We model strong growth for Semtech's analog content in optical modules, contemplating a ~26% CAGR to 2030 and expanding to exceed \$900M over that time. This is a material tailwind for Semtech's revenue growth, and while we see LPO and LRO as a boost to this growth, we highlight that Semtech benefits from broad optics growth overall, not just from those linear solutions.

Interconnect Systems: Diversified Industrials & Automation – Joe Giordano

High powered interconnects are essential to the performance, reliability, and scalability of hyperscale data centers as they provide the physical pathways for both data and power. Every server, switch, and storage device depends on these high-speed, mission critical components. Generative AI oriented data centers differ from traditional facilities primarily in the scale, speed, and architecture of their interconnect systems.

As we examined high-performance data center infrastructure, it became abundantly clear that information on competing architectures is fragmented and often inconsistent. We believe this reflects the industry's rapid evolution and the modular, iterative nature of upgrades. In this section, we aim to synthesize these approaches and highlight mission-critical solutions from leading vendors.

Companies in our coverage most exposed to these products include **Amphenol (APH)** and **TE Connectivity (TEL)**:

Amphenol (APH), ~33% of total revenues (IT & Data Com Business) (Product list [Link](#)):

Very broad portfolio covering nearly all aspects of a data center. Key components in power infrastructure include busbars, busyways, power distribution units "PDUs", remote power panels, and automatic transfer switches. APH also supplies a wide variety of cabling and assemblies and supporting components (terminal blocks, sensors, communication and control harnesses). The company recently bolstered its fiber exposure through the CommScope CCS acquisition.

APH Organic Analysis [HERE](#)

IT & Data Com related revenues have increased to ~33% of total sales FY25E (doesn't include CommScope impact) vs. ~19% in 2019. In our recent note [HERE](#), we map out IT & Data Com's organic contribution to total revenues, which has been the majority of total company organic growth for the past 5 quarters.

TE Connectivity (TEL), ~13% of total revenues (Digital Data Networks Business) (Product list [Link](#)):

TEL supplies a broad portfolio of high-speed backplane connectors (leading in orthogonal and cabled backplanes), mezzanine and board-to-board solutions, I/O connectors, power connectors, copper and fiber cable assemblies, and active optical modules.

Digital Data Networks related revenues have increased to ~13% of total sales FY25E vs. ~7% in 2019. The growth in this business has lagged APH over the past couple years due to two main reasons: (1) TEL relationships are more tied to the hyperscalers and custom ASICS providers (Broadcom, Marvell etc.), which have yet to reach the same scale as GPUs, and (2) power interconnects (where TEL historically has had a strong positioning) represent less than 25% of the content found on server racks vs. networking interconnects (according to our expert calls, though today the company's current offerings provide a holistic interconnect suite).

Connecting Our TD Cowen Semi Forecast

Both APH and TEL offer hundreds of products related to data centers which can be located within and outside of the server room and amongst networking and power infrastructure. While this versatility clearly has its benefits from a business perspective, from the outside it makes forecasting these developments with any precision near impossible, though we reference broad trends for directional movements.

According to our TD Cowen Semi Team's Datacenter (ex-Storage) Build-Up & Forecast, we see our current APH and TEL datacom estimates to be well supported (if not conservative) as the model implies 28% y/y growth in 2026 vs. our ~15% and ~35% estimates for APH and TEL's datacenter businesses, respectively. Importantly, we estimate APH's datacenter business will grow nearly 200% to almost \$4B and TEL's business from ~\$300MM to over \$800MM in 2025 – both well in excess of the build up forecast.

Notably, Broadcom and Marvell AI ASIC related revenues are expected to start scaling meaningfully in 2026, which we expect to have an outsized positive contribution for TEL, though both TEL and APH are well positioned as the broader ecosystem builds out.

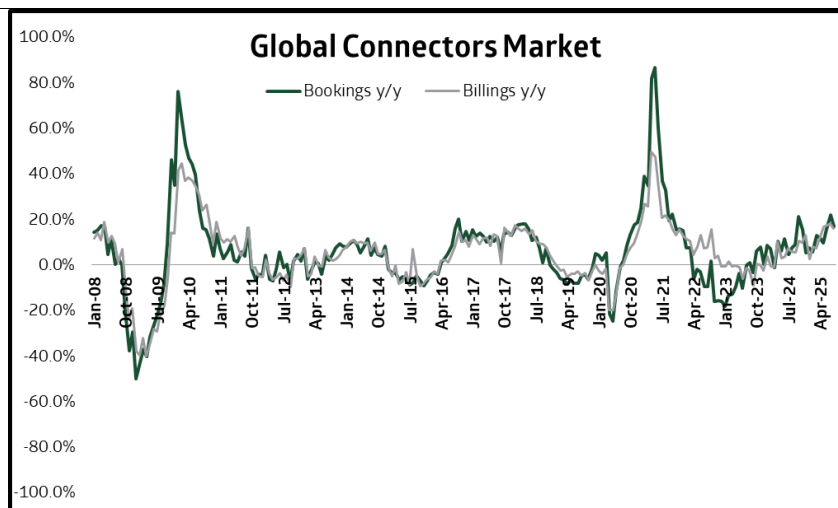
Connectors Market Trends

According to Bishop & Associates, the Global Connectors market (all end markets) booking and billings (orders & sales) trends remain constructive in at least the short-to-medium term after coming off late 2022 / early 2023 lows.

Amphenol and TE Connectivity maintain premium market share in all regions globally and typically trade-off between first and second ranking by end market.

Monthly connector growth data and market leader information available upon request.

Figure 38 Global Connectors Bookings (Orders) and Billings (Sales) y/y % chg.



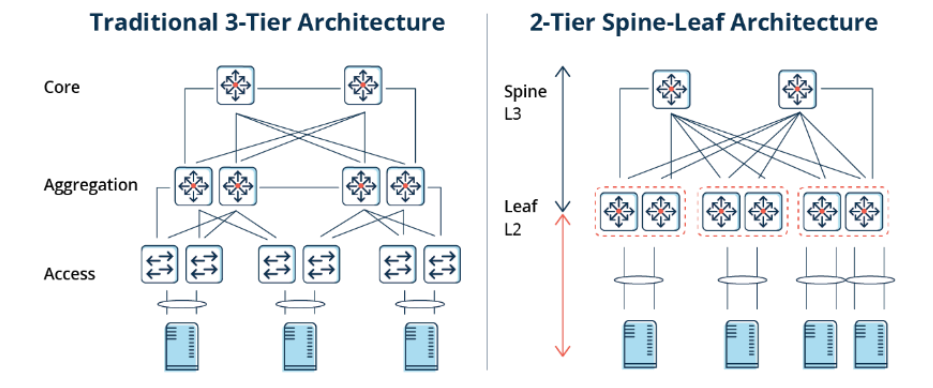
Source: Bishop & Associates

End markets: Computers, Business/retail/Ed, Instruments, Medical, Industrial, Auto, Transportation, Military, Telecom/Datacom, Consumer Electronics.

Interconnect Networking Equipment

According to a recent Amphenol white paper ([link](#)), datacenter architectural design can be broken into two different styles: 3-tiered and 2-tiered configurations. Our TD Cowen Semi team's May 2024 Ahead of the Curve ([HERE](#)) provides a primer on these topologies and deep dive into cabling architectures on pgs. 116-141.

Figure 39 Datacenter Topologies



Source: HP Enterprise

The first configuration is primarily utilized by traditional data centers (cloud and other internet capabilities) and inference facilities (which run AI workloads). The two-tiered configuration is used for AI model training data centers (massive GPU clusters that are connected through a “leaf-spine” fabric enabling “east-west” traffic). These differing network topologies also have other physical characteristics:

Transmission protocols: high-performance AI clusters rely on low-latency, high-throughput protocols such as Remote Direct Memory Access (RDMA) over Converged Ethernet (RoCE), enabling faster speeds across compute nodes. This format bypasses traditional Transmission Control Protocol (TCP/IP) stacks to provide direct access between servers.

RDMA protocol requires precisely tuned networks for operation, meaning cabling, connectors and switch configurations must meet extremely tight signal integrity requirements. Any loss of efficiency can dramatically slow training jobs that can run days or weeks.

Physical Connectivity: high-speed and high-density interconnects support +400G and beyond (400G and below can be orchestrated by direct attach copper or active copper), and often needs optical cables, pluggable optics, and structured fiber management systems to maintain signal integrity and simplify deployment (though advancements in active copper has seen 800G and 1.6T speeds in some cases). High density fiber management is critical due to congestion concerns and associate cooling implications.

Power Distribution: significantly higher power must be delivered to the rack to feed dense GPU configurations. This requires robust busway systems, high-tech power distribution units (PDUs) and redundancies. These configurations provide medium voltage level power directly to the rack/rack adjacent areas vs. low voltage alternatives.

AI racks can draw power >100kW (i.e. NVDA Blackwell) far exceeding what's typical (<50kW) in legacy racks – there is currently a roadmap by NVDA to get to ~600 kw per rack by the end of 2027. This requires busways, PDUs and connector systems to deliver medium voltage closer to the rack than in traditional system designs while including redundancy to prevent downtime.

Thermal Management: racks can exceed >100kW in AI training configurations making advanced cooling essential.

According to another Amphenol white paper ([HERE](#)), latency sensitivity and limited reach are defining characteristics of AI clusters. These constraints are best served by high-density fiber management platforms. Without these systems the maximization of port density, signal integrity, and scalability wouldn't be possible. Core components include frames/racks (could be thousands of connectors per rack vs. ~100 in typical configurations), modules & cassettes, patch panels/adaptor plates, cable management accessories, and connectivity hardware (active optics etc.). There are several key suppliers of these systems: **Amphenol** (HyperX Optical Distribution Frame specs [HERE](#)), **Corning** (EDGE and EDGE8 specs [HERE](#)), **CommScope** (now part of APH, NG4access specs [HERE](#) and [HERE](#)).

TE Connectivity (detail [HERE](#)) and **Molex** (detail [HERE](#)) also have exposure to these high density fiber systems, however more on the component level rather than full systems.

Emerging Networking Trends

Parallel optics are critical for next generation data center interconnects, especially as workloads for AI training reach terabit-scale bandwidth (vs. hundreds of gigabytes used in intensity-modulation with direct-detection (IM-DD) architectures) – and are used with Multi-Fiber Push On (MPO) connectors (more detail [HERE](#)). Instead of a single fiber, parallel optics split the signal across multiple fibers in “parallel” lanes, thereby increasing throughput for short-reach inside the rack. APH and TEL's Octal Small Form-Factor Pluggable (OSFD/QSFD-DD) products cater to this trend.

Coherent optics encode information not just on intensity but also phase amplitude and polarization by using a local oscillating laser and digital signal processing (DSP) to recover signal. This has been typically used in long haul applications but more recently has moved to shorter-reach “coherent lite” modules for inside data center use supporting terabit-scale bandwidth.

Co-packaged optics (CPO) moves the optics right next to / inside the switch ASIC package, eliminating long electrical traces. This reduces power per bit, improves signal integrity and much higher port densities. As speed climbs to terabit speeds the electrical traces between the switch ASIC and pluggable connections become a bottleneck (power consumption rises, thermal density becomes unmanageable etc.)

Based on conversations with our TD Cowen Semis team these architectures will most likely initially impact scale out (connections between racks and switches, which is already largely optical). A move to replace copper with CPO on scale up (connections between GPUs) – if it happens – would likely not be until the later part of the decade. We dive into this topic in more detail in our recent note [HERE](#).

Direct Attach Copper (DAC) and Active Optical Cables (AOCs) are more conventional means of transporting networking signals within data centers. APH has historically been strong in this area in particular.

Recent Collaboration on CPO With Our TD Cowen Semis Team [HERE](#), as well as broader discussion in this report [here](#).

Interconnect Power Equipment

The difference in server rack power components in the most advanced AI applications comes down to power density, current handling, and thermal management. The most advanced AI training racks can draw power in excess of 200kW, which means these racks require connectors and busbars that can handle 2-10x the current of traditional racks (further power comparisons [HERE](#)). Component differences boil down into the following four areas:

Backplane & Drawer connectors like TEL's CROWN CLIP / CROWN Edge or APH's RADSOK technology both support amperage in the hundreds with lower resistance.

Busbars & Distribution need to scale to +1K amps, often with liquid cooled or hybrid designs to manage heat. TEL's pluggable busbar assemblies and APH's high-current RADSOK busbars are tailored for these responsibilities.

Rack-Level Power Shelves can be enabled by AC/DC input connectors.

Thermal & Safety Design low-resistance and high-surface area contacts, which enhance creepage and clearance for safety at higher currents.

Recent Deal Activity - APH + COMM (CCS)

In August, Amphenol announced it had signed a definitive agreement to acquire CommScope's (COMM) Connectivity and Cable Solutions (CCS) business for \$10.5B in cash. The deal expands APH's content in data center, broadband and building infrastructure. Most importantly the deal gives APH exposure to networking the entire data hall related to DC operations.

Data center related products are roughly ~40% of sales and comprise mainly "rack-to-rack" solutions and include fiber optic cables and cable assemblies, optical distribution equipment, fiber panels, passive optical devices, and raceways.

Please find the deep dive on the CCS asset and financial contributions from our recent note [HERE](#).

Emerging Debates—Scale Up & Photonics

But Will It Scale: Whose Open Scale Up Protocol Will Prevail?

Scale up has become increasingly important to AI networking (and to investors) as model sizes continue to increase far faster than accelerator designs can add high-bandwidth memory. Scale up networks address this bottleneck by creating a virtual shared memory domain across multiple physically discrete processors.

While scale up networking has largely been dominated by NVIDIA's closed proprietary standard NVLink, the industry is developing open standard alternatives for non-NVIDIA accelerators. As a result, we have seen new standards developed, like Ultra Accelerator Link (UALink), as well as older standards expanded or adapted as in the case of Ethernet (Scale Up Ethernet, or SUE) and PCI Express (PCIe).

We see scale up networking and specifically scale up switching as a new greenfield TAM in the datacenter networking infrastructure market, which we model growing at a ~40% CAGR from ~\$4.5B in 2025E to almost \$25B in 2030E. Scale up switching is **one of the fastest growing buckets in our model** and one in which there is **no true incumbent** outside of NVIDIA, which we see commanding ~70% share by the end of the decade (though with little-to-no adoption outside of NVIDIA accelerators). The non-NVLink scale up switch market is therefore **potentially a \$8B opportunity that is effectively a jump ball** for Broadcom, Astera Labs, and Marvell.

In our view, the market is most likely to converge around Scale Up Ethernet—or said more precisely, **is likely to rely upon Ethernet's physical layer and hardware**, regardless of the transport layer protocol that is implemented. This is effectively the schema AMD is implementing in its forthcoming Helios rack—while the data-link and transport layers are UALink, the physical layer relies on Ethernet (Broadcom's Tomahawk switches). While AMD (accurately) calls this UALink (nee Infinity Fabric), from a hardware vendor perspective it is Ethernet—and we struggle to identify a single competitor who has outperformed Broadcom in Ethernet switching hardware (other than *perhaps*...NVIDIA).

While initial specs for SUE submitted to the Open Compute Project lagged UALink's in XPU-to-XPU latency, Broadcom's Tomahawk Ultra has largely closed this hardware gap... at a faster time to market versus UALink-native switches. In an environment where performance reigns supreme and time to market is key, we believe the push for "UALink-native" hardware will fully depend on the ability to deliver superior latency and throughput. **If Broadcom can deliver better specs, we believe AMD is likely to continue to iterate on its UALink/Infinity Fabric roadmap while leveraging Ethernet PHYs**—just as it has done for the first 4 generations of Infinity Fabric (which has always used either PCIe or Ethernet physical layer) and plans to do on the Helios rack.

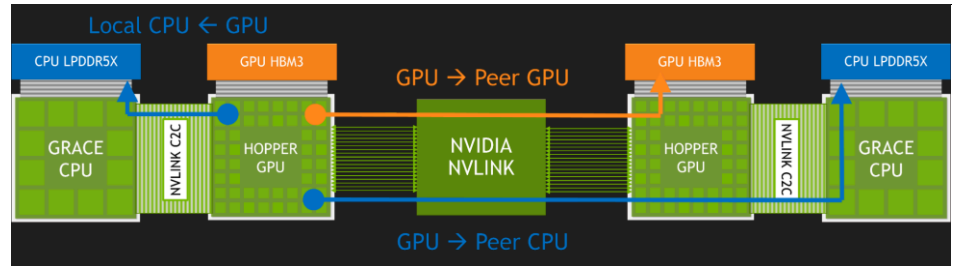
More importantly, we believe **Amazon is likely to eventually transition its PCIe-based NeuronLink to either Scale Up Ethernet or UALink**—and will face a similar decision matrix that will AMD, in which performance is paramount. Regardless of the protocol used, Amazon would need to expand its communication kernel, which is the true workhorse of the shared memory space abstraction. While some of this comes with UALink adoption, some does not—and like AMD, we believe Amazon would be willing to leverage a hybrid approach leveraging Ethernet PHY and UALink transport...Or "SUE by another name."

Scale Up: Becoming Bigger and Better

A more technical deep dive into what is meant by "scale up" vs "scale out" can be found in the appendix, [here](#)

In short, scale up fabrics connect accelerators within a server, rack-scale system, or across racks using a high-speed, high-bandwidth, and low-latency technology. As a result, the accelerators are able to appear as a larger, single system to the higher-level software, coordinating computations for workloads and having access to a unified virtual memory space.

Figure 40 A Scale Up Domain is Defined by Memory Access



Source: NVIDIA

Scale up has become an increasingly important and prominent part of datacenter connectivity as AI workloads begin to scale to hundreds (and later, thousands) of accelerators in and across compute trays and racks. As more and faster HBM gets attached to each accelerator and the number of accelerators in a single, scale up cluster both grow, the importance and criticality of scale up protocols rises.

A Brief Aside: An Industry at Crossroads

Just as NVIDIA established its leadership position in acceleration with the debut of the Tensor Core in 2017 (which we discuss [here](#) and [here](#)), it also laid the foundations for leadership in scale up networking through NVLink in 2016 after recognizing the limitations of traditional interconnects in meeting increasing demands of high-performance computing and AI workloads. Over successive generations, NVLink has been one of the many key factors in enabling NVIDIA's dominance in AI acceleration, enabling high-bandwidth, low-latency chip-to-chip communication that could scale to larger clusters.

While NVLink and the subsequent NVLink Switch dominated the market's early days, its proprietary nature is driving increased focus and development of open standards by both hyperscale customers and vendors for a variety of reasons, including avoiding vendor lock-in, broadening adoption, increasing interoperability, and reducing costs. Despite NVIDIA's more recent announcement of NVLink Fusion, companies must still connect their semi-custom ASIC or CPU with NVIDIA's technology.

As a result, the industry has seen increased efforts to further develop or create open standards to better compete with NVLink, such as PCI-Express (PCIe), the Ultra Accelerator Link (UALink) Consortium and Scale up Ethernet (SUE). For a more detailed explanation of the technology and different protocols in scale up connectivity, we encourage readers to refer to the appendix section, **Scale Up Domains: NUMA NUMA Yay** ([here](#)).

Packet with Care: Key Success Factors for Scale Up Protocols

With the express purpose of facilitating connection between accelerators within a single or rack-scale system, protocols should be optimized around certain key features in order to maximize bandwidth with minimal latency between accelerators: high bandwidth and bandwidth efficiency, ultra-low latency, shared memory semantics, losslessness, and best power performance area.

- **High Bandwidth and Bandwidth Efficiency:** Amdahl's law states that the "overall performance improvement gained by optimizing a single part of a system is limited by the fraction of time that the improved part is actually used." As a result, interconnect speeds must be able to handle heavy communication traffic between accelerators efficiently to prevent networking infrastructure from becoming a bottleneck. Further, with the growing amount of tensor or parallel data being exchanged, bandwidth efficiency (i.e., the number of bits in a data frame that carry the actual payload data) should be as close to 100% as possible with minimal overhead.
- **Ultra-Low Latency:** Related to the previous criteria, end-to-end (i.e., XPU to XPU) should be as low as possible despite the increasing traffic from communication between the accelerators with computational work. Low latency is key in managing the non-uniform memory access (NUMA) domain—for the higher level abstractions to "just work," each location in the virtual memory should have relatively similar latencies. In a rack-scale scale up domain, this means that top to bottom GPU to GPU connectivity needs to be as fast as adjacent GPU to GPU—a tall order.
- **Shared Memory Semantics:** As mentioned above, in a scale up fabric, the accelerators are able to act as a larger, single system. As a result, the aggregate memory of all the accelerators should also appear as a single pool for each accelerator to load and store operations as if they were locally attached. The load/store semantic is distinct from a networking framework which creates *connections* (i.e. a TCP session) rather than maintains memory *addresses*.
- **Lossless & Non-Blocking Fabric:** Scale up fabrics must be lossless and provide reliable links; this is seemingly a given, but it is particularly important because the load/store memory semantics cannot tolerate packet loss for any reason, including congestion/oversubscription. In a traditional Ethernet set-up, the transport layer typically manages packet re-sends when faced with packet loss or corruption; however, retransmissions at Layer 4 require a full roundtrip through the stack to retransmit the full segment, adding (1) a significant amount of time to tail latency due to congestion/head-of-line blocking and (2) disruptions to memory semantics. As a result, scale up fabrics need to be non-blocking – that is, packets containing requests and responses should be able to be sent and received without blocking each other by supporting hop-by-hop control on a per-port or per-traffic-class basis, thus eliminating oversubscription-induced bottlenecks and keeping latency low in a distributed computing environment. Further, necessary retransmissions should be done *before* reaching the transport layer to maintain a tight tail-latency and create minimal overhead; link-layer retransmission/retry (LLR) sends a single frame immediately upon noticing an error at the NIC-switch level, isolating and correcting the erroneous frame in microseconds versus a transport-layer resend, which requires a retransmission of an entire segment across the stack at 10-100x the time.
- **Best Power Performance Area:** Just as it is true for many components in other applications, achieving the best power per performance area is also true for any switch. As accelerators, rack-scale systems, and AI datacenters overall continue to increase necessary power draws, it is important to optimize for power draw in the chassis and cost across components, including switches. Not only does this improve the cost of silicon, but it also yields a better TCO from an improved cost of thermal management.

Currently, there are five main scale up protocols, two of which are proprietary and the remaining three consortia-based: NVLink (NVIDIA proprietary), Inter-Chip Interconnect (ICI, Google proprietary), UALink, Scale Up Ethernet, and PCIe. Each scale up domain has different points of optimization based on its heritage. In the figure below, we attempt to compare each standard based on the major criteria described above based on publicly available information:

Figure 41 Scale Up Protocols Are Largely Differentiated in Per-Chip Bandwidth and Latency Specs

Protocol	Open/Closed	Main Sponsor	Generation	Line Rate Speed	Per-Link Bandwidth (Unidirectional)	SerDes Speed	Per-Chip Bandwidth	Fabric Radix	Latency Targets (XPU-to-XPU)	Platforms
NVLink	Closed	NVIDIA	5th Gen	100 GT/s	200 Gb/s	224G PAM4	1,800 Gb/s 9x Lanes	57.6	100-150 nanoseconds	NVIDIA-based Platforms
PCIe	Open	Intel	0.6	64 GT/s	16 Gb/s	PAM4	256 Gb/s 16x Lanes	Switch-Dependent (Trainium 2 Scales to 64 Accelerators)	<100 nanoseconds (x16 lanes)	Trainium (NeuronLink)
UALink	Open	AMD	0.1	212.5 GT/s	200 Gb/s	224G PAM4	800 Gb/s 4x Lanes	102.4	350-550 nanoseconds	MI500-Series Platforms
Scale Up Ethernet	Open	Broadcom	N/A		200 Gb/s	224G PAM4	800 Gb/s 4x Lanes	102.4	<2 microseconds (<2,000 nanoseconds)	
Tomahawk Ultra	Open	Broadcom	0.7		100 Gb/s	100G PAM4	800 Gb/s 8x Lanes	25.6	<400 nanoseconds	AMD's Helios Rack
Inter-Chip Interconnect (ICI)	Closed	Google	v7		1,344 Gb/s		5,378 Gb/s 4x Lanes	921.6		TPU-based Pods

Source: TD Cowen Emphasis and Hypothesis, Company Data

The Route to Success: Who Wins the Open Standards Debate?

While NVLink remains the dominant scale up protocol in AI environments today due to the ubiquity of NVIDIA's systems in the market, we think it is early days in the open standards market as consortiums work to bring products to market. Of the three open standards we've outlined previously, we see most of the investor debate revolve around UALink and SUE.

Led by AMD with promoting partners such as Astera Labs, Intel, Google, and Microsoft, UALink has established its 1.0 specification in April 2025 detailing competitive line rate speeds and per-link bandwidth, with a higher fabric radix than NVLink. Its published latency targets are lower than SUE's performance, though still trails NVLink's GPU-to-GPU latency. UALink was specifically designed for scale up for AI accelerators, optimizing for factors such as overhead. In fact, "UALink" is largely AMD's proprietary Infinity Fabric protocol that has been implemented in the past three generations of AMD's cache-coherent APUs and accelerators. In order to foster the open standard, AMD simply donated this IP to the UALink consortium (similar to how Intel donated the original PCIe spec in the 90s).

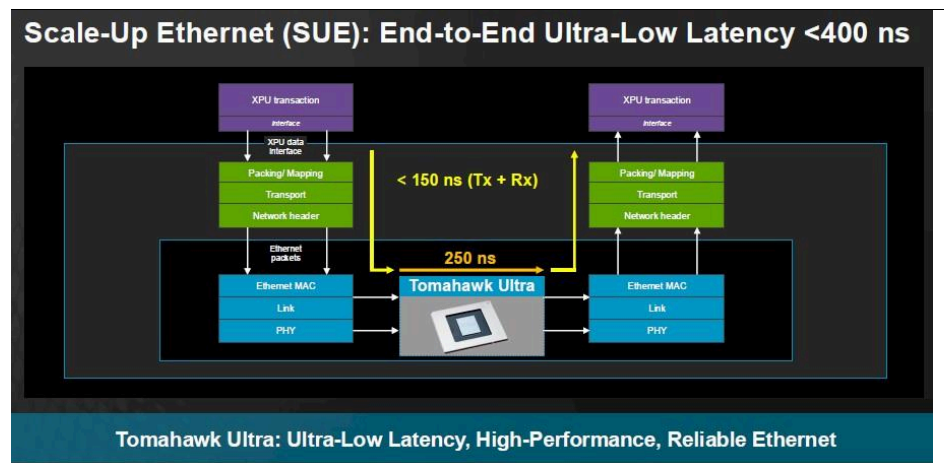
Stepping back, its initial specifications for industry partners is a compelling open-source alternative; however, given the recent publication, we expect it would take 12-24 months for UALink-native products from member companies to enter the market. Case in point, UALink-native products will not come to market in time for the launch of AMD's Helios rack-scale solution in 2026, which expects to implement UALink tunneled over Ethernet (which is the same as how AMD has implemented past generations of Infinity Fabric).

While Ethernet has been more widely used in scale out networks, Broadcom has led SUE's efforts. Once a founding member of the UALink consortium, Broadcom believes Ethernet is more than capable of getting products to market while unifying AI infrastructure under one network using the same interconnects; this would theoretically result in lower system complexity, reduced system design time, and reduced TCO. While initial specifications submitted to the Open Compute Project (OCP) were fairly comparable with UALink, the most notable area where it lags UALink is in latency, citing round-trip latency under 2 microseconds compared to UALink's estimated ~350-550 nanoseconds (based on a 128-512 radix). Recall that ultra-low latency is a critical factor for scale up fabrics to enable interactions as close to real-time as possible for AI applications.

Based on initial specifications, UALink has a technical advantage over SUE given it was designed specifically for scale up applications. However, with the release of Broadcom's Tomahawk Ultra switch in July 2025, only several months after these standards were published, we think SUE's value proposition has become much more compelling and is now on more even footing with UALink due to the significantly improved latency specs.

Detailed in the figures below, Tomahawk Ultra details port-to-port (or GPU-to-switch) latency of <150 nanoseconds for both transmit and receive ends, with a 250-nanosecond pin-to-pin hop across the switch when paired with SUE-compliant hardware. Net, this results in XPU-to-XPU communication of <400 nanoseconds, putting it squarely within range of UALink's 350-550 nanosecond estimate. Broadcom also notes the product is pin-compatible with Tomahawk 5 to ensure a quicker time to market, and the product is shipping now for deployment in rack-scale AI training clusters and supercomputing environments.

Figure 42 Broadcom's Tomahawk Ultra Introduces Materially Lower Latency at <400ns...



Source: Broadcom, courtesy of Next Platform

Figure 43 ...Putting SUE In Serious Contention As a Leading Domain for Scale Up

Protocol	Open/Closed	Main Sponsor	Generation	Line Rate Speed	Per-Link Bandwidth (Unidirectional)	SerDes Speed	Per-Chip Bandwidth	Fabric Radix	Latency Targets (XPU-to-XPU)	Platforms
PCIe	Open	Intel	0.6	64 GT/s	16 Gb/s	PAM4	256 Gb/s 16x Lanes	Switch-Dependent (Trainium 2 Scales to 64 Accelerators)	<100 nanoseconds (x16 lanes)	Trainium (NeuronLink)
UALink	Open	AMD	0.1	212.5 GT/s	200 Gb/s	224G PAM4	800 Gb/s 4x Lanes	102.4	350-550 nanoseconds	MI500-Series Platforms
Scale Up Ethernet	Open	Broadcom	N/A		200 Gb/s	224G PAM4	800 Gb/s 4x Lanes	102.4	<2 microseconds (<2,000 nanoseconds)	
Tomahawk Ultra	Open	Broadcom	0.7		100 Gb/s	100G PAM4	800 Gb/s 8x Lanes	25.6	<400 nanoseconds	AMD's Helios Rack

Source: TD Cowen Emphasis and Hypothesis, Company Data

As we had discussed in the context of accelerators in our recent Ahead of the Curve report ([here](#)), at the end of the day in GenAI acceleration, **performance is king**. Taken in conjunction with Amdahl's law and the full-stack nature of GenAI applications, we believe that market leadership will be defined by companies that drive the most innovation and best performance in physical networking infrastructure in this case.

Longer term, we believe the market is likely to converge around one open standard as the protocol of choice. In our view, SUE is likely to pull ahead in the scale up switching market as the open protocol of choice given Tomahawk Ultra is shipping today, over a year ahead of when UALink-specific switches are expected to come to market. In fact, we think AMD's Helios rack could be using Tomahawk Ultra at launch to tunnel UALink over Ethernet. Further, Tomahawk Ultra has closed the gap with UALink's 1.0 specifications, particularly on latency, but we remind investors that these are currently only targets and specs; it is difficult to predict what materialized product performance will emerge until UALink-specific switches come to market in volume and are tested in production environments.

We acknowledge that there may be shortcomings with SUE; while Tomahawk Ultra has largely solved the hardware limitations around latency, we note that **Ethernet in and of itself does not manage coherent memory pools**—at its core, it is the protocol to send the message, which is only one step of many in creating a scale up domain. Scale Up Ethernet will still require drivers and runtimes on top of the protocol to orchestrate and manage memory traffic between XPU's. While this may change in the future, we believe vanilla SUE offerings today still require the significant software required for memory access and semantics, which, as discussed above and expanded upon in the appendix, is key to defining a scale up system. That said, we believe SUE will still present a competitive and viable solution with a first mover advantage, with flexibility for other customers (e.g., AMD) to use their own interconnect fabric on top.

SUE does not need to see broad adoption for Ethernet hardware to "win" in scale up—performance will dictate what switch is used, and there are very few switch ASICs that have been able to compete with Broadcom

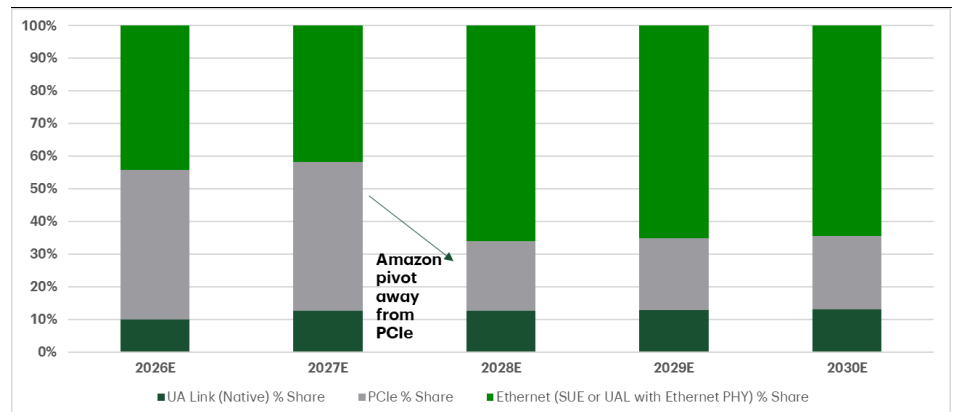
This final point is a nuance worth emphasizing. Our model is focused on the silicon—the physical layer. Scale up protocols span four layers of the network, and as AMD is demonstrating with its plans to implement a hybrid UALink/Ethernet architecture in the forthcoming Helios racks, UALink benefits from having Ethernet as the PHY. A switch is not a pure Layer 1 device (hence why they are called Layer 2 switches), but it is clear some adaptation can occur within the Infinity Fabric/UALink stack to allow for Ethernet L2. **Our call is not on AMD abandoning UALink**. Rather, we believe that the "penalties" for using Ethernet hardware, specifically switches, are so low that the debate is **less about protocol and much more about performance**. With that in mind...show us a better switching ASIC than Broadcom.

Modeling Discussion & Company Implications

We see scale up switching as a meaningful greenfield TAM for datacenter networking infrastructure, with meaningful revenue just beginning to accrue in 2025E. Our model calls for growth at a 38% CAGR through 2030E, reaching almost \$23B in revenue by the end of the decade from just ~\$400M in 2024. This growth is largely driven by unit growth (~25% CAGR 2025E-2030E) as XPU clusters continue to grow in size and bolstered by continued ASP expansion.

Given the proliferation of NVIDIA systems in the industry, we see NVLink (the incumbent) maintaining majority share of the scale up switch market, exiting the decade with still ~70% share. While this constitutes the lion's share of the market, we still see a very healthy ~\$7B market ex-NVLink up for grabs for other protocols (e.g., UALink, Ethernet, PCIe/CXL) and other companies (e.g., Astera Labs, Broadcom, Marvell). We take the view that Scale Up Ethernet is likely to see broad adoption and model it growing to be >25% of the market by the end of the decade, with the remainder of the market split between UALink, PCIe/CXL, and other protocols (e.g., Google's ICI). That said, we acknowledge that it is early days for the scale up market and the trajectory of the market could very well change direction, and we have relatively lower conviction levels on the prevailing (open) protocol by the end of the decade.

Figure 44 Scale Up Switch ASIC TAM Through 2030E



Source: 650 Group, TD Cowen estimates

Company Implications

Based on our extended model, we see the future of the scale up market as being largely positive for **NVIDIA** and **Broadcom** and come away from our findings neutral to negative for **Astera Labs** and **Marvell**. Our views on the companies are discussed below, in alphabetical order:

Astera Labs (ALAB, O'Loughlin): In our view, Astera is well positioned in scale up switching as the market exists today, but we see significant threat from Broadcom should the market move to Ethernet as we expect. We see Astera Labs as the early leader in PCIe for scale up switching and putative leader in bringing UALink scale up switching to market, with our model estimating that Astera will hold ~60% and ~90% share in those markets, respectively. We forecast Astera Labs growing scale up switch-specific revenue from ~\$100M in 2025E to almost \$700M by the end of the decade. While this represents a healthy growth vector, we see this as largely already incorporated into investor expectations (or is perhaps even lower than investor's expectations). More importantly, we see a real risk that this growth is capped by the emergence of Scale Up Ethernet or UALink-over-Ethernet. Further, while we currently model UALink share of 90%, we believe AMD specifically is engaged with **all** of

Astera's competitors on UALink switching and if that open protocol is ultimately adopted, we would anticipate fierce competition from Broadcom and Marvell to capture that market given its size.

Broadcom (AVGO, Buchalter): As the leader in Ethernet and PCIe switching (for front-end applications traditionally), we model Broadcom accruing over 20% share of the scale up switching market by the end of the decade, with market share only second to NVIDIA. We believe Broadcom will emerge as the leading open-standard Ethernet (and PCIe) switch player given its early time-to-market versus UALink and decades of undisputed switching leadership. Our model forecasts Broadcom taking ~75% share of the combined markets, representing almost \$5B of revenue. At a hardware GM% of 67.5%, we estimate scale up switching could add ~\$0.35 to EPS estimates long-term.

NVIDIA (NVDA, Buchalter): NVIDIA is the market incumbent with its NVLink offering, and we see the proliferation of NVIDIA-based systems as a wide moat for the company in the scale up switch market. Net, even as other offerings come to market, we see NVIDIA maintaining its market leadership at ~70% share through the end of the decade, with scale up switching revenue growing to almost \$16B (~30% CAGR 2025E-2030E).

Marvell (MRVL, Buchalter): We see Marvell as a limited participant in the scale up market, with minority share in UALink, PCIe, and Ethernet switches (~10-20%). That said, we still expect revenue to grow at a healthy clip (off limited contributions) through the end of the decade to >\$700M, primarily as a result of participation in the Ethernet market. While we estimate this as ~20% of Marvell's networking infrastructure silicon revenue, we note this would only represent ~5% of total datacenter revenue in 2030E, limiting its impact on Marvell's overall revenue and earnings power.

Advanced Micro Devices (AMD, Buchalter): We see AMD as less impacted by the switch market given switches are not currently part of AMD's core product portfolio and is thus not a direct participant of the scale up market for either UALink or SUE. In the near-medium term, we think the Helios rack will implement UALink tunneled over Ethernet using Broadcom's Ethernet-based Tomahawk switches for scale up given it is expected to go to market ahead of UALink-native switches. This, in our view, will not be as big an execution risk as scaling up to 72 GPUs given its Infinity Fabric (which was donated to the UALink Consortium) has historically been tunneled over PCIe or Ethernet.

Given that AMD is a founding member of the UALink consortium and one of its loudest advocates, this merits emphasizing—we believe AMD is **largely agnostic to the broader adoption of UALink beyond its own roadmap** and if UALink-native silicon does not adequately compete with Ethernet offerings such as Tomahawk Ultra, we believe **AMD will continue to use UALink at the data-link and transport layer, on top of Ethernet physical layer switches**, as they are doing with the MI400-based Helios rack.

Longer-term, while we expect SUE to emerge as the market-leading open standard for scale up, we expect AMD to continue advancing "UALink" given its foundations in AMD's own Infinity Fabric, which will remain in development to support its server portfolio and continue to be donated to the consortium. While this may cause potential investor confusion on its accelerator and rack-scale roadmap (as this first generation has done so), we believe AMD could continue to use SUE at the physical layer with fewer disruptions than investors appreciate, with the ultimate deciding factor (for AMD and its customers) being (as always) performance.

Thus far in our discussion of connectivity technology, we have been largely focused on the electrical side of the ledger—those components focused on converting a digital bit stream to an analog waveform. In a fiber link, that waveform must be then further transformed into a light wave. The optical front-end is a distinct supply chain and ecosystem from the electrical side and is likewise undergoing tectonic shifts precipitated by the demands of AI interconnect. Of greatest relevance to our coverage is the emergence of silicon photonics as a method by which many optical components are being integrated into single-chip solutions—disrupting the established supply chain while delivering cost savings and improving manufacturability.

Longer term, we are monitoring the developments from a number of smaller public companies and private start-ups whose innovation in photonics shows early promise, but still need to be proved out. Among these are **Avicena**, **Ayar Labs**, **Celestial AI**, and **Lightmatter**.

The core promise of silicon photonics is to leverage the established (and depreciated) silicon manufacturing supply chain to replace the complex and expensive III-V group-based photonics prevalent today such as indium phosphide (InP) and indium gallium arsenide (InGaAs).

The diagram is divided into two main sections by a large grey arrow pointing from left to right.

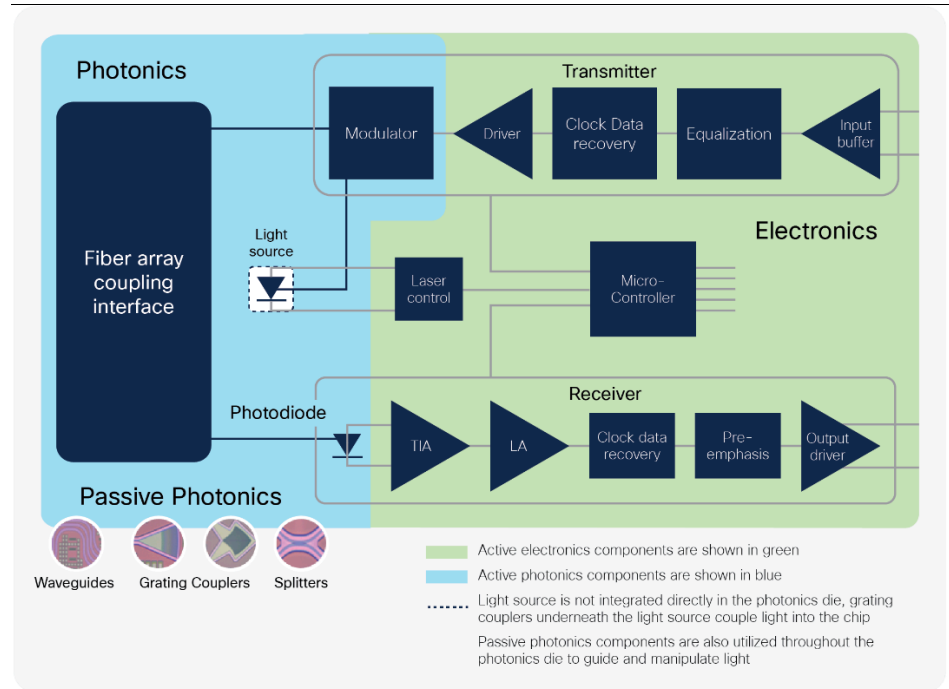
Conventional Module Design: This section shows a complex assembly of various components. Labels include: TOSA Top Cover, Epoxy TOSA Top Cover, Epoxy TOSA Heat Sink, TOSA LASER SUB-ASSEMBLY, Epoxy TOSA Fiber Strain Relief, TOSA SI Base, Optical Tap Plate, Fiber Female Collar, Epoxy TOSA SI Base, TEC Thin Sheet, Epoxy TEC Thin Sheet, ROSA EMR SHIELD HOOD, ROSA EMR ABSORBER, ROSA Jumper City, EPDXY TOSA BASE to PCB, TOP LENS, BOTTOM LENS, EPOXY ROSA LENS, FIBER FEMELLE COLLAR, FIBER FEMELLE, PDI, and TX. The components are shown in a disassembled or partially assembled state, highlighting the complexity and multiple steps involved in conventional module design.

Modules with Silicon Photonics: This section shows a more integrated design. Labels include: SiPh Chiplets in Module, Phy IC, and Fiber Jumper. The components are shown as a single, more compact unit, indicating a reduction in complexity and a move towards higher integration.

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The major photonic components in an optical link (whether pluggable or co-packaged) are the light source (often a laser), a modulator, the coupling interface, and the photodiode or photodetector. Additional passive components such as waveguides and couplers round out what is required to send and receive optical signals. A photonic integrated circuit (PIC) combines these functions on a single component.

Figure 46 Photonics Deals With the Modulation and Transmission of Light (Photons) Much As Electronics Deals With the Modulation and Transmission of Electricity (Electrons)



Source: Cisco

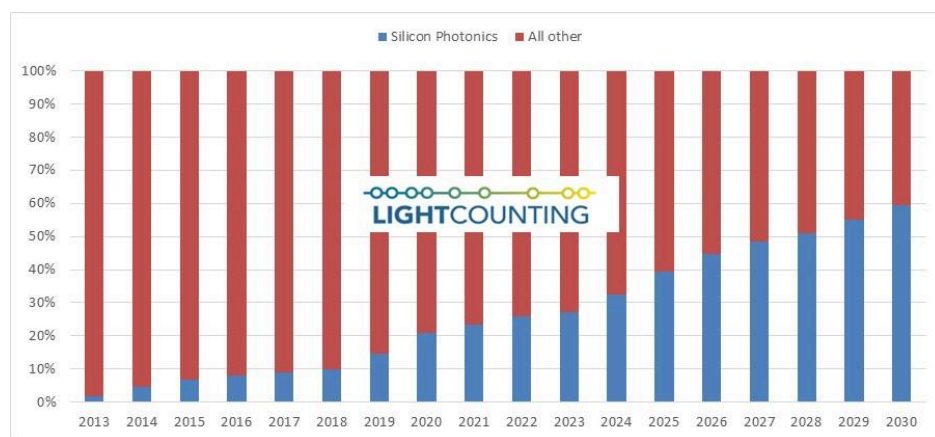
The integration of the photonic components into a single chip brings material benefits in simplicity of transceiver design and manufacture. In a discrete implementation of an 8-lane, single-mode link, 8 separate waveguides and modulators are required, all of which requires rigorous testing.

PICs can and have been developed using non-silicon semiconducting materials such as indium phosphide (InP). However, the InP manufacturing process is extremely complex and expensive, even in the context of semiconductor manufacturing. InP crystal growth is complex, the wafers are brittle, and epitaxial structures can be hard to align properly to the resulting crystal lattice. Thus, while InP is acknowledged as the best *performing* substrate material for photonic applications, scaling this supply chain is difficult (especially scaling a PIC given the yield implications of the larger die). As an illustrative example, in 2024 Coherent announced the world's first-ever 6-inch (150mm) InP wafer fab ([here](#)). Consider this is some **45 years after** the first introduction of 150mm silicon wafers, which came in 1980.

Simply put, silicon photonics enables a much greater level of integration and scale as compared to InP-based solutions. SiPho benefits from the established CMOS supply chain (though the unique requirements of photonics makes the manufacturing process itself a specialty process at foundries such as Tower Semi and Global Foundries), in addition to the smaller process geometries achievable in silicon (to a point).

Due to the benefits of SiPho, especially at higher lane rates and lane aggregation counts, third-party industry analysts estimate the penetration of SiPho-based transceivers is likely to grow materially. Lightcounting, for example, estimates that revenue share of SiPho-based transceivers is likely to expand from ~32% in 2024 to nearly 60% in 2030, due to the penetration of units as well as SiPho's over-indexing to higher performance applications (such as 800G+).

Figure 47 Silicon Photonics-Based Transceivers Are Gaining Share

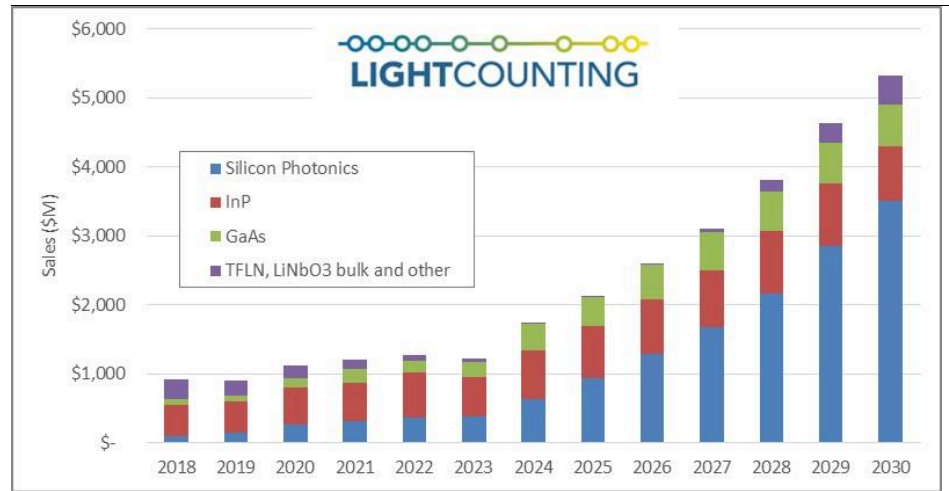


Source: Lightcounting

In our view, the growth of SiPho is likely to benefit a few key suppliers in our coverage. **Broadcom**, **Marvell**, and **Intel** have all developed an extensive portfolio of silicon-based PICs that are being increasingly integrated into high-speed transceiver applications. Marvell has been investing in the technology for over ten years (dating back to early efforts inside then-separate Inphi). Likewise, Broadcom's silicon photonics heritage stems from its long history as a leader in optical interconnect, and it has been an early leader in co-packaging as enabled by its silicon PICs. Finally, though Intel exited the *transceiver* business in 2023, it continues to invest in the PIC side of silicon photonics and has in fact shipped more than 8 million PICs since 2016 making it technically the incumbent.

Importance within the market notwithstanding, it is key to contextualize the opportunity for Broadcom and Intel especially as largely consigned to more of a "drop in the bucket" type category than a true growth opportunity. In terms of total market size, Lightcounting estimates SiPho chipsets could generate ~\$3.5B by 2030, from only ~\$635M in 2024. Significant growth to be sure, but likely not impactful to Broadcom or Intel's >\$50B revenue base today. However, for **Marvell**, gaining material share in SiPho could be important, especially in the context of their potentially declining share in the overall DSP market and the threat posed to DSPs from LPO. While the dollar content would certainly decline going from a full DSP plus integrated TIA/driver and SiPho PIC to just the PIC (or even PIC + TIA/driver), retaining any dollar content in an LPO module would likely be considered a "win" from the Marvell perspective.

Figure 48 Lightcounting Estimates Material Dollar Growth for SiPho Chipsets



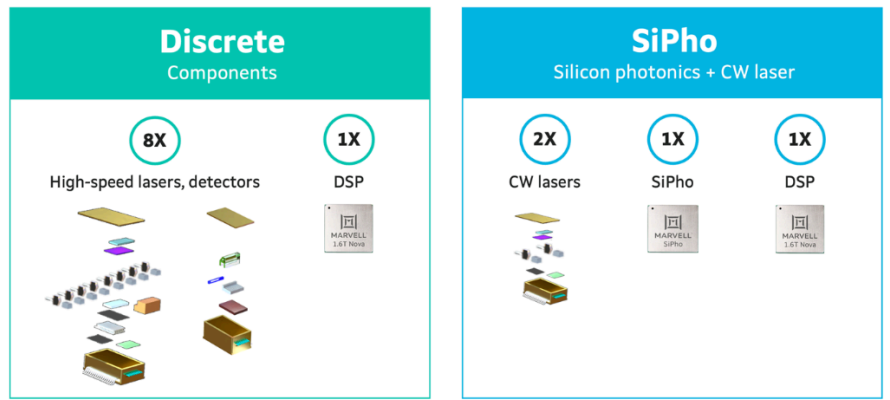
Source: Lightcounting

Continuous wave (CW) lasers are often the lasers of choice for SiPho applications due to their cost and their ability to split into multiple lanes from a single laser source. Silicon itself cannot generate photons directly, and thus an external non-Si based laser must still be used (often packaged directly with the silicon). InP-based optics typically use either single-mode electro-absorption modulated lasers (EMLs) or multi-mode vertical-cavity surface-emitting lasers (VCSELs) that have modulation integrated with the light source. EMLs are typically much more expensive than CW lasers, while VCSELs have a shorter reach. Because the integration of SiPho means the modulators are already aligned with the waveguides and other passive components in the PIC, the integration of modulators on the EML become less important or even redundant. Further, the un-modulated CW laser can be multiplexed (split) before it is modulated, meaning a single laser source can be used for as many as four lanes.

On the laser side, we believe Broadcom's portfolio of CW lasers is likely to benefit, though again is likely approaching a rounding error in overall Broadcom revenue model (and is potentially offset by declining VCSEL units). However, we see **MACOM's** CW laser portfolio as potentially participating meaningfully in the growth of single mode and SiPho-based transceivers (although we leave this largely as upside to our modeled numbers given the competitive environment).

Figure 49 SiPho Integrates Optical Components and Often Uses Fewer Lasers as Well

Discrete vs SiPho (1.6T pluggable)



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MARVELL Essential technology, done right™

Source: Marvell

Laser ASPs vary greatly by application and power, but we estimate a CW laser for DR8 applications could carry an ASP of ~\$3, with each transceiver requiring 4 lasers (each split into 2 channels for a total of 8 channels). In the context of LightCounting's 100M+ >100G optical transceivers based on SiPho in 2030, the CW laser market could represent a **multiple-hundred-million-dollar TAM for MACOM** by the end of the decade (though one with *many* competitors). While this may seem paltry given the numbers thrown around elsewhere in this report, we see this as solidly aligned with MACOM's broader strategy of diversified growth.

Start-Ups Taking Photonics Integration and Turning It Up to Eleven

In addition to the large public incumbents, there are smaller ventures (both public and private) that are innovating in unique photonic applications we are monitoring. While still very early days, these solutions are positioning themselves as the next generation in the datacenter interconnect market and will thus be key for investors to track and understand as they come to market.

Broadly, we would categorize these start-ups into two buckets: those that are seeking to fold into the existing ecosystem such as **Avicena** and **Ayar Labs**; and those that are seeking to upend the existing ecosystem such as **Celestial AI** and **Lightmatter**. For simplicity, we'll simply call them the Evolutionaries and the Revolutionaries.

The Evolutionaries: Better Integration in the Existing System Designs

Ayar Labs operates at the intersection of silicon photonics and advanced semiconductor packaging, positioning itself to address bandwidth and power bottlenecks in high-performance computing systems. The company focuses specifically on developing optical I/O solutions designed to replace traditional electrical interconnects.

Central to Ayar Labs' strategy is the TeraPHY Optical I/O chiplet, a silicon photonics-based device engineered for seamless integration with ASICs, CPUs, or GPUs via standard 2.5D/3D packaging interfaces like UCle (Universal Chiplet Interconnect Express). This chiplet utilizes wavelength division multiplexing (WDM) over a single fiber to achieve aggregate bandwidths scaling to multiple terabits per second (with initial deployments targeting 2.4 Tbps), while significantly reducing power consumption per bit—reportedly operating below 5 pJ/bit at scale (this would draw ~8W for a 1.6T link). Crucially, TeraPHY is designed as a disaggregated component, allowing system architects to pair it with separate optical engines (such as Ayar's SuperNova laser source) and leverage existing CMOS fabrication processes, thereby lowering the barriers to integration and positioning itself squarely amid the silicon photonics trend described above.

Figure 50 Ayar Labs Looking to Integrate CPO-Like Packaging via Chiplet Ecosystem

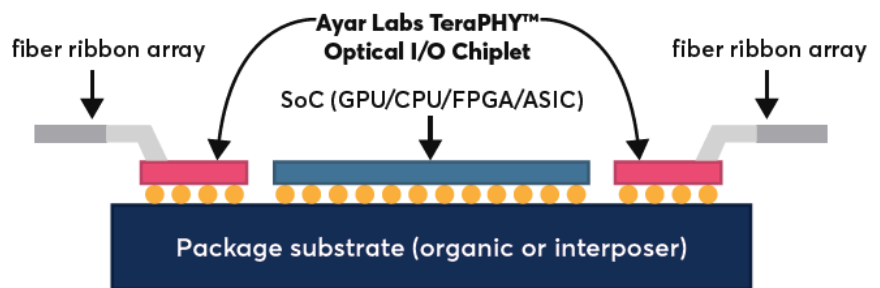


Figure 4 - Flip-Chip Attached for High Volume Manufacturing

Source: Ayar Labs

In contrast to Ayar Labs, which is primarily focused on using optics to replace electrical interconnect, **Avicena** seeks to obviate the need for SerDes entirely through its unique microLED array. It is pursuing a fundamentally distinct pathway centered on visible-light communications, leveraging gallium nitride (GaN) microLED technology to address ultra-short-reach data movement challenges in computing systems.

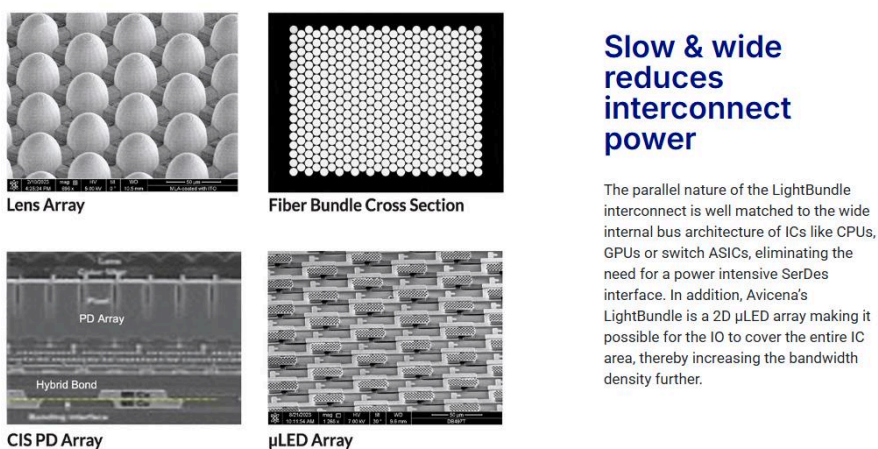
The company targets the chip-to-chip and die-to-die connections within computing packages where traditional SerDes circuits dominate. The company's approach centers on eliminating the serialization step by directly aligning its optical data transmission mechanism with the parallel structure of processor internal buses. This offers an alternative pathway to silicon photonics-based co-packaged optics solutions that still rely on serial transmission schemes.

The core innovation enabling this architecture is Avicena's LightBundle platform, which integrates dense arrays of individually addressable gallium nitride (GaN) microLEDs (reportedly exceeding 10,000 emitters/mm²) with CMOS driver electronics to create massively parallel optical channels. Unlike serial optical solutions that convert parallel bus data into high-speed serial streams for transmission, LightBundle transmits data in parallel across its microLED array, with each emitter or group of emitters directly corresponding to a bit lane in the processor's native bus structure. This direct mapping allows data to flow from the processor's parallel bus through the microLED array into multimode fibers or free-space channels without serialization, achieving demonstrated aggregate bandwidths >1 Tbps while operating at sub-1 pJ/bit energy efficiency in laboratory settings.

Within the broader photonics ecosystem, Avicena's bus-aligned microLED strategy represents a departure from industry-standard optical interconnect approaches, targeting a niche where parallelism—not serial speed—is the optimal solution for overcoming the electrical interconnect bottleneck. Avicena's LightBundle aims to render SerDes obsolete entirely for specific ultra-short-reach applications by leveraging the processor's inherent parallel data pathways.

However, its ecosystem viability hinges on overcoming significant challenges: achieving reliable high-speed modulation across thousands of microLEDs simultaneously, managing thermal density in emitter arrays, and establishing compatibility with semiconductor packaging and communication standards. Interestingly, we highlight Credo's recent acquisition of Hyperlume ([here](#)) as a validation of the potential merits a microLED approach may offer for high-speed interconnects and we look forward to learning more as these technologies mature.

Figure 51 Avicena MicroLED Array Obviates Need for SerDes



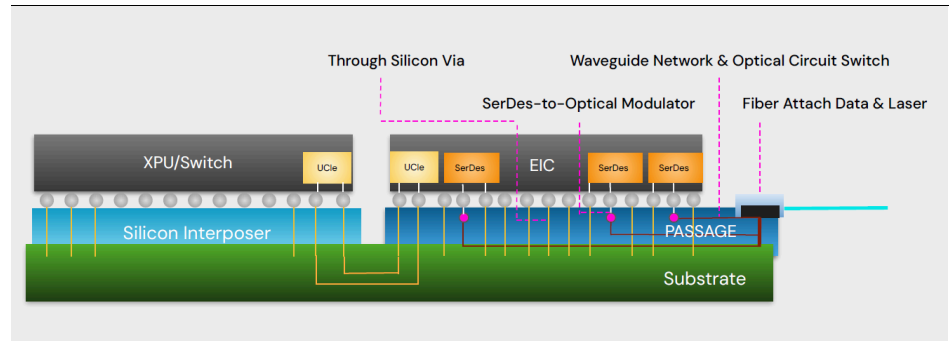
Source: Avicena

The Revolutionaries: Moving to 3D Photonic Integration to Solve the Beachfront Problem

Moving from evolution to revolution, **LightMatter** and **CelestialAI** represent convergent approaches to the integration of optical data movement directly into the foundational layers of semiconductor packages. Rather than treating optics as discrete chiplets or peripheral modules, both companies envision an "optical fabric" where photonic waveguides, modulators, and detectors are monolithically embedded within silicon interposers or 3D-stacked substrates—effectively collapsing the distinction between electronic and photonic layers. This approach targets the elimination of SerDes-intensive electrical interfaces for die-to-die and chip-to-chip communication, positioning optics as a native component of the compute architecture itself. Within the broader landscape, they occupy the extreme end of co-packaging ambition, distinct from hybrid chiplet solutions (e.g., Ayar Labs) or parallel microLED approaches (e.g., Avicena).

From a product perspective, LightMatter offers its Passage M1000 and L200, both designed for direct 3D integration through hybrid bonding, solving the "beachfront" problem of traditional I/O. Similarly, CelestialAI's Photonic Fabric leverages a wafer-scale integrated optical architecture, embedding thousands of photonic transceivers directly into silicon interposers. Both solutions bypass traditional I/O methods (in which SerDes is placed on the outside of the chip and data exits laterally) by layering photonic integrated circuits directly under the electrical interface on the processor.

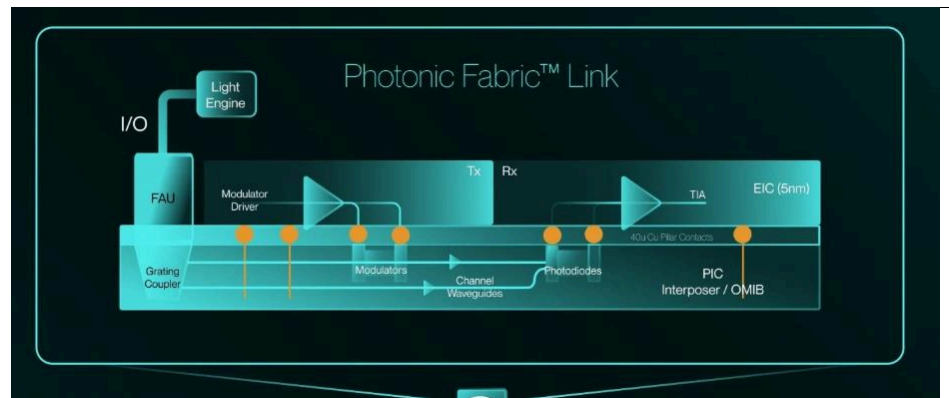
Figure 52 Lightmatter Envisions 3D Photonic Integration



Source: Lightmatter

The pursuit of this "ultimate co-packaging" paradigm introduces formidable manufacturing hurdles that underscore the nascent nature of both approaches. However, the promised specs are impressive—for example, the **1,000 Tbps of total bandwidth** offered by Lightmatter's M1000 in a switch application.

Figure 53 CelestialAI's Photonic Fabric



Source: CelestialAI

FRAME CHECK SEQUENCES: APPENDIX

Ethernet Dataframe Format



Information Theory: Bits, Noise, and Bandwidth

Signal Modulation & Loss: Who is PAM??

Networks Are Like Onions: They Have Layers

SerDes: Nuts and Bolts of Communications

Scale up & Scale Out: Let's Get Some Things Straight

Cheat Sheets: Sound Smart

Frame Check Sequence: Appendices

The final section in the standard Ethernet dataframe is the four-byte frame check sequence, which enables the receiver to identify bit-level errors. We close with our own frame check sequence, a brief overview of fundamental technologies in datacenter connectivity. These act as a foundation against which we can compare the above payload to check for errors in our thinking. Investors can largely treat the below as a primer, and we intend for this section to be largely expository in nature.

Information Theory and Physics: Bits, Noise, and Bandwidth

*"The fundamental problem of communication is that of reproducing at one point, either exactly or approximately, a message selected at another point."
-Claude Shannon*

What is a bit?

For most semis investors, a bit describes the state of an individual gate—on or off, 1 or 0. The fundamental building block that underpins every processor and the Boolean-logic enabled world of digital electronics. The word, however, representing "binary digit," pre-dates the first microprocessor by some 20 years. Its first use was to describe a unit for measuring information, by Claude Shannon in his seminal *A Mathematical Theory of Communication* (though he does characteristically assign credit for the term itself to his Bell Labs colleague J.W. Tukey).

Disclaimer: This is, at best, a cursory summary of key information theory principles relevant to our discussion. The field has broad implications well beyond this in areas such as data compression and more that we make no attempt to unpack

The fundamental insight of information theory, a field of study effectively established with the publication of Shannon's paper, is that all communication can be described with a common framework and language. Regardless of the specific message being sent, the format of the message, or even the physical medium used, all communication can be abstracted into a source, a transmitter, a signal, a channel, a receiver, and a destination (shown in Fig. 54 below). Further, every message can itself be described—and therefore transmitted—digitally via coding as digital logic.

Figure 54 Any Communication Link Can Be Described by This Fundamental Schema

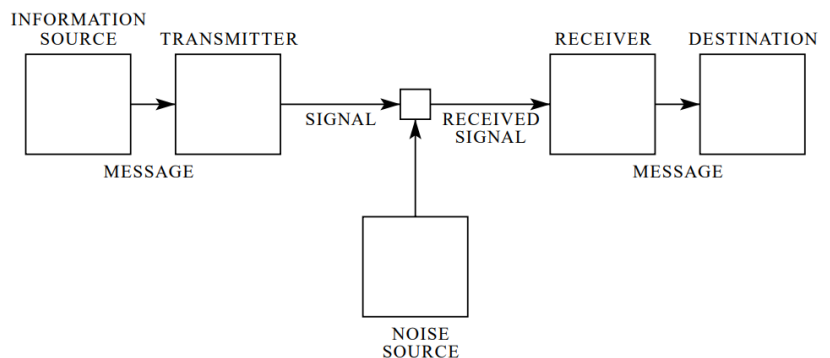


Fig. 1 — Schematic diagram of a general communication system.

Source: Shannon, Claude; *A Mathematical Theory of Communication*, 1948.

Once distilled into this framework, Shannon and others over the years have described a series of fundamental formulae that describe and frame communication in the presence of "noise"—the stochastic perturbations in the medium and environment that prevent a message from being precisely "heard" by the receiver and destination.

One such formula, called the Shannon Limit or the Shannon-Hartley theorem, is especially instructive in helping to frame how to think about digital communication. It states:

$$C = B \log_2(1 + S/N) ,$$

where:

C = Channel capacity (Bits per second)

B = Channel bandwidth (Hz)

S/N = Ratio of signal power to noise power

A similar predecessor to Shannon-Hartley is the Nyquist formula for communication in a noiseless channel:

$$C = B \log_2 M ,$$

where:

C = Channel capacity (Bits per second)

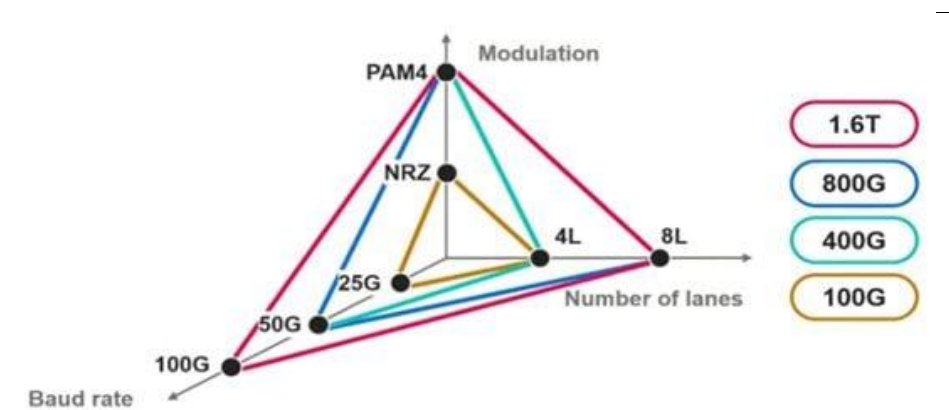
B = Channel bandwidth (Hz)

M = Bits communicated per symbol

Taken together, we can observe that there are three interconnected levers that systems designers can pull to increase maximum throughput of a communications system: bandwidth, density of bits per symbol, and increasing the ratio of signal to noise (by either increasing signal power or decreasing noise power). Those interested in falling down the rabbit hole can explore these dynamics directly [here](#).

This framing of communications throughput having multiple levers has a direct corollary in how we consider achieving a specified line rate speed in a datacenter interconnect. From Marvell in Fig. 55 below, we can increase the baud rate (how frequently a symbol flips), we can increase the modulation (how many bits are carried per symbol) or increase the number of lanes over which a signal can travel (parallelizing the signal).

Figure 55 Sources of Speed: Symbol Rate, Bits per Symbol, Number of Paths



Source: Marvell

"From RF to Light" is more than just a tag line—it's an allusion to the universal nature of digital communication

Fundamentally this framing is important because it provides a mental model with which to think about *all* digital communication. This applies to not only the debates within the datacenter between copper and fiber, but further to include cellular, microwave backhaul, and satcom. **Each of these physical media in the specific appears to be radically different in application. However, when viewed through the lens of Shannon's framework these differences are abstracted away, and we can make concrete and informative conclusions about the trade-offs associated with each.**

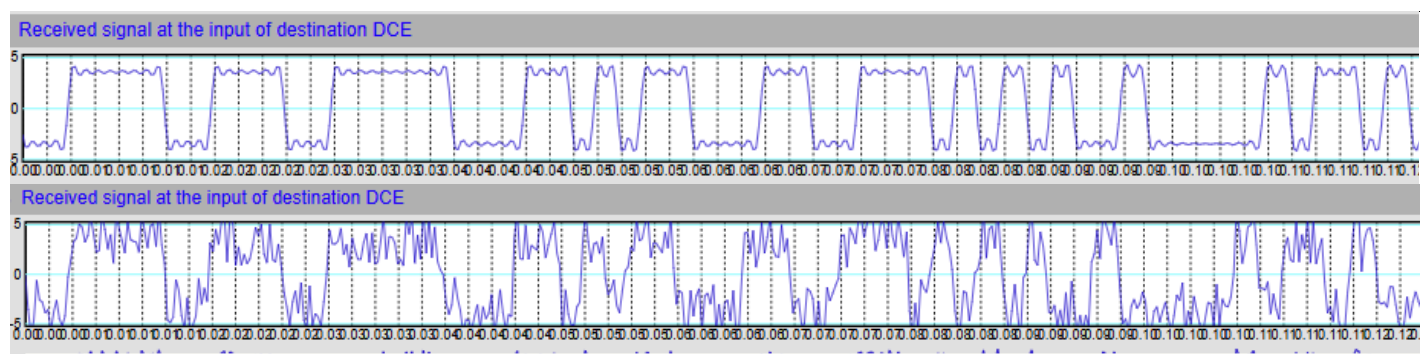
The universality of digital communications extends into the signal itself—whether over copper, fiber, or wireless, the signal being transmitted takes the form of an electromagnetic (EM) wave. Digital signals traveling through copper (say, from the Ethernet cable plugged into your laptop to the server in your firm's datacenter) appear instantaneous because the signal is in fact an EM wave "traveling along" (being guided by) the cable, moving at nearly (90%+) the speed of light. Similarly (and perhaps more intuitively), fiber is the transmission of (visible) light waves, while wireless communications use various EM spectra depending on the geography and standard used. Notably, this is the theory behind MACOM Technology's slogan, "From RF to light." The tagline alludes to the insight that RF and optical communications, especially at the component level, have many similarities and overlap because they are fundamentally linked.

Back to Reality: Noise and Attenuation Matter

There are limits to the universality of this model, however—while the physical medium does not matter in the framework above, it does matter very much in application. The two primary variables we have ignored thus far are noise and attenuation.

Noise, which we have encountered already in Shannon-Hartley's formulation, is simply any unwanted modifications that may impact the clarity of a signal at the receiver. There are many (many) different sources of noise, as well as other noise-like impacts to signal fidelity such as reflections and electromagnetic interference (EMI). We will not attempt to exhaustively describe noise. In this context, it is sufficient to note that **wireless communication is most susceptible to noise, copper less so, and fiber is the least susceptible to noise** (all else equal, which it never is).

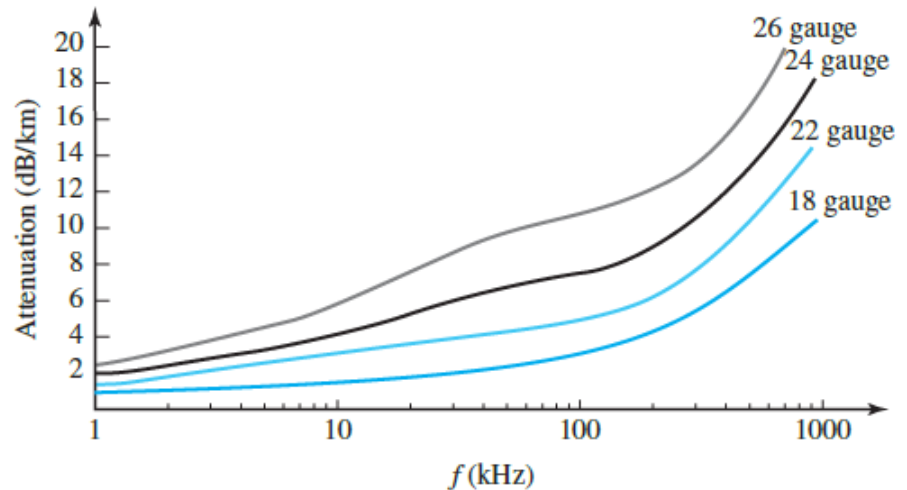
Figure 56 The Impact of Noise in a Channel on Received Signal (SNR of 500 [Top] vs. SNR of 5 [Bottom])



Source: TD Cowen, CodSim 2.0.5, Universidad de Malaga

If we consider noise a negative impact to signal fidelity, we can think of attenuation as a negative impact to signal strength, or amplitude. Attenuation is relatively intuitive—the further away you are from a noisy colleague (transmitter), the greater that "signal" is attenuated by the time it reaches your ear (receiver). Like noise, **attenuation is generally greatest in wireless, less but still material in copper, and minimal in fiber.**

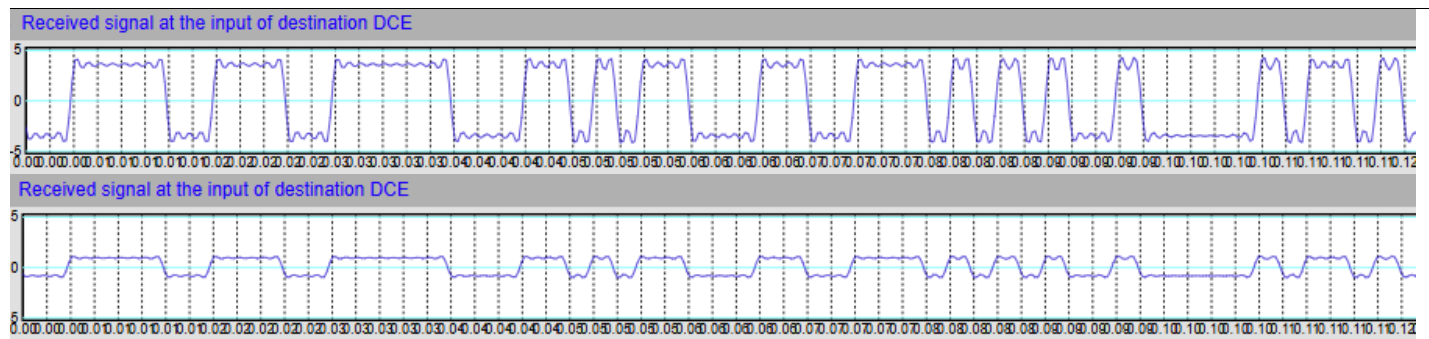
Figure 57 Higher Frequencies Attenuate Faster in Copper



Source: Forouzan, *Data Communications and Networking with TCP/IP Protocol Suite*, 6th Ed. 2022.

Importantly, attenuation has a direct relationship with frequency—a higher frequency signal will have a greater coefficient of attenuation (all else equal) (see Fig. 57, above). A higher signal rate requires more high frequency harmonics (see below), thus attenuates faster.

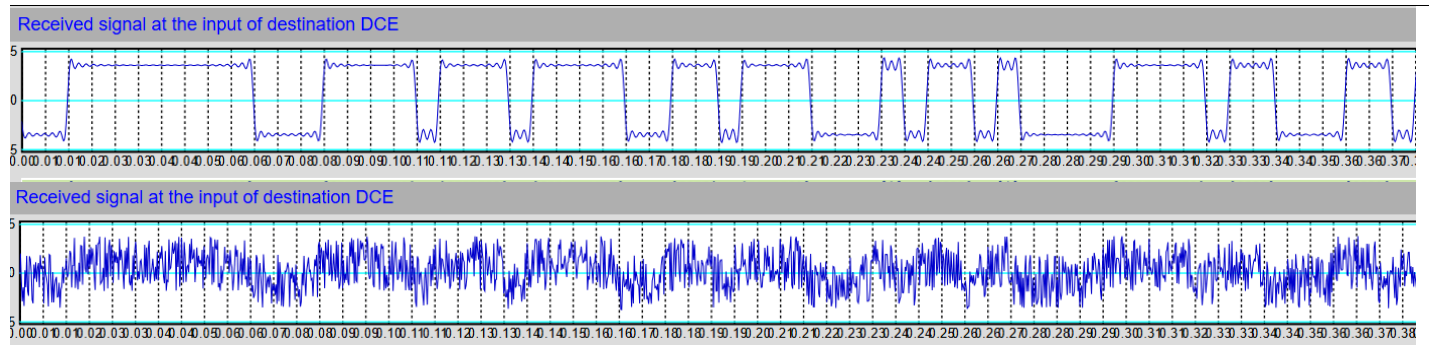
Figure 58 The Impact of Attenuation in a Channel on Received Signal (A_{coef} of 1 dB/Km [Top] vs. A_{coef} of 5 dB/Km [Bottom])



Source: TD Cowen, CodSim 2.0.5, Universidad de Malaga

When noise is added to an attenuated signal, the problem compounds. In Fig. 56 above, the signal is still largely recognizable despite the noise. Likewise, while the attenuation in the channel in Fig. 58 decreases the amplitude, the signal remains distinct. However, consider Fig. 59, in which the two are combined at the same magnitude as in the previous examples. The resultant signal has a bit error rate (BER) of ~0.35. That is, on average every third bit would be incorrectly interpreted at the receiver.

Figure 59 The Combined Impact of Attenuation and Noise (SNR of 500 & A_{coef} of 1 dB/Km [Top] vs. SNR of 5 & A_{coef} of 5 dB/Km [Bottom])



Source: TD Cowen, CodSim 2.0.5, Universidad de Malaga

Whether in the AI datacenter, in telecom, or beyond, **the challenge for engineers is to overcome noise and attenuation** through a complex menagerie of tools in both physical- and encoding-level domains. This fundamental struggle against physics is the *raison d'être* for companies such as MACOM, Semtech, Credo, Aster Labs, Marvell, Broadcom, and many more.

Signal Modulation & Loss: Who Is PAM?

*"Any sufficiently advanced technology is indistinguishable from magic."
-Arthur C. Clarke*

A changing electrical impulse transmitted over a wire takes the form of an electromagnetic wave. This is a fundamental point worth repeating, and it must be understood (or at the very least accepted) to fully harmonize our previous discussion of information theory with that of signal modulation and loss.

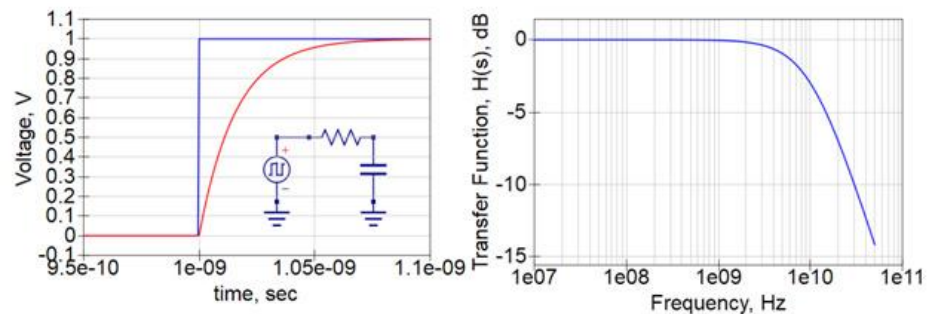
We tend to think of electricity as a "flow"—power is "flowing" from the wall outlet into the TV, for instance. This mental model works when **transmission line effects** can be ignored—that is, when the fundamental nature of electricity described above has a sufficiently small impact that it can be assumed away. The rule of thumb is typically transmission line effects can be ignored when the line length is less than ~10% the wavelength. We do not need to worry about the signal propagation associated with plugging in our vacuum cleaner—it just turns on. This is because the power grid (in the US) is delivering alternating current at 60 Hz—a wavelength of ~5 km.

Digital communications modulate at a much higher frequency than 60 Hz and have correspondingly significantly smaller wavelengths. Transmission line effects and the wave-like nature of electricity play a key role in understanding digital communications.

The most straightforward method to transmit a bit through an electric channel is by manipulating voltage levels. Simplistically, imagine that to send a 1, +1V is used, and to send a 0, +0V is used. In practice, different voltage levels are used and there are a variety of different coding manipulations used to adjust for the practicalities of such a method (for example, how to send a string of consecutive 1s or 0s?), but for our purposes let's ignore these complexities for now.

An idealized message using this method might look like the blue line in Fig. 60 below, with a distinct and instantaneous square shape. However, in reality, voltage levels do not change instantaneously—as shown in red, it takes a non-zero amount of time for the voltage levels to reach the "1" point (and likewise to return to zero). This is called "rise time" and can lead to intersymbol interference (ISI)—if the symbol rate is very high, the rise times can begin to overlap.

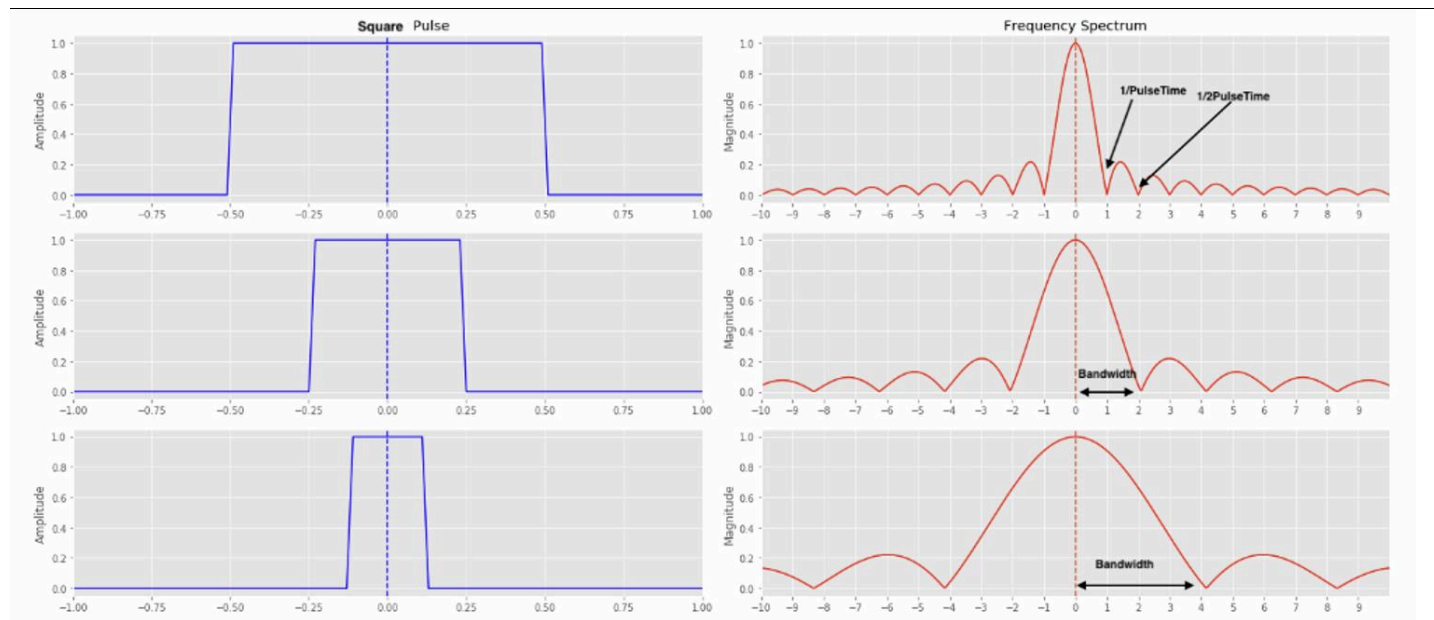
Figure 60 Idealized Square Impulse Shapes Do Not Account for "Rise Time"



Source: Signal Integrity Journal

The left side of Fig. 60, above, shows the pulse in the time domain (seconds in time are the x-axis), but we can also analyze a pulsed signal in the frequency domain (frequency bands are on the x-axis). The time domain shows the *message*, while the frequency domain shows the *waveforms the message is using*. Fig. 61 compares the time and frequency domains side by side.

Figure 61 The Shorter the Pulse (Higher the Symbol Rate), the More Frequency Bandwidth is Required

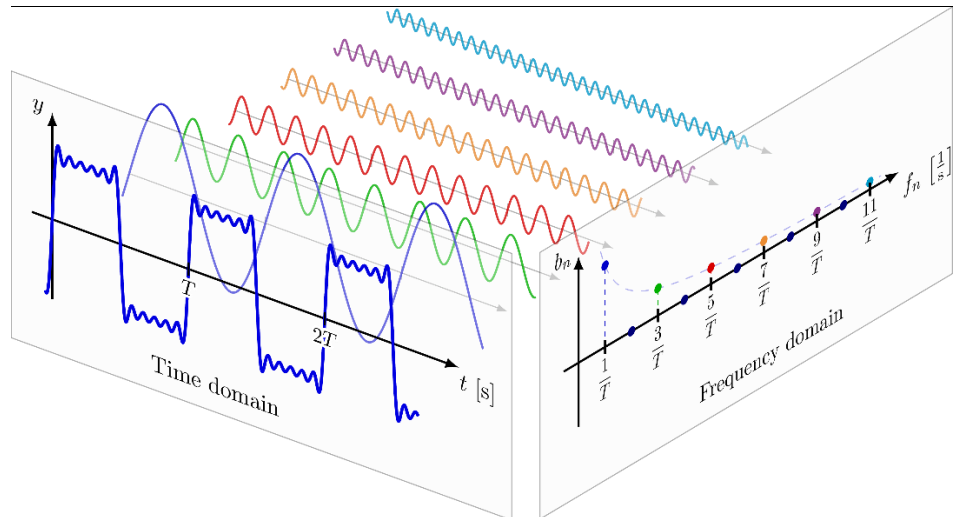


Source: Dip Singh

Fig. 61 above further demonstrates a fundamental property of modulation—the faster the symbol changes (which is the symbol rate, or baud rate), the more bandwidth in the frequency domain is required to transmit the square pulse. A decomposition of a square wave (in the time domain) using a Fourier analysis may assist in understanding why this is the case.

Briefly, Fourier showed that any waveform can be decomposed into a series of sinusoidal waves at increasingly higher harmonic frequencies (Fig. 62, below). An ideal square wave pulse is composed of an infinite number of higher and higher frequency waveforms to create the sharp angles in the time domain (the animation [here](#) is instructive and frankly we are indebted to Mr. Singh for expanding our understanding of this topic).

Figure 62 Contemplating a Square Pulse Modulation's Fourier Composition May Aid in Understanding (Or May Further Confuse the Issue)

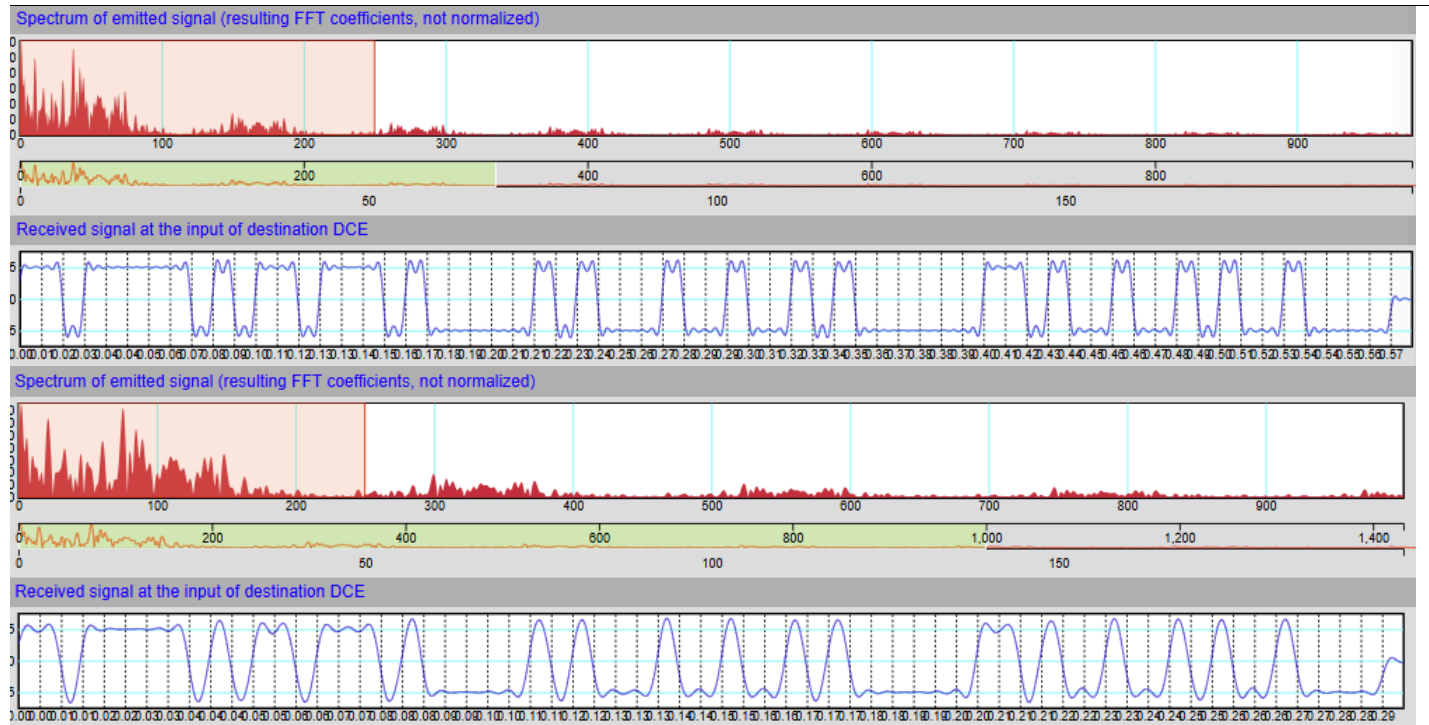


Source: tikz.net/fourier_series

We will conclude this confusing foray into Fourier (puns always intended) by connecting this back to the Shannon-Hartley theorem. Recall that Shannon-Hartley states that a channel's **theoretical** capacity depends on bandwidth (in the frequency domain) and the signal-to-noise ratio. This is true because to increase the symbol rate and therefore the data rate, the wave has to be "more square" to prevent ISI—and you therefore must "use" more frequency bandwidth to better square the corners. I cannot emphasize enough how "in layman's terms" that description just was.

Consider the two examples below, which show the frequency domain diagram and received signal for two bandwidth-limited channels. The shaded red area in the frequency domain is the available bandwidth, with the spectrum that is fully attenuated showing in the white. The first example shows a signal rate of 112 bps, and the second doubles that to 224 bps. Note the relative clarity of the received signal between the two, and compare how significant the higher frequency harmonics are in the spectral analysis. Once again leveraging extreme layman's terms—compare how "square" the signal is at 112 bps, and how small the red stuff is in the white background compared to the rounded edges of the 224 bps example and the much larger red stuff in the white background.

Figure 63 Increasing the Signal Rate Requires More Bandwidth for a Cleaner Signal (112 bps baud rate vs. 224 bps baud rate)

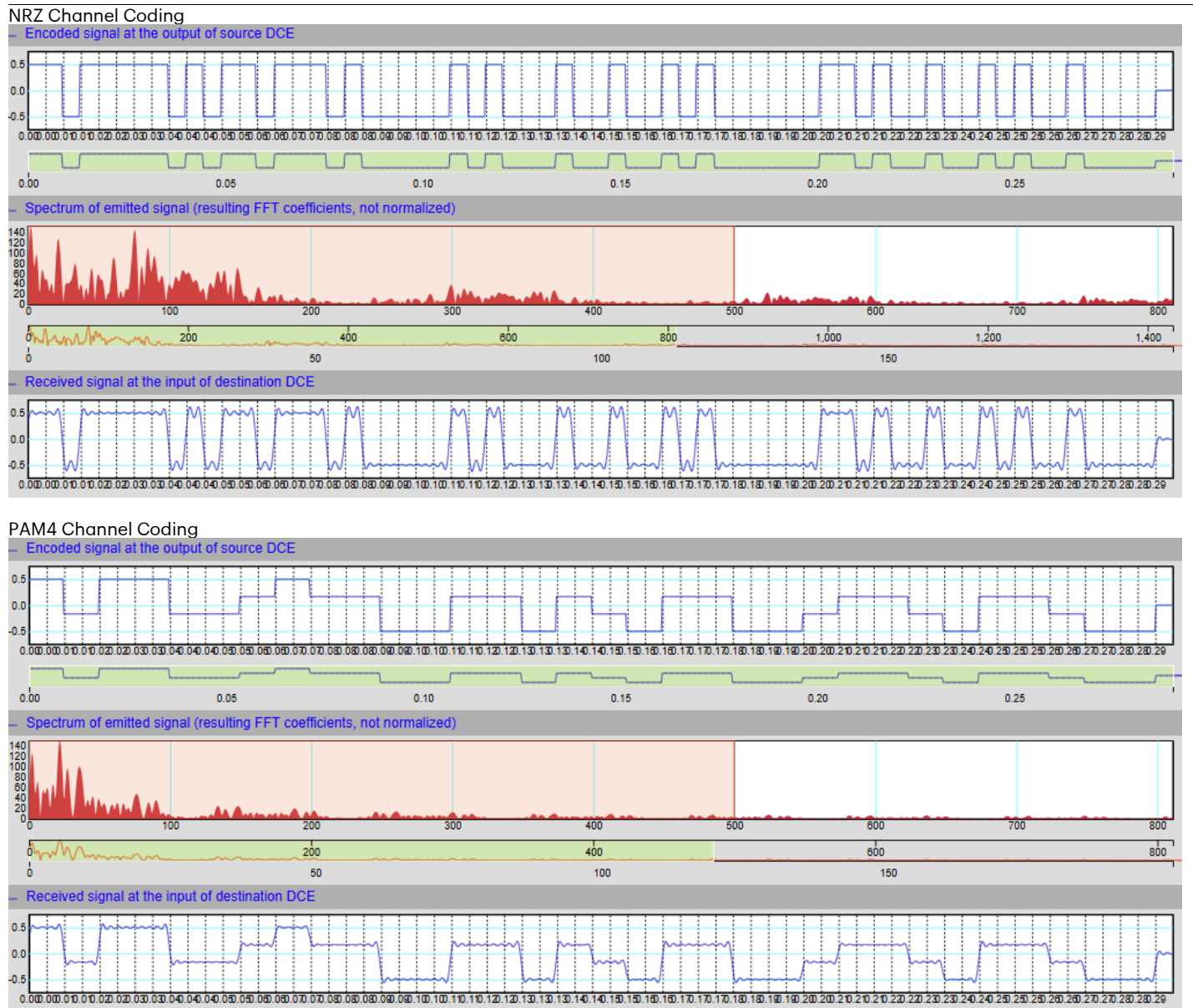


Source: TD Cowen, CodSim 2.0.5, Universidad de Malaga

To fully bring this discussion together, consider that all else equal, bandwidth availability in guided transmission (copper or fiber) is relatively fixed for a given cable—the bandwidth is dependent on the physical properties of the cable used, and cannot be "selected," if you will. Increasing the symbol rate over the same cable reduces signal integrity because the higher frequencies (that are required to cleanly transmit the signal) are attenuated—limiting the signal's reach (or increasing the processing required on the other end to achieve an acceptable post-error correction bit error rate).

One way to overcome these bandwidth limitations is to increase the number of bits that are communicated per symbol. Recall, this is the insight from Nyquist's formula for communication in a noiseless channel. The examples above transmit one bit per symbol: there are two voltage levels, and each level represents a single bit. Broadly, this type of signaling is referred to as **"non-return to zero" (NRZ) modulation**. If we add an additional two voltage levels, we can double the number of bits communicated each cycle (High could represent 11, Medium-High could represent 10, etc). This is known as **Pulse Amplitude Modulation-4 (PAM4)**. Fig. 64 below demonstrates the difference between NRZ and PAM4 in a noiseless channel.

Figure 64 PAM4 Encoding Decreases Bandwidth Demands by Lowering the Symbol Rate for the Same Bit Rate



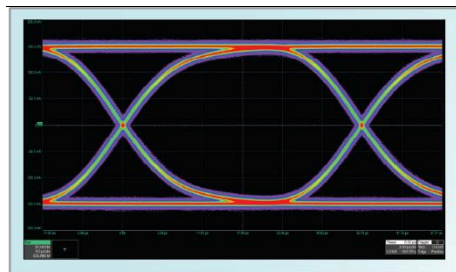
Source: TD Cowen, CodSim 2.0.5, Universidad de Malaga

In the comparison above, consider the difference in spectrum at higher frequencies for NRZ (top) and PAM4 (bottom). Because the voltage levels in the PAM4 example are changing half as frequently as in NRZ (to accomplish the same bit rate), its effective symbol rate has halved. More commonly, PAM4 is used to achieve double the bit rate at the same symbol rate—or said differently, leverage the same frequency bandwidth to double throughput.

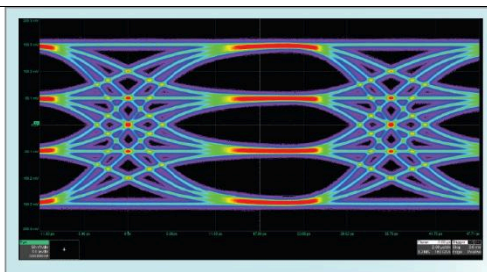
However, the transition from PAM4 to NRZ is not delivered as a free lunch—the added complexity of the four-level modulation comes with overhead in terms of receiver sensitivity required to now interpret four different voltage levels compared to two. This can be inferred from the above Fig. 64, but is most frequently visualized with an "eye diagram." A typical eye diagram plots a repeated signal's voltage level on the y-axis and time on the x-axis, overlaying many multiple signals on top of one another

until a heat map of sorts is formed. The size of each "eye" represents the clarity of the signal. In NRZ modulation, because there are only two levels, the eyes are relatively "clearer," demonstrating the lower sensitivity required to interpret NRZ signaling as compared to PAM4.

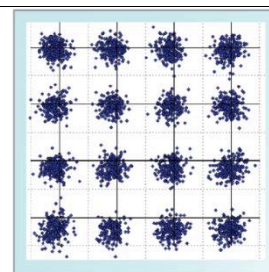
Figure 65 Higher Level Signaling Methods Can Transmit >1 Bit Per Symbol, At The Cost of Higher Complexity



NRZ



PAM4



Coherent



Source: Ethernet Alliance

Overcoming Noise and Attenuation: Filtering, Equalization, and Digital Signal Processing

Effective digital communication in the presence of noise and facing the realities of attenuation is made possible through (1) from a system design perspective, by making the signal more predictable; and (2) from a physical implementation perspective, by eliminating predictable or repetitive sources of noise through techniques such as equalization.

At a systems level, digital (discrete) communication itself is a simplified distillation of analog (continuous) communication. Bits are predictable—there is a relatively small defined set of outcomes. A message can either be a 0 or a 1. This allows for a degraded signal to be reasonably "guessed" by the receiver. Simplified, if the received message has a "value" of 0.83, the receiver can be reasonably sure the intended signal was 1, as 0.83 is not among the set of possible outcomes.

This can be extended to series of multiple bits, much like the English language has a defined set of outcomes for series of letters, albeit a much larger set. Thus, if a received message were to be interpreted as "semiconductor," the receiver could be reasonably sure the message was intended to be "semiconductor" as the former is not within the defined set of outcomes (fundamentally this is how line coding schemes such as 8b/10b function).

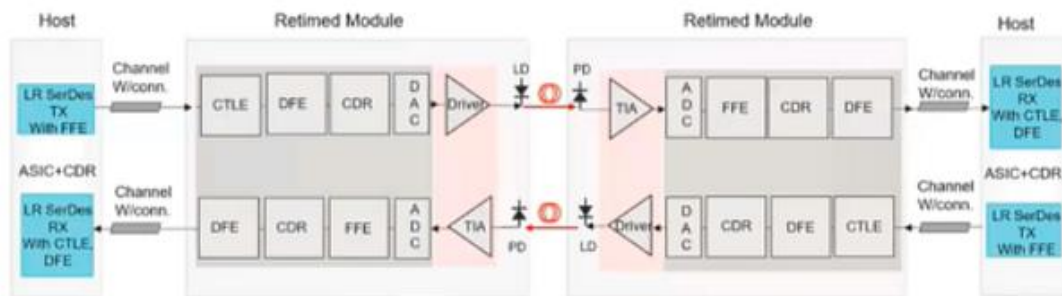
At the physical implementation level, there are steps that can be taken on the transmit side and the receive side to improve the probability of a successfully received message. On the **transmit side**, a **clock data recovery circuit** can be used to re-sync the signal after its transmission through the PCB trace (which itself is subject to the same noise and attenuation challenges as interconnect). The incoming electrical signal must be amplified from the relatively low power levels that processors operate at (often sub-1V) to a sufficient level to power a signal with an integrated circuit called a **driver** or **driver amplifier**. Transmit-side equalization techniques can then be applied, such as **continuous time linear equalization (CTLE)**. As the phrase "equalization" implies, this circuit pattern attempts to "equalize" the amplitude of the frequency ranges that arrive at the receiver. Recall that higher frequencies attenuate greater than lower frequencies. CTLE boosts these higher frequencies such that the amplitude of each frequency at the receiver is closer to equal. Finally, in the case of an optical channel, the electrical signal must be converted to light via a **laser** (either VCSEL, EML or CW and then modulated—either integrated with the laser itself or via a separate modulator).

At **the receiver**, the process described above is reversed. A **photodiode** receives the signal (in the case of an optical link) and generates an electrical current. A **transimpedance amplifier** converts this current to voltage via what is known in the industry as "the dark arts" of analog circuit design. Similar to the transmit side, **equalization** circuits can be applied to compensate for the different attenuations in the channel at different frequencies. Finally, the **clock data recovery** circuits can be applied to fully recover the signal.

(Very) briefly, in a non-guided digital communication link (aka wireless), an additional "step" is required in place of the optical components described above. Collectively this is referred to as the RFFE (Radio Frequency Front End). Because the signal is not guided, there is effectively an unlimited amount of other electromagnetic waves that could be arriving at the receiver at any given time. Filter technology (such as BAW, SAW, and TC-SAW) and antenna tuning mitigate this free space problem, but these are beyond the scope of this report.

Notably missing from this summary for those familiar with modern optical communication is any mention of a digital signal processor (DSP). Deliberately, we have largely described the digital communication channel's **functions** rather than the specific **components** that comprise a modern optical module. The DSP **performs some or all the functions described above** and adds additional capabilities such as adaptive equalization, re-timing, and forward error correction (as well as auxiliary functionality that adds value such as telemetry). For example, observe in the top half of Fig. 66 (below) the CDR, CTLE and DFE functional blocks *inside* the retimer. The DSP has not displaced the techniques described above but rather subsumes them.

Figure 66 A Digital Signal Processor Integrates and Improves Analog Functions... But Adds Latency and Power



Source: Gigalight

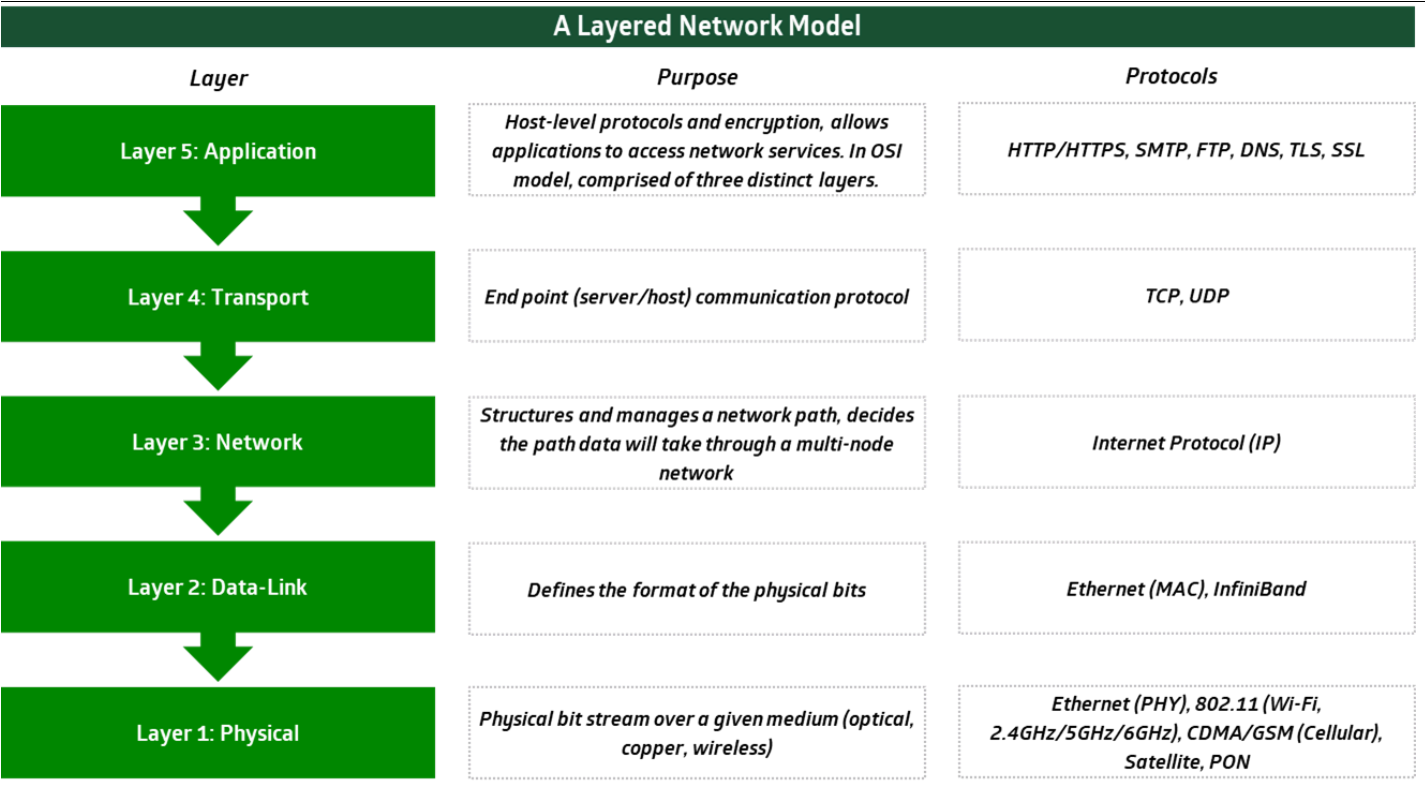
Networks Are Like Onions (They Have Layers)

*"A journey of a thousand miles begins with a single step."
-Traditional Chinese Proverb*

Like onions, not only do they at times smell bad, and *frequently* make people cry, but networks have layers—like the various layers of abstraction in the silicon/software stack, each level of the network model has a specific set of "responsibilities." Sometimes this is framed as each layer provides a service to that of the layer above, enabling the hardware and software of that particular layer to abstract away the nitty gritty details of the layers below (we acknowledge Kurose & Ross's work, [here](#), as immensely informing our understanding of this topic and many others).

The original network models were developed alongside two competing protocols—OSI and TCP/IP. While the TCP/IP *protocol* eventually won out in terms of adoption, the OSI *model* is frequently still taught and used conceptually (for instance, in describing Ethernet as a "Layer 1 & 2" protocol—TCP/IP would simply refer to it as Layer 1). We (again) borrow from Kurose & Ross and smash the two together for our purposes, with the resultant five-layer model seen in Fig. 67 below.

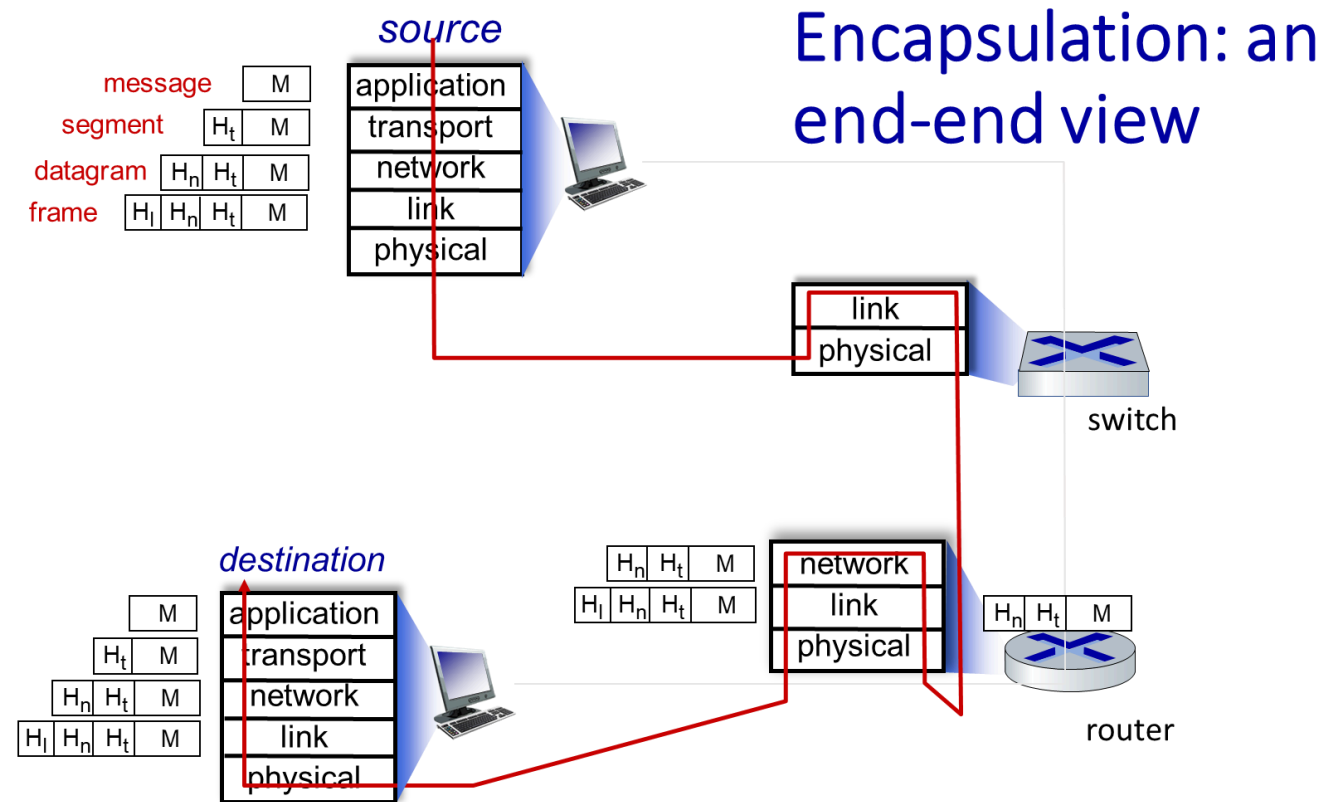
Figure 67 A Hybrid of the OSI Model and the TCP/IP Model, We Use a Five Layer Model to Contextualize Network Functions



Source: TD Cowen, Kurose & Ross Computer Networking: A Top-Down Approach

A key concept that the layered network model informs is that of encapsulation—a message sent across a network (or network of networks) does not necessarily engage each layer at every point along the way. Consider the simple example below in Fig. 68. A source end point (perhaps an investor) sends an email to a destination (perhaps a sell-side analyst). The Layer 5 protocol used is SMTP—but once the message is *encapsulated* by the lower levels, that fact does not matter again until it reaches the destination. The Ethernet "message" (technically the Ethernet *frame*) is sent to the investor's network's switch (a Layer 2 device). From the information in that frame, the switch in turn sends it to a router (a Layer 3 device), at which point the IP *datagram* is sent along to the destination sell-side analyst's computer (which for simplicity's sake we're imagining is physically connected directly to the investor's router, unrealistic to say the least). Only in the destination's NIC is the message un-encapsulated back to the application layer for the CPU to interpret as an incoming email.

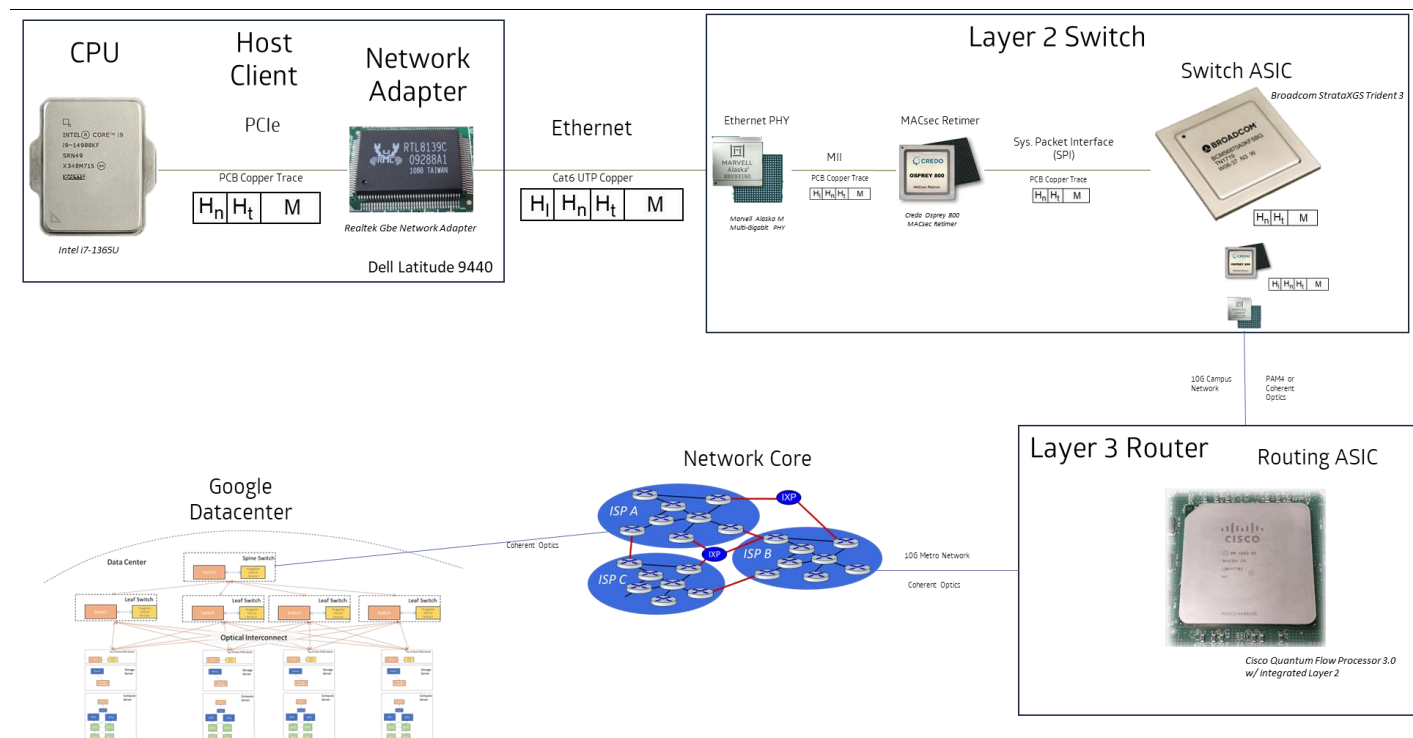
Figure 68 A Journey of A Thousand Miles Begins With a Single Step



Source: Kurose & Ross Computer Networking: A Top-Down Approach

Importantly for our purposes, **different pieces of hardware and software perform these different encapsulation tasks**. This gives rise to the seemingly endless acronyms and hardware that comprise networking connectivity companies' portfolios. Potentially the best (only?) way to explore this dynamic is through an overly ambitious example.

Figure 69 An Overly Ambitious Example of Network Encapsulation and Layering



Source: TD Cowen

What actually takes place when we access the Internet? To search for cat videos, I open a browser and enter a URL into the address bar, say "www.google.com." Within my CPU, the browser and operating system work together to create an HTTP message ("I would like to access the website located at 142.250.69.68"), which is subsequently encapsulated in a TCP, transport-layer header ("I would like to use TCP, socket 80"). Finally, the IP packet ("Please send cat videos to IP address 192.168.1.201") is transmitted over the PCIe bus to an external network adapter, also called a Network Interface Card when it is a discrete add-in card (and which, confusingly, is sometimes virtualized in software).

The network adapter has a physical address, or Media Access Control (MAC) address, unique to each *adapter*—so often the same machine will have at least two different MAC addresses: one for wired, and one for WiFi (if you don't believe us, open Command Prompt and type "ipconfig -all" to view all your different physical addresses). In the network adapter, the final header of the eventual network message is added, the Ethernet link-layer information. *Everything* else (IP address, transport protocol instructions, and the HTTP GET message itself) is encapsulated inside the Ethernet payload.

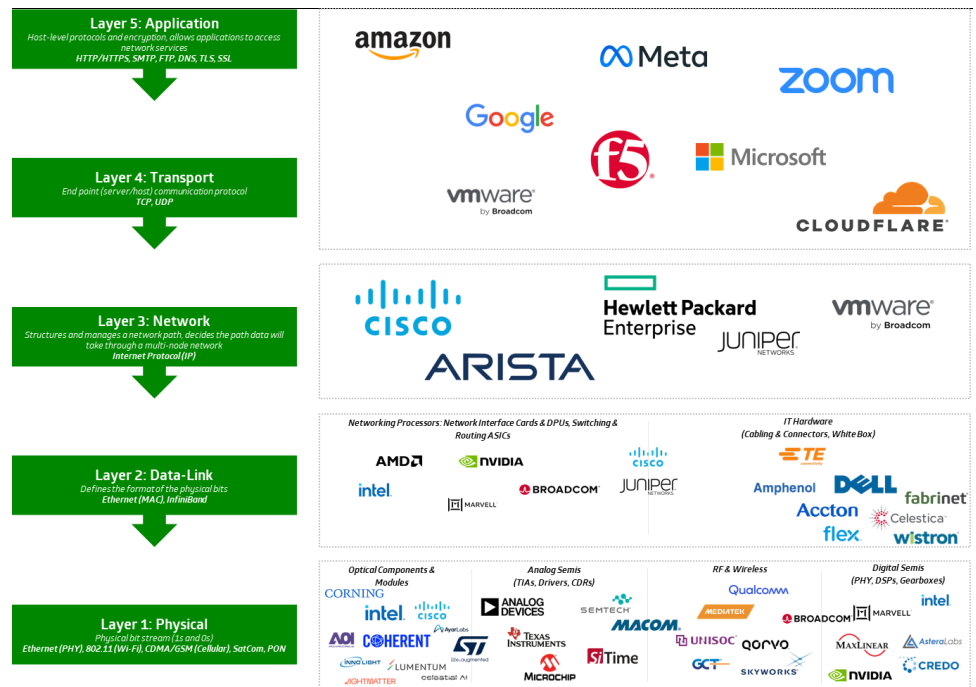
The network adapter also contains the Ethernet physical layer (PHY), which translates the abstract Ethernet frame into a tangible bit stream (much more on that below). The signal is sent across what would colloquially be called an Ethernet cable (specifically, an Unshielded Twisted Pair Cat6 copper wire with RJ-45 connectors) to a Layer 2 switch somewhere inside TD Bank's campus network (perhaps an unmanaged switch that aggregates a portion of the Research Department's client systems). Because this is a Layer 2 switch, *only the Ethernet header* is "read" at the switch level. However, that still requires the translation from physical signal to digital bits, which takes place in the PHY (often a part of the faceplate). MACsec retimers can be integrated or separate from this PHY, and the frame is sent across an internal bus called System Packet Interface (SPI, similar in nature to PCIe). The switching ASIC "reads" the destination physical address and the sending physical address and

forwards the frame to the corresponding port—in the example above, to the Layer 3 router. When leaving the switch, the message is again translated into the physical domain by a similar MACsec/PHY silicon combo.

At the Layer 3 router, this is the first time the IP address is processed—hence the nomenclature, "Layer 3." The processor reads the MAC address and says, "well, that's me!" and keeps processing the frame. The IP address tells the router which of its ports it should send the message along and—you guessed it—the IP packet is then re-encapsulated into a new Ethernet frame, which now contains the next router's MAC information, and so one until the frame reaches Google's datacenter and the message is fully read ("Hello, I am at 96.174.23.468, please send me cat videos"), and the process begins again in reverse—this time, with cat videos.

This is a very simple example, and in fact, we ignored a lot of things! Topics such as DNS, NAT, and routing protocols were completely glossed over in this brief summary. The takeaway is that each layer of the network does a small slice of the job to get from Point A to Point B—which is why the Internet can be decentralized. We do not need to know every step of the way to get from our computers to Google's cat video server—we just need to know the first step.

Figure 70 Not Even Close to Comprehensive Snapshot of the Network

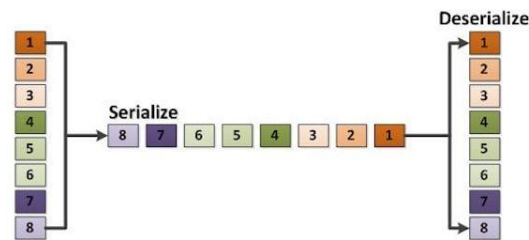


Source: TD Cowen

SerDes: Nuts and Bolts of I/O

Briefly, a serializer/deserializer function takes a bit stream that is traveling in parallel (such as the 64-bit channel common in many high-end processors) and transforms that bit stream into a linear (or "serial") series of bits that can then be sent in a narrower channel such as a PCB copper trace, a copper wire, or a fiber optic cable. On the other end, the serialized bits are transformed back into parallel to be further processed by the destination. You can perform your own SerDes live demonstration using the "transpose" function in Excel—though this example would be quite a bit slower than the requirements of high-speed datacenter connectivity.

Figure 71 SerDes: A Really Complicated "Transpose" Function



Source: Texas Instruments

SerDes IP is fundamental to high-speed connectivity. Modern processors leverage 64-bit wide pipelines to allow for all the complex instructions and data forms that enable today's computing. Within a processor, these 64 "wires" are cheap, in terms of power and space. However, for data to leave the processor, practical considerations require it to be serialized—imagine for example, having 64 distinct copper traces in a PCB for one single memory channel. Even ignoring the electromagnetic interference that would be generated, it is difficult to see how there would be sufficient physical space for such a layout in a modern server. Due to these considerations, there are SerDes functional blocks *everywhere*. Consider Fig. 72, below—across just one path from processor through a top-of-rack switch requires >10 serializations or deserializations. Each of these has the potential to act as a bottleneck.

Figure 72 An Illustrative Data Path Highlights the Ubiquity of SerDes in the Datacenter

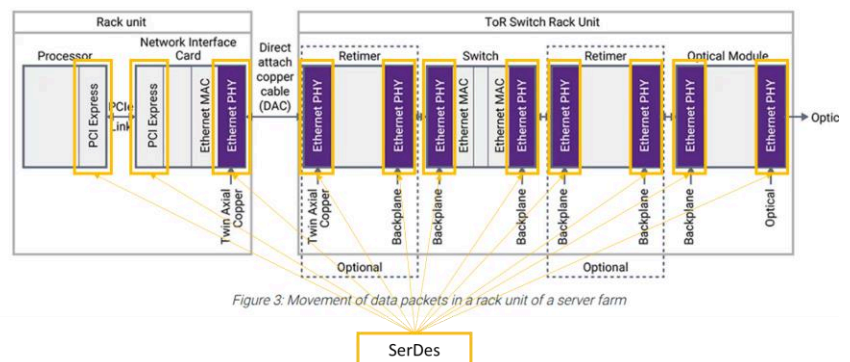


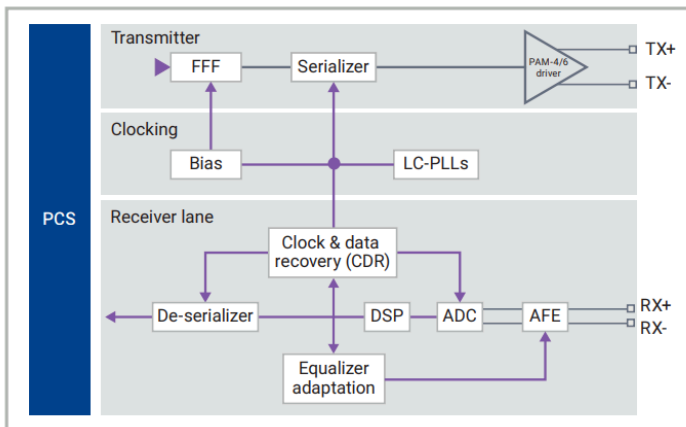
Figure 3: Movement of data packets in a rack unit of a server farm

Source: Synopsys, TD Cowen annotation added

Efficient high-speed SerDes is also very difficult. Consider that to maintain the same information density as an x bit parallel stream that is running at y GHz clock speed, the serialized bits must travel at $x*y$ Gbps. For a given linear improvement processing speed, the SerDes speed has to increase by $64x$ to keep pace (for a 64-bit processor).

In addition to the core serialization/deserialization, there are other functions that a typical "SerDes IP" block accomplishes. As shown in Fig. 73 below, these include clock recovery functions (CDR), digital signal processing (DSP), analog-to-digital conversion (ADC), and analog front-end (AFE).

Figure 73 SerDes IP is More Than Just Turning Bits Sideways



Source: Synopsys

This IP block is the transition point between the four top layers of the networking model and the final, physical layer. This is the point at which computer science passes the baton to electrical engineering—and things get weird.

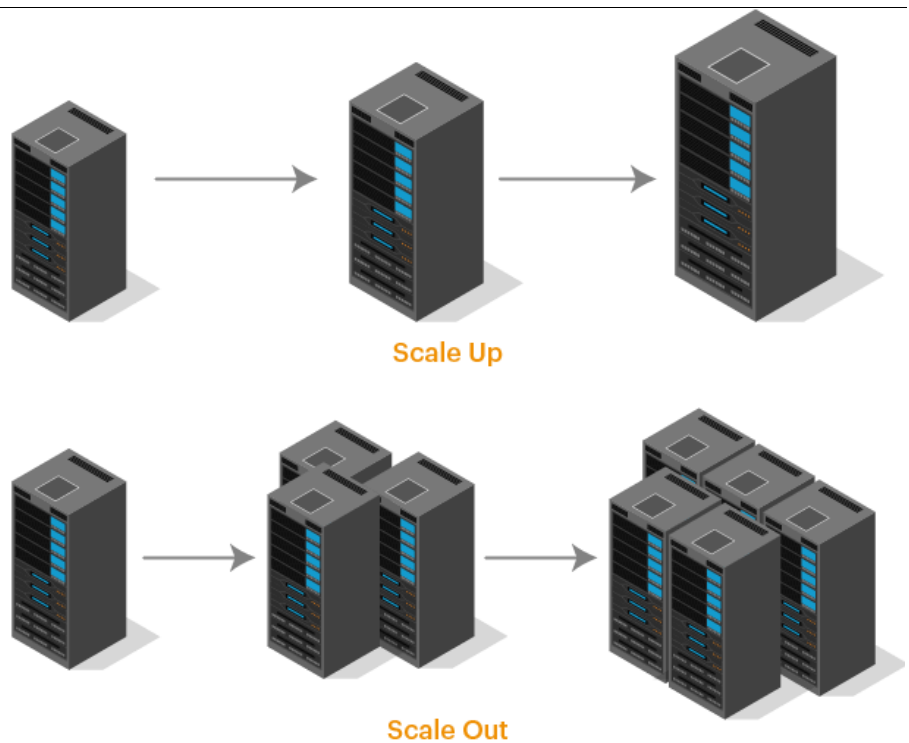
Scale Up Domains: NUMA NUMA Yay

*"Never trust a computer you can't throw out a window."
Steve Wozniak, Apple*

The scale requirements of training and inference for the largest cutting-edge AI models has caused the concepts of "scale up" and "scale out" to explode into the investor zeitgeist in recent years. Each is driving demand for networking connectivity, and while there are similarities between the two (for example, physics remains physics for both), they are sufficiently distinct such that it is worth detailing each.

Let's start with some definitions. Traditionally, scale up and scale out were phrases used to describe a decision-making paradigm: should infrastructure managers scale "up" their footprint by buying bigger/more powerful servers? Or should they scale "out" their footprint by buying more, but smaller servers. This intuition is key in understanding the difference between the scale up and the scale out domains: "scale up" means "bigger, single system."

Figure 74 Traditional View of Scale Up and Scale Out: Bigger or More?



Source: eG Innovations

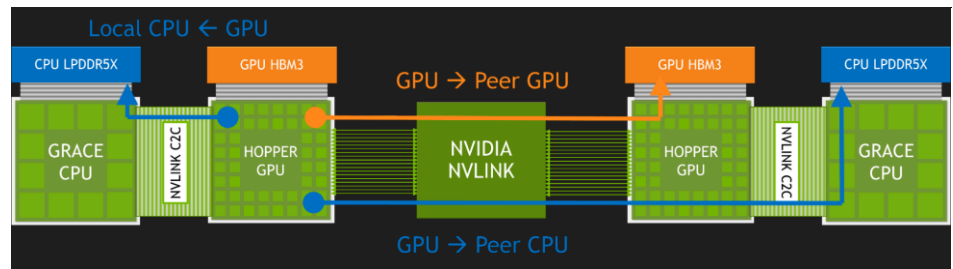
More recently, scale up has broken the constraints of a single hardware enclosure and now includes "scale up networking" to allow for a single system to span multiple compute trays in a rack or even in some cases multiple racks. This has led to a blurring between scale up and scale out and in our view no small amount of investor confusion regarding the difference between the two. This is the reason we began with recalling the traditional meaning of scale up: a larger, single system. Scale up is defined by the system presenting itself to higher-level software as a single system—regardless of whether there is one compute tray or 18.

Memory access and semantics are key to defining a multi-node system as "scale up." Consider a traditional server, equipped with a multi-core CPU and multiple slots of DRAM. Each core in the CPU has direct access to the system memory, managed by the Memory Management Unit (MMU) within the processor. To scale up this system, more DIMMs are added. In a two-socket server, both CPUs can access the entire system memory—with nuance.

The scale up in this instance is enabled by a serial bus interconnect protocol. Each DIMM is associated directly via memory channel to one of the two CPUs, and the indirect (remote) memory can be accessed via this interconnect. Intel's interconnect for this purpose is called Ultra Path Interconnect, while AMD CPUs use Infinity Fabric. However, the CPU simply has a single contiguous virtual address space, equal to the size of *both* CPU's memory. That is, when a CPU reads from a given virtual address, it does not know if it is requesting data from a direct physical address or a remote physical address—this is entirely managed in the MMU and the memory controllers.

Scale up for accelerators is similarly defined and implemented—multiple compute nodes have access to a unified virtual memory space, managed by a runtime layer of low-level software. To "higher" layers of software (i.e. compiled languages such as C/C++ and above), the physical system is represented as a pool of available processing elements and memory addresses (though CUDA C/C++ does still *allow* for the manual management of physical memory).

Figure 75 A Scale Up Domain is Defined by Memory Access



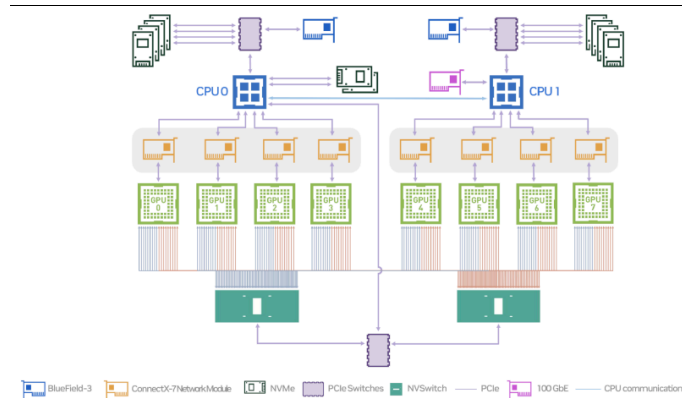
Source: NVIDIA

Scale Up vs. Scale Out in the Wild

We have found that clearly defining the boundary between scale up and scale out is often a point of confusion for investors. An example is potentially useful in fully contemplating the scale up vs. scale out delineation, but it is worth keeping in mind that there can be many different implementations of scale up (and scale out, for that matter). **The defining factor in determining a scale up domain is the unified virtual memory address space.**

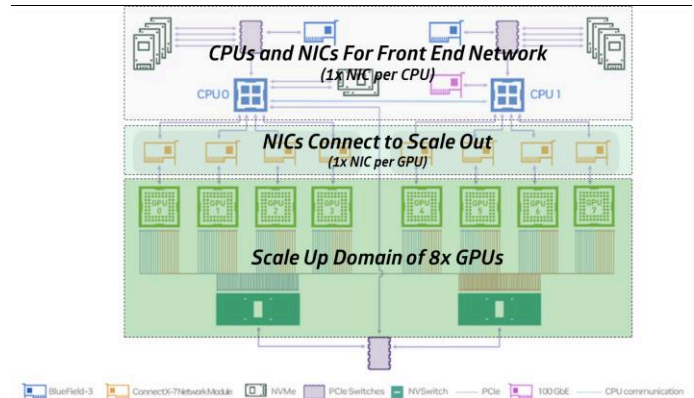
In a DGX or HGX implementation of NVIDIA scale up, eight GPUs are connected in a single domain leveraging NVIDIA's proprietary scale up protocol, NVLink. NVIDIA leverages two NVLink switches to create all-to-all connectivity across 8x GPUs—this is the scale up domain. The NVLink protocol is applied *in the GPU*, and there is no separate hardware encapsulation (see above discussion of encapsulation and layering). The low latency NVLink interconnect bus allows the runtime to manage a unified memory address space across these GPUs (specifically in the case of NVLink the runtime is an interplay of CUDA, NVIDIA Fabric Manager, and NVLink Subnet Manager).

Figure 76 A B200 System Topology...



Source: NVIDIA with TD Cowen Annotation

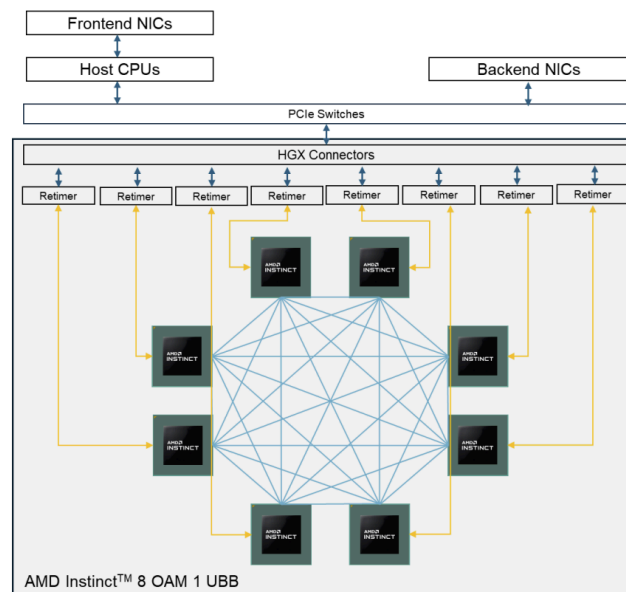
Figure 77 ...with Scale Up and Scale Out Labeled



Consider that leveraging two NVLink switches as does the B200 system is not the only way to create an 8x GPU scale up domain. For example, AMD's Infinity Fabric (its own scale up protocol that it has in turn donated to the UALink Consortium) frequently leverages an all-to-all mesh topology, as in Fig. 78 below. To this point, this mesh topology has obviated the need for distinct switch silicon within AMD's scale up domain, though we would expect that the forthcoming Helios rack will leverage UALink switches to achieve its goal of creating a >1,000 GPU scale up domain ([here](#)).

Figure 78 AMD's Scale Up Domain Forgoes Switches in Favor of Mesh Topology...For Now

Figure 3.2: Accelerator Mesh Network with MI300 Series Accelerators



Source: AMD

Critical Characteristics For Scale Up Protocols

With the express purpose of facilitating connection between accelerators within a single or rack-scale system, protocols should be optimized around certain key features in order to maximize bandwidth with minimal latency between accelerators: high bandwidth and bandwidth efficiency, ultra-low latency, shared memory semantics, losslessness, and best power performance area.

- **High Bandwidth and Bandwidth Efficiency:** Amdahl's law states that the "overall performance improvement gained by optimizing a single part of a system is limited by the fraction of time that the improved part is actually used." As a result, interconnect speeds must be able to handle heavy communication traffic between accelerators efficiently to prevent networking infrastructure from becoming a bottleneck. Further, with the growing amount of tensor or parallel data being exchanged, bandwidth efficiency (i.e., the number of bits in a data frame that carry the actual payload data) should be as close to 100% as possible with minimal overhead.
- **Ultra-Low Latency:** Related to the previous criteria, end-to-end (i.e., XPU to XPU) should be as low as possible despite the increasing traffic from communication between the accelerators with computational work. This is particularly important in end-user cases, such as chain-of-thought reasoning and Mixture of Experts (MoE), where latency to compute and generate tokens can become noticeable to users.
- **Shared Memory Semantics:** As mentioned above, in a scale up fabric, the accelerators are able to act as a larger, single system. As a result, the aggregate memory of all the accelerators should also appear as a single pool for each accelerator to load and store operations as if they were locally attached.
- **Lossless & Non-Blocking Fabric:** Scale up fabrics must be lossless and provide reliable links; this is seemingly a given, but it is particularly important because the load/store memory semantics cannot tolerate packet loss for any reason, including congestion/oversubscription. In a traditional Ethernet set-up, the transport layer typically manages packet re-sends when faced with packet loss or corruption; however, retransmissions at Layer 4 require a full roundtrip through the stack to retransmit the full segment, adding (1) a significant amount of time to tail latency due to congestion/head-of-line blocking and (2) disruptions to memory semantics. As a result, scale up fabrics need to be non-blocking – that is, packets containing requests and responses should be able to be sent and received without blocking each other by supporting hop-by-hop control on a per-port or per-traffic-class basis, thus eliminating oversubscription-induced bottlenecks and keeping latency low in a distributed computing environment. Further, necessary retransmissions should be done *before* reaching the transport layer to maintain a tight tail-latency and create minimal overhead; link-layer retransmission/retry (LLR) sends a single frame immediately upon noticing an error at the NIC-switch level, isolating and correcting the erroneous frame in microseconds versus a transport-layer resend, which requires a retransmission of an entire segment across the stack at 10-100x the time.
- **Best Power Performance Area:** Just as it is true for many components in other applications, achieving the best power per performance area is also true for any switch. As accelerators, rack-scale systems, and AI datacenters overall continue to increase necessary power draws, it is important to optimize for power draw in the chassis and cost across components, including switches. Not only does this improve the cost of silicon, but it also yields a better TCO from an improved cost of thermal management.

Currently there are five main scale up protocols, two of which are proprietary and the rest consortia-based: NVLink (NVIDIA proprietary), Inter-Chip Interconnect (ICI, Google proprietary), UALink, Scale Up Ethernet, and PCIe. Each scale up domain have different points of optimization based on their heritage. In the figure below, we attempt to compare each standard based on the major criteria described above based on publicly available information:

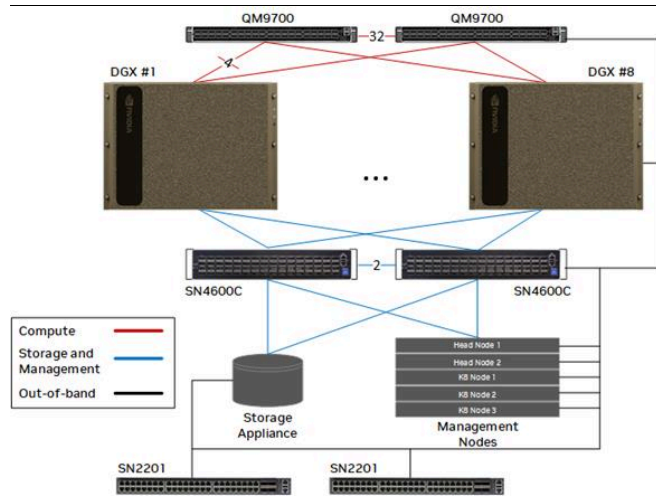
Figure 79 Scale Up Domains Have Different Points of Optimization Based on Heritage

Protocol	Open/Closed	Main Sponsor	Generation	Line Rate Speed	Per-Link Bandwidth (Bidirectional)	SerDes Speed	Per-Chip Bandwidth	Fabric Radix	Latency Targets (XPU-to-XPU)	Platforms
NVLink	Closed	NVIDIA	5th Gen	100 GT/s	100 Gb/s	224G PAM4	1,800 Gb/s 18x Lanes	576	100-150 nanoseconds	Blackwell & Blackwell Ultra-based Platforms
PCIe	Open	Intel	6.0	64 GT/s	15 Gb/s	PAM4	484 Gb/s 16x Lanes	Switch-Dependent Trainium 2 Scales to 64 Accelerators	<100 nanoseconds (x16 lanes)	Trainium (NeuronLink)
UALink	Open	AMD	1.0	212.5 GT/s	200 Gb/s	224G PAM4	800 Gb/s 4x Lanes	1,024	350-550 nanoseconds	MI400-Series Platforms
Scale Up Ethernet	Open	Broadcom	N/A		200 Gb/s	224G PAM4	800 Gb/s 4x Lanes	1,024	<2 microseconds (<2,000 nanoseconds)	
Inter-Chip Interconnect (ICI)	Closed	Google	v7		1,344 Gb/s		5,378 Gb/s 4x Lanes	9,216		TPU-based Pods

Source: TD Cowen, Company Data

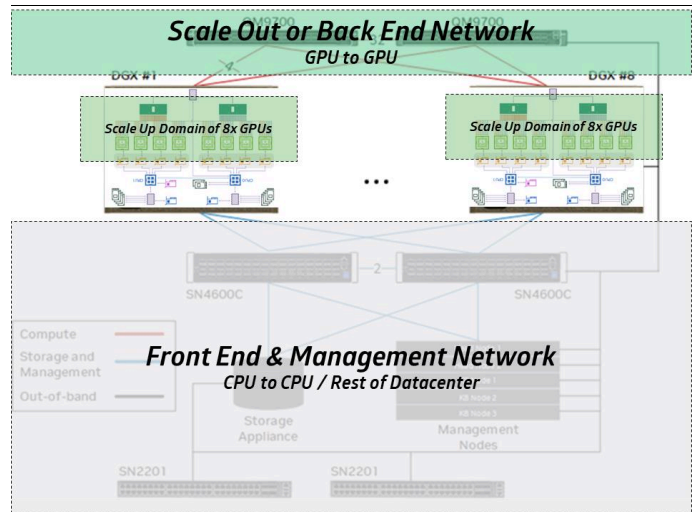
Zooming out, these scale up domains (which in this case are contained in a single server configuration) are connected to one another via scale out, or the backend network. For example, NVIDIA provides a reference architecture for an 8-server (64-GPU) BasePOD architecture. For scale out networking this reference architecture leverages two Quantum Infiniband switches, in addition to four Spectrum Ethernet switches for managing the storage and head node servers as well as providing connectivity to rest of the datacenter. Note in Fig. 81 we have replaced the picture of the physical DGX server with the diagram from Fig. 76 above, highlighting the location of the scale up domain in the context of the broader front and back end networks.

Figure 80 A BasePOD Reference Architecture...



Source: NVIDIA with TD Cowen Annotation

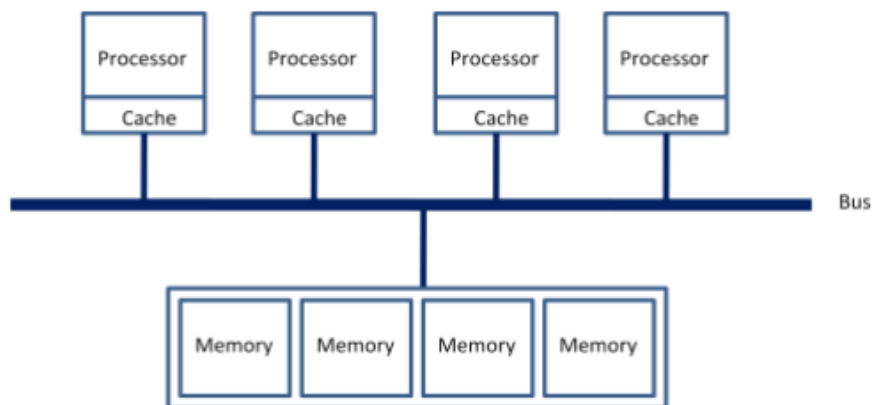
Figure 81 ...with Scale Up and Scale Out Labeled



NUMA: Scale Up is a Full Stack Problem

A final general concept worth highlighting in the context of scale up domains is that of Non-Uniform Memory Access (NUMA). NUMA is an architectural structure in which some physical memory locations have different latencies associated with them. Consider Fig. 82, below—this represents a Uniform Memory Access architecture, as all processors traverse a universal bus to access the shared memory pool. In a *non*-UMA setup, some processors will have access to local "fast" memory as well as remote "slow" memory—some memory reads will be slower than others. While an UMA architecture is attractive in its simplicity, a NUMA architecture (such as that depicted in Fig. 75, above) can actually lower *average* time for memory reads when managed properly. The OS or runtime, which typically manages the NUMA hierarchy, can build in efficiencies such as processor affinity and local memory placement to decrease average time spent calling to memory.

Figure 82 A Uniform Memory Access Architecture Is Simple, But Inefficient



Source: Intel

In the context of scale up, the NUMA hierarchy concept is important because it underscores how difficult it is to create a scale up domain and perhaps more importantly, it demonstrates that scale up is not an interconnects-only problem. A high-speed interconnect bus is a necessary but insufficient condition for the creation of a coherent scale up domain that truly acts as a single system. Close integration with runtime software and the communication kernel that leverages this interconnect is critical.

Scale Up Connectors: Copper (For Now)

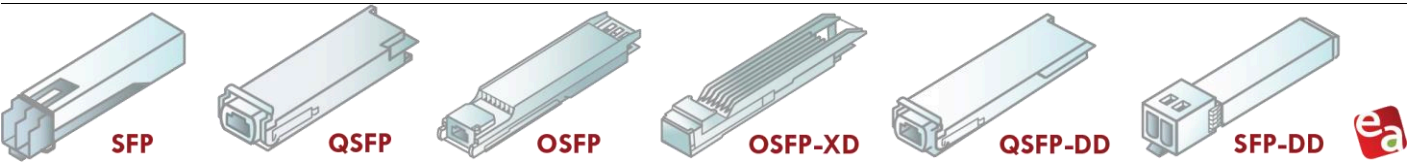
This is a datacenter connectivity report after all, so how is scale up connected? Scale up primarily leverages either **passive or retimed copper...for now**. However, it is possible that eventually, scale up fabrics could transition to either pluggable or co-packaged optics (as some have already demonstrated with their PCIe over Optics products).

Keeping in mind that the connectivity rule of thumb is "copper when you can, optics when you must," there are some key elements and requirements of a scale up fabric that have to this point let themselves primarily to **analog copper** solutions (though there are exceptions and caveats to this characterization).

We believe these elements will continue to impact the decision-making surrounding scale up physical media into the future:

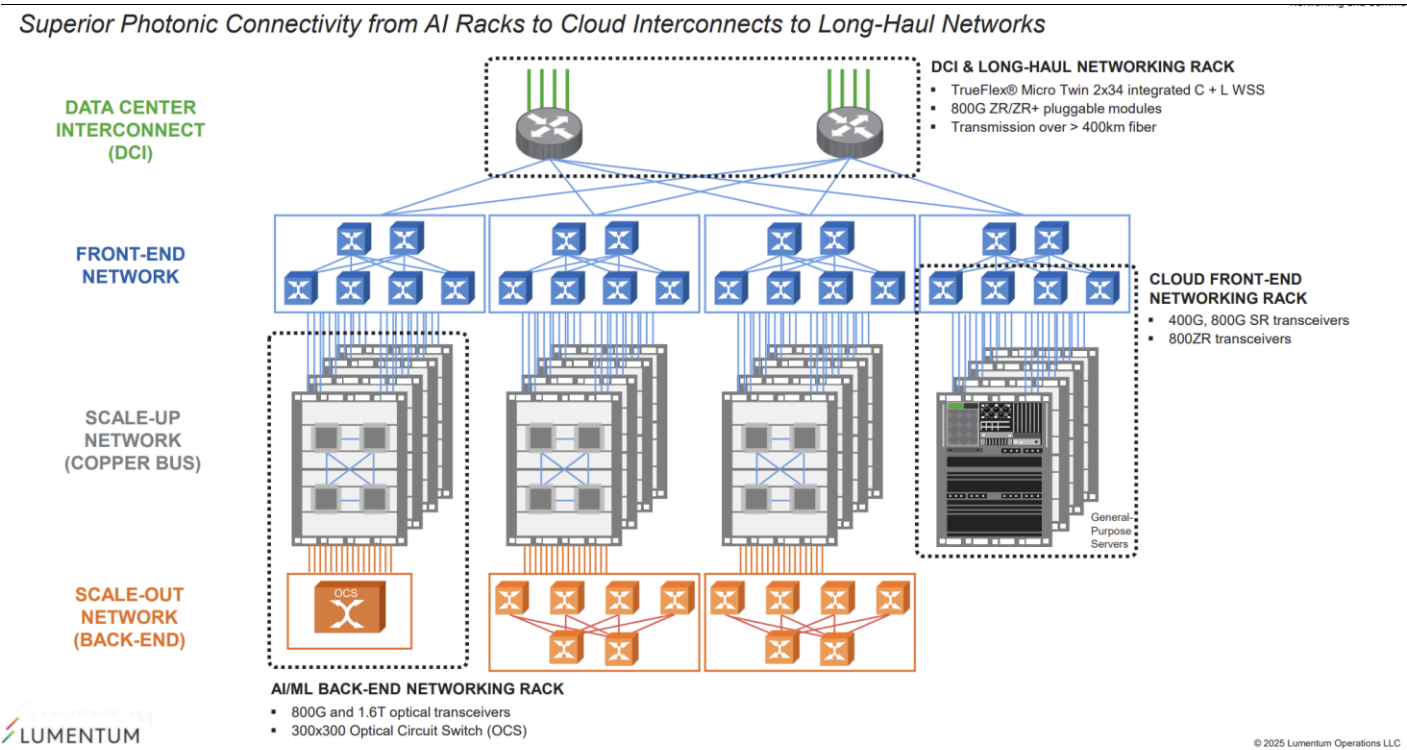
1. **Distance.** Prior to the development of rack-scale systems, a scale up domain was typically contained within a single server tray. This allowed for higher-attenuation copper to be leveraged as the distances involved were being measured in centimeters, rather than meters. As both speeds and distances have increased, the attenuative properties of passive copper have necessitated adding redrivers and retimers in many configurations.
2. **Latency.** The latency requirements for a scale up fabric are far greater than scale out due to the NUMA problem (above). Consider that in a rack scale solution, some physical memory address locations centimeters from the processor loading them and some will be meters. Each additional nanosecond of latency adds to the complexity of the NUMA issue the runtime must abstract away. All else equal, an analog solution that remains electrical is going to have lower latency because no time will be added in digital processing (i.e. bounded by clock speed) or medium conversation (i.e. having to transition from optics to electrics and back).
3. **Radix.** As with scale out, radix (the number of connections a switch specifically can support) is a key variable when designing all-to-all networks. All else equal, a higher radix can support a "flatter" network (i.e. fewer tiers of switches). This improves latency, as the flatter a network, the fewer hops required for XPU to XPU communication.

Figure 87 Flavors of Pluggable Modules



Source: Ethernet Alliance

Figure 88 The Networks Within the Datacenter



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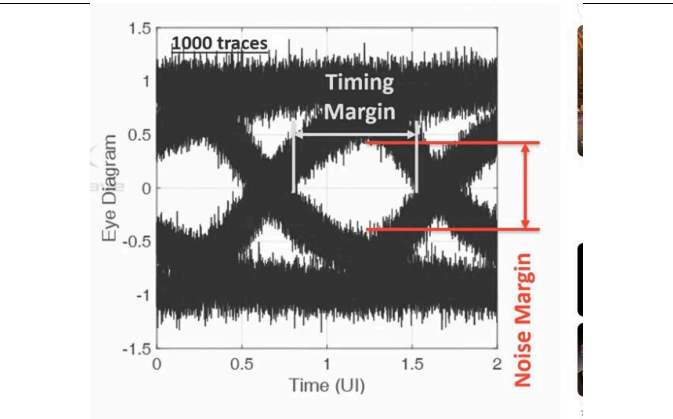
Source: Lumentum

Figure 89 Redrivers vs. Retimers

	Redriver	Retimer
Protocol	Agnostic	Aware
External clock	No	Yes
Cascading	No	Yes
Latency	~100ps	64ns based on PCIe® 4 & 5 Spec
Protocol link training	No	Yes
Power and cost	Low	Much higher
Equalizer	CTLE	CTLE, DFE and transmitter FIR
Jitter	Reduce deterministic jitter (Dj), but also add random jitter (Rj)	Eliminate Dj and completely reset Rj
Debug mode	None	Eye diagram monitoring, link state status, receiver margining

Source: Allaboutcircuits.com

Figure 90 Interpreting an Eye Diagram



Source: AlphawaveSemi

Figure 91 Scale Up Protocols at a Glance

Protocol	Open/Closed	Main Sponsor	Generation	Line Rate Speed	Per-Link Bandwidth (Unidirectional)	SerDes Speed	Per-Chip Bandwidth	Fabric Radix	Latency Targets (XPU-to-XPU)	Platforms
NVLink	Closed	NVIDIA	5th Gen	100 GT/s	200 Gb/s	224G PAM4	1,800 Gb/s 9x Lanes	57.6	100-150 nanoseconds	NVIDIA-based Platforms
PCIe	Open	Intel	0.6	64 GT/s	16 Gb/s	PAM4	256 Gb/s 16x Lanes	Switch-Dependent (Trainium 2 Scales to 64 Accelerators)	<100 nanoseconds (x16 lanes)	Trainium (NeuronLink)
UALink	Open	AMD	0.1	212.5 GT/s	200 Gb/s	224G PAM4	800 Gb/s 4x Lanes	102.4	350-550 nanoseconds	MI500-Series Platforms
Scale Up Ethernet	Open	Broadcom	N/A		200 Gb/s	224G PAM4	800 Gb/s 4x Lanes	102.4	<2 microseconds (<2,000 nanoseconds)	
Tomahawk Ultra	Open	Broadcom	0.7		100 Gb/s	100G PAM4	800 Gb/s 8x Lanes	25.6	<400 nanoseconds	AMD's Helios Rac
Inter-Chip Interconnect (ICI)	Closed	Google	v7		1,344 Gb/s		5,378 Gb/s 4x Lanes	921.6		TPU-based Pods

Source: TD Cowen

Glossary

ACC Active Copper Cable incorporates signal conditioning electronics within copper cabling to extend reach beyond passive DAC limits while maintaining lower power consumption than optical solutions for short-reach interconnects.

ADC Analog-to-Digital Converter transforms continuous analog signals into discrete digital values, essential for digitizing received optical or electrical waveforms in high-speed transceivers and SerDes interfaces.

AEC Active Electrical Cable uses retiming to condition signals over copper, extending reach beyond passive DAC limitations while maintaining lower power than optical solutions.

AFE Analog Front-End circuitry processes analog signals in transceivers or SerDes, handling tasks like amplification, filtering, and analog-to-digital conversion for data integrity.

AOC Active Optical Cable integrates optical transceivers within the cable assembly, enabling high-bandwidth, low-latency connections over longer distances without the manual fiber alignment associated with discrete pluggable modules.

API Application Programming Interface defines protocols for software components to communicate, critical for automating network configuration and integrating infrastructure management systems.

ASIC Application-Specific Integrated Circuit is a custom silicon chip designed for dedicated networking functions like packet processing, offering higher performance and efficiency than general-purpose hardware.

BAW Bulk Acoustic Wave technology enables high-frequency filtering in RF components, improving signal integrity for wireless infrastructure within datacenter environments.

BER Bit Error Rate quantifies transmission reliability as the ratio of erroneous bits to total bits transmitted, a fundamental metric for validating physical layer performance.

BGP Border Gateway Protocol governs routing between autonomous systems, essential for managing large-scale datacenter interconnects and internet edge routing policies.

CDR Clock and Data Recovery circuitry extracts timing information from serial data streams, synchronizing receivers to compensate for jitter in high-speed links.

CMOS Complementary Metal-Oxide-Semiconductor is the dominant semiconductor fabrication technology for digital logic circuits, including CPUs, NICs, and network switch ASICs.

CPC Co-Packaged Copper integrates electrical I/O circuitry directly with switch ASICs using copper interconnects, reducing latency and power by eliminating traditional pluggable module interfaces for specific high-bandwidth applications.

CPO Co-Packaged Optics integrates optical engines directly with switch ASICs, reducing power and latency by eliminating electrical I/O bottlenecks in high-bandwidth systems.

CPU Central Processing Unit executes general-purpose computing instructions, serving as the primary compute engine for servers and control-plane functions in network devices.

CTLE Continuous-Time Linear Equalization compensates for channel-induced signal degradation (e.g., ISI) in high-speed serial links through adaptive analog filtering.

CUDA Compute Unified Device Architecture is NVIDIA's parallel computing platform and API model for general-purpose processing on GPUs, accelerating AI and HPC workloads.

CXL Compute Express Link is a high-speed interconnect protocol enabling cache-coherent memory expansion and between CPUs, GPUs, and accelerators.

DAC Direct Attach Copper cable provides cost-effective, low-latency connectivity for short-reach switch-to-server or switch-to-switch links within racks.

DFE Decision Feedback Equalization is a nonlinear equalization technique in SerDes that mitigates post-cursor intersymbol interference using past symbol decisions.

DIMM Dual In-line Memory Module is the standard form factor for volatile memory (DRAM) in servers, providing the primary working memory for CPUs and accelerators.

DNS Domain Name System translates human-readable domain names to IP addresses, a foundational service for application discovery and load balancing in datacenters.

DRAM Dynamic Random-Access Memory is the dominant volatile semiconductor memory technology used for main system memory in servers due to its density and cost-effectiveness.

DSP Digital Signal Processor implements algorithms for signal conditioning (e.g., equalization, FEC decoding) in modern optical transceivers and high-speed SerDes.

EMI Electromagnetic Interference refers to noise disrupting electronic circuits, a critical design constraint mitigated through shielding and layout in high-density datacenter hardware.

EML Electro-absorption Modulated Laser is a high-speed optical transmitter technology used in coherent and direct-detect modules for 100G+ datacenter interconnects.

FIT Failures in Time quantifies component reliability as the number of failures per billion device-hours, a standard metric for semiconductor lifetime estimation.

GPU Graphics Processing Unit is a massively parallel processor optimized for accelerating compute-intensive tasks like AI training, scientific simulation, and rendering.

HBM High Bandwidth Memory stacks DRAM dies vertically with a wide interface, providing significantly higher bandwidth and lower latency than traditional DIMMs for accelerators.

HTTP Hypertext Transfer Protocol is the application-layer protocol for communication on the World Wide Web, fundamental for RESTful APIs in cloud services.

ICI Inter-Channel Interference occurs in wavelength-division multiplexing (WDM) systems when signals on adjacent wavelengths interfere, degrading optical link performance.

IEEE Institute of Electrical and Electronics Engineers develops and maintains critical networking standards, including Ethernet (802.3) and Wi-Fi (802.11).

ISI Intersymbol Interference distorts digital signals as energy from one symbol spills into adjacent symbol periods, primarily caused by channel bandwidth limitations.

LLR Link-Level Retry is an error recovery mechanism that retransmits corrupted data packets at the physical layer without involving higher protocol stacks, minimizing latency impact in high-performance networks.

LPO Linear-drive Pluggable Optics employs linear signaling without DSP-based modulation, targeting lower power consumption for specific short-reach optical modules.

LRO Linear Receive Optics employs linear signaling in the receive channel only, while maintaining digital signal processing in the transmit channel.

MAC Media Access Control is the sublayer of the data link layer (OSI Layer 2) defining how devices access shared physical media and frame data for transmission.

MMU Memory Management Unit handles virtual-to-physical address translation and memory protection, crucial for operating system and hypervisor functionality in servers.

MSA Multi-Source Agreement is a vendor consortium specification ensuring interoperability for pluggable optical modules (e.g., QSFP-DD, OSFP form factors).

MTBF Mean Time Between Failures is a reliability metric estimating the average operational time between hardware failures, often used for disk drives and power supplies.

MTTR Mean Time to Repair quantifies the average duration required to restore failed hardware components to operational status, impacting datacenter availability.

NAT Network Address Translation maps private IP addresses to public ones, commonly used at datacenter perimeters for security and IP address conservation.

NCCL NVIDIA Collective Communications Library optimizes multi-GPU and multi-node communication patterns (e.g., all-reduce) for distributed deep learning training.

NIC Network Interface Controller connects a server to the network, handling packet transmission/reception and increasingly offloading tasks network-related tasks.

NRZ Non-Return-to-Zero is a binary signaling scheme where voltage levels represent bits directly, commonly used in lower-speed Ethernet (e.g., 10G) before PAM4.

NUMA Non-Uniform Memory Access describes multi-socket server architectures where memory access latency depends on the CPU socket and memory location.

OCP Open Compute Project is an industry initiative developing open-source, efficient hardware designs for datacenter servers, storage, and networking.

ODM Original Design Manufacturer produces hardware (e.g., switches, servers) to specifications defined by cloud operators or networking vendors.

OIF Optical Internetworking Forum develops implementation agreements for interoperable optical components and networking technologies.

OSA Optical Sub-Assembly is the core optical component within a transceiver module, containing the laser, photodiode, and associated optical elements.

OSFP Octal Small Form Factor Pluggable is a high-density, 8-lane electrical interface and module form factor supporting 400G/800G optical transceivers.

OSI Open Systems Interconnection model is the conceptual framework defining seven layers of network protocol functions, though modern implementations often deviate.

PAM Pulse Amplitude Modulation encodes multiple bits per symbol (e.g., PAM4 uses 4 levels for 2 bits/symbol), doubling data rates compared to NRZ.

PCB Printed Circuit Board provides the physical and electrical substrate for mounting and interconnecting electronic components in switches, servers, and modules.

PCI Peripheral Component Interconnect is a legacy computer bus standard, largely superseded by PCI Express (PCIe) for high-speed device connectivity.

PCS Physical Coding Sublayer (in Ethernet) handles functions like 64b/66b encoding and aligns data for transmission over the physical medium.

PHY Physical Layer device implements the electrical/optical signaling and low-level protocols for a network interface, sitting between the MAC and the physical medium.

PIC Photonic Integrated Circuit integrates multiple optical components (lasers, modulators, detectors) onto a single chip, enabling advanced optical transceivers.

PMA Physical Medium Attachment sublayer (in Ethernet) manages serialization/deserialization and clocking for the physical medium interface.

QSFP Quad Small Form Factor Pluggable is a transceiver form factor supporting 4 electrical lanes, commonly used for 40G/100G/400G interfaces.

RFFE RF Front-End refers to the circuitry (amplifiers, filters, switches) handling radio frequency signals in wireless infrastructure, relevant for datacenter edge connectivity.

SAW Surface Acoustic Wave devices provide filtering and timing functions in RF components, though less common than BAW in modern high-frequency applications.

SDK Software Development Kit provides tools, libraries, and documentation for developing applications targeting specific hardware platforms or APIs.

SMTP Simple Mail Transfer Protocol is the standard for sending email messages across networks, used internally for system alerts and notifications.

SNR Signal-to-Noise Ratio measures the strength of a desired signal relative to background noise, a critical factor determining achievable data rates in communication channels.

SOI Silicon-on-Insulator is a semiconductor substrate technology improving performance and reducing power in advanced CMOS processes for high-speed networking ASICs.

SWAG Scientific Wild Guess is informal engineering jargon for a rough, order-of-magnitude estimate made when precise data is unavailable.

TCO Total Cost of Ownership encompasses all direct and indirect costs (acquisition, operation, maintenance, power, cooling) associated with datacenter infrastructure over its lifecycle.

TCP Transmission Control Protocol provides reliable, connection-oriented data delivery at the transport layer (OSI Layer 4), fundamental for most datacenter applications.

TDP Thermal Design Power specifies the maximum heat a cooling system must dissipate for a component (e.g., CPU, GPU) under typical workload conditions.

TIA Transimpedance Amplifier converts the photodiode's current output in an optical receiver into a usable voltage signal, a critical element in optical modules.

TPU Tensor Processing Unit is Google's custom ASIC accelerator designed specifically for high-performance machine learning inference and training workloads.

UMA Uniform Memory Access describes symmetric multiprocessing architectures where all CPUs access shared memory with identical latency, common in single-socket servers.

VCSEL Vertical-Cavity Surface-Emitting Laser is a low-cost, energy-efficient semiconductor laser widely used in multimode optical transceivers for short-reach datacenter links.

VLAN Virtual Local Area Network logically segments a physical network into isolated broadcast domains at Layer 2, enhancing security and traffic management.

WDM Wavelength Division Multiplexing combines multiple optical signals on different wavelengths onto a single fiber, dramatically increasing fiber capacity (e.g., CWDM, DWDM).

XPU A heterogeneous computing architecture term encompassing diverse processing units (CPUs, GPUs, TPUs, FPGAs) working together, often managed via unified APIs.

Datacenter Silicon Model: Connectivity Expansion

Figure 92 Datacenter Silicon Model: Networking Infrastructure Summary & Company Revenues

	2021	2022	2023	2024	2025E	2026E	2027E	2028E	2029E	2030E	CAGR (25-30)
Networking Infrastructure <i>Includes switching/routing silicon, electro-optics, active cabling, NIC/HBA, storage controllers, DPUs, IPUs, and networking-related ASIC</i>											
Summary											
Total Networking Infrastructure Silicon	2021	2022	2023	2024	2025E	2026E	2027E	2028E	2029E	2030E	CAGR (25-30)
Section B: Layer 1 Physical Connectivity											
Cable Chipsets (ACC, AEC, AOC)	\$ 129	\$ 236	\$ 225	\$ 378	\$ 1,235	\$ 1,792	\$ 1,993	\$ 2,124	\$ 2,219	\$ 2,376	14.0%
Optical Chipsets (DSP + Analog incl LRO, LPO, CPO)	\$ 635	\$ 785	\$ 935	\$ 1,680	\$ 2,828	\$ 3,974	\$ 4,570	\$ 5,010	\$ 5,828	\$ 6,713	18.9%
Line Cards & Retimers (Ethernet and PCIe)	\$ 1,109	\$ 1,131	\$ 1,489	\$ 2,186	\$ 2,766	\$ 3,454	\$ 3,770	\$ 4,227	\$ 4,700	\$ 5,129	13.1%
Total Layer 1 Physical Connectivity	\$ 1,873	\$ 2,151	\$ 2,648	\$ 4,244	\$ 6,828	\$ 9,220	\$ 10,332	\$ 11,361	\$ 12,747	\$ 14,217	15.8%
Section C: Layer 2 & 3 Data & Network Layer Processing											
Ethernet Switch ASICs & NICs	\$ 6,445	\$ 7,865	\$ 9,276	\$ 10,715	\$ 15,421	\$ 21,410	\$ 24,761	\$ 29,296	\$ 33,307	\$ 37,185	19.2%
Infiniband Switch ASICs & NICs	\$ 314	\$ 538	\$ 2,159	\$ 3,198	\$ 3,624	\$ 3,306	\$ 3,204	\$ 3,009	\$ 2,820	\$ 2,671	-5.9%
Scale-Up Switch ASICs	\$ -	\$ -	\$ -	\$ 385	\$ 4,278	\$ 7,080	\$ 9,828	\$ 13,868	\$ 19,103	\$ 24,046	41.2%
Total Layer 2 & 3 Data & Network Layer Processing	\$ 6,759	\$ 8,404	\$ 11,435	\$ 14,299	\$ 23,323	\$ 31,796	\$ 37,793	\$ 46,173	\$ 55,231	\$ 63,901	22.3%
Total Networking Infrastructure Silicon Revenue	\$ 8,633	\$ 10,555	\$ 14,083	\$ 18,543	\$ 30,151	\$ 41,017	\$ 48,125	\$ 57,534	\$ 67,978	\$ 78,119	21.0%
Additional Systems-Level Revenue (Switch Boxes, Cables, etc.)	\$ 6,587.19	\$ 9,002.55	\$ 14,168.82	\$ 21,978.28	\$ 29,171.83	\$ 45,752.29	\$ 54,958.34	\$ 67,711.84	\$ 81,839.34	\$ 96,122.40	26.9%
Less: AI ASICs (Broadcom & Marvell)	\$ (1,500.00)	\$ (1,893.59)	\$ (3,144.28)	\$ (10,042.52)	\$ (16,295.30)	\$ (31,796.05)	\$ (40,847.13)	\$ (53,909.70)	\$ (69,096.98)	\$ (84,790.85)	39.1%
Total Networking Infrastructure Revenue (\$M)	\$ 13,720	\$ 17,664	\$ 25,107	\$ 30,479	\$ 43,028	\$ 54,973	\$ 62,236	\$ 71,336	\$ 80,721	\$ 89,450	15.8%
y/y Growth (%)		29%	42%	21%	41%	28%	13%	15%	13%	11%	
Broadcom											
Total Broadcom Networking Revenue (Reported/Est.)	\$ 7,382.28	\$ 9,300.00	\$ 11,825.50	\$ 17,101.59	\$ 25,158.94	\$ 45,166.01	\$ 55,301.16	\$ 71,360.66	\$ 88,546.78	\$ 104,653.74	33.0%
Marvell											
Total Marvell Datacenter Revenue (Reported/Est.)	\$ 1,785	\$ 2,409	\$ 2,217	\$ 4,151	\$ 6,009	\$ 6,853	\$ 8,185	\$ 9,497	\$ 11,144	\$ 13,754	18.0%
NVIDIA											
Total NVIDIA Networking Revenue	\$ 1,566	\$ 2,718	\$ 8,575	\$ 12,990	\$ 19,673	\$ 21,781	\$ 25,299	\$ 28,655	\$ 32,620	\$ 37,778	13.9%
Credo											
Total Credo Datacenter Revenue	\$ 57	\$ 143	\$ 128	\$ 288	\$ 839	\$ 1,263	\$ 1,577	\$ 1,873	\$ 2,147	\$ 2,394	23.3%
Astera											
Total Astera Datacenter Revenue	\$ -	\$ 81	\$ 117	\$ 396	\$ 775	\$ 1,287	\$ 1,472	\$ 1,487	\$ 1,806	\$ 2,012	21.0%
Semtech											
Total Potential Semtech Datacenter Revenue	\$ 44.94	\$ 58.70	\$ 74.77	\$ 142.13	\$ 278.63	\$ 443.28	\$ 590.44	\$ 735.81	\$ 869.97	\$ 1,017.18	29.6%
MACOM											
Total MACOM Datacenter Revenue	\$ 133.70	\$ 149.00	\$ 157.06	\$ 213.60	\$ 313.15	\$ 436.92	\$ 529.45	\$ 605.93	\$ 705.00	\$ 811.03	21.0%

Source: TD Cowen

Figure 93 Section A: Ports & Plugs (Total Ethernet Connections by Type)

Ports	2021	2022	2023	2024	2025E	2026E	2027E	2028E	2029E	2030E
Ethernet Switch Ports--By Speed										
1G	5,048,611	4,653,846	4,123,571	4,138,030	3,517,326	2,643,819	2,667,276	2,691,321	2,714,915	2,738,716
10G	7,609,581	7,140,849	6,735,054	8,384,154	6,733,460	-	-	-	-	-
25G	9,647,125	10,948,363	11,563,497	10,167,187	8,642,109	14,517,058	12,583,242	10,143,989	7,576,911	5,659,469
40G	1,320,811	1,122,885	644,140	700,376	480,276	-	-	-	-	-
50G	41,904	49,170	90,923	68,583	61,725	72,585	85,026	98,879	114,369	122,374
100G	32,841,221	40,053,247	41,468,088	41,355,217	31,016,412	28,699,596	24,774,213	22,897,815	22,169,645	21,464,630
200G	940,643	5,491,029	6,685,000	9,454,872	8,982,128	7,317,217	5,515,484	4,261,382	3,372,374	2,668,831
400G	8,029,373	12,853,001	15,397,583	26,243,140	35,428,239	36,405,670	34,904,469	29,904,279	29,093,756	28,305,202
800G	-	40,640	532,352	2,652,160	27,847,680	45,948,672	48,863,731	45,931,907	43,635,312	41,453,546
1600G	-	-	-	-	770,000	9,394,000	27,675,804	39,443,992	44,013,604	51,055,780
3200G	-	-	-	-	-	-	1,280,000	9,656,000	23,388,000	35,082,000
Total	65,479,269	82,353,029	87,240,209	103,163,719	123,479,355	144,998,617	158,349,244	165,029,565	176,078,886	188,550,549
y/y Growth	13%	26%	6%	18%	20%	17%	9%	4%	7%	7%
y/y Growth Ethernet Switch Ports--By Speed										
1G	-2%	-8%	-11%	0%	-15%	-25%	1%	1%	1%	1%
10G	-5%	-6%	-6%	24%	-20%	-100%				
25G	8%	13%	6%	-12%	-15%	68%	-13%	-19%	-25%	-25%
40G	-22%	-15%	-43%	9%	-31%	-100%				
50G	292%	17%	85%	-25%	-10%	18%	17%	16%	16%	7%
100G	11%	22%	4%	0%	-25%	-7%	-14%	-8%	-3%	-3%
200G	238%	484%	22%	41%	-5%	-19%	-25%	-23%	-21%	-21%
400G	73%	60%	20%	70%	35%	3%	-4%	-14%	-3%	-3%
800G				398%	950%	65%	6%	-6%	-5%	-5%
1600G						1120%	195%	43%	12%	16%
3200G								654%	142%	50%
Total										
800G and Above Relative to Accelerator Units										
Accelerator Units (Total) (M)	1.9	1.5	3.9	7.9	9.6	12.1	12.8	14.2	15.8	17.1
800G & Above Ports (M)	-	0.1	0.7	3.4	36.9	73.2	99.8	121.8	140.2	159.8
Ports per Accelerator		0.0	0.2	0.4	3.8	6.0	7.8	8.6	8.9	9.4

Source: 650 Group, TD Cowen Estimates.

Figure 94 Section B: Layer 1 Cables & Link-Layer Processing (DSPs and Analog Components)

Cables & Transceivers	2021	2022	2023	2024	2025E	2026E	2027E	2028E	2029E	2030E	CAGR (25-30)
Chipsets include electrical interface components, including both digital (DSP) and analog (drivers/amplifiers). Optical/photonic components (lasers, modulators) are <u>not</u> included, even if silicon-based											
Totals from Section A											
Passive Copper	20.19	25.20	33.43	38.38	25.31	22.53	18.29	17.42	14.93	13.00	-12.5%
ACC	-	-	0.16	0.35	2.79	5.63	8.04	11.67	13.36	15.27	40.5%
AEC	0.43	1.43	1.06	2.44	9.05	13.14	15.35	16.91	17.77	18.94	15.9%
AOC	10.68	17.17	15.14	18.54	31.23	35.68	32.01	28.69	27.61	26.75	-3.0%
Optical Module (Fully Retimed)	62.86	66.51	65.50	72.72	93.52	107.13	104.01	93.72	94.06	94.39	0.2%
Optical Module (Half-Retimed)	-	-	-	-	4.43	11.79	17.71	25.57	32.10	38.09	53.8%
Optical Module (LPO)	-	-	-	0.02	0.45	7.28	12.95	17.22	18.11	19.53	112.9%
Co-Packaged	-	-	-	0.14	0.09	0.48	1.14	4.63	8.77	12.60	169.9%
Total	94.16	110.30	115.29	132.60	166.87	203.65	209.50	215.84	226.71	238.57	7.4%
Integrated Cable Chipsets Revenues (\$M)	2021	2022	2023	2024	2025E	2026E	2027E	2028E	2029E	2030E	CAGR (25-30)
ACCs	\$ -	\$ -	\$ 2.21	\$ 5.03	\$ 79.87	\$ 152.98	\$ 207.76	\$ 286.44	\$ 311.60	\$ 338.31	33.5%
AECs	\$ 27.63	\$ 92.67	\$ 68.74	\$ 158.63	\$ 678.47	\$ 983.48	\$ 1,146.64	\$ 1,260.62	\$ 1,321.89	\$ 1,406.10	15.7%
AOCs	\$ 101.75	\$ 143.20	\$ 153.83	\$ 214.00	\$ 476.67	\$ 655.09	\$ 638.42	\$ 576.87	\$ 585.92	\$ 631.68	5.8%
Total Cables Chipset Revenue (\$M)	\$ 129.38	\$ 235.87	\$ 224.78	\$ 377.65	\$ 1,235.01	\$ 1,791.54	\$ 1,992.82	\$ 2,123.94	\$ 2,219.41	\$ 2,376.08	14.0%
Total Cables Chipset Units (M)	11.11	18.59	16.35	21.34	43.07	54.44	55.40	57.27	58.74	60.96	
Pluggable Fully Retimed Transceivers (incl. TIAs/Drivers)	2021	2022	2023	2024	2025E	2026E	2027E	2028E	2029E	2030E	CAGR (25-30)
Units (M)	62.86	66.51	65.50	72.72	93.52	107.13	104.01	93.72	94.06	94.39	0.2%
y/y growth		6%	-2%	11%	29%	15%	-3%	-10%	0%	0%	
ASP (\$)	\$ 10.10	\$ 11.80	\$ 14.27	\$ 23.10	\$ 28.80	\$ 32.54	\$ 35.80	\$ 38.66	\$ 41.37	\$ 43.85	8.8%
y/y growth		17%	21%	62%	25%	13%	10%	8%	7%	6%	
Revenue (\$M)	\$ 634.78	\$ 784.74	\$ 934.86	\$ 1,679.58	\$ 2,693.42	\$ 3,486.42	\$ 3,723.26	\$ 3,623.39	\$ 3,891.10	\$ 4,139.01	9.0%
y/y growth		24%	19%	80%	60%	29%	7%	-3%	7%	6%	
Linear Receive Transceivers (LRO) (incl. TIAs/Drivers)											
Units (M)	-	-	-	-	4.43	11.79	17.71	25.57	32.10	38.09	53.8%
y/y growth						166%	50%	44%	26%	19%	
% Share of Retimed Transceivers					5%	10%	15%	21%	25%	29%	
ASP (\$)	\$ -	\$ -	\$ -	\$ -	\$ 28.80	\$ 32.54	\$ 35.80	\$ 38.66	\$ 41.37	\$ 43.85	8.8%
y/y growth						13%	10%	8%	7%	6%	
Revenue (\$M)	\$ -	\$ -	\$ -	\$ -	\$ 127.64	\$ 383.53	\$ 634.08	\$ 988.62	\$ 1,327.93	\$ 1,670.22	67.2%
y/y growth						200%	65%	56%	34%	26%	
Linear Pluggable Transceivers (LPO) (only TIAs/Drivers)											
Units (M)				0.02	0.45	7.28	12.95	17.22	18.11	19.53	112.9%
y/y growth					1858%	1530%	78%	33%	5%	8%	
ASP (\$)				\$ 9.24	\$ 11.52	\$ 13.02	\$ 14.32	\$ 15.46	\$ 16.55	\$ 17.54	8.8%
y/y growth					25%	13%	10%	8%	7%	6%	
Revenue (\$M)				\$ 0.21	\$ 5.15	\$ 94.76	\$ 185.38	\$ 266.24	\$ 299.68	\$ 342.50	131.6%
y/y growth					2341%	1742%	96%	44%	13%	14%	
Co-Packaged Optics											
Units (M)				0.14	0.09	0.48	1.14	4.63	8.77	12.60	169.9%
y/y growth					-36%	451%	135%	307%	89%	44%	
ASP (\$)				\$ 2.93	\$ 16.64	\$ 20.20	\$ 23.65	\$ 28.41	\$ 35.32	\$ 44.52	21.7%
y/y growth					28%	21%	17%	20%	24%	26%	
Revenue (\$M)				\$ 0.40	\$ 1.46	\$ 9.79	\$ 26.90	\$ 131.61	\$ 309.75	\$ 561.02	228.6%

Source: Lightcounting, 650 Group, TD Cowen estimates

Figure 95 Section B (cont): Other Physical Layer Processing

Other Physical Layer Processing	2021	2022	2023	2024	2025E	2026E	2027E	2028E	2029E	2030E	CAGR (25-30)
Ethernet Line Cards, PHYs, Gearboxes											
Units (M)	25.60	29.26	25.98	31.48	41.87	53.17	63.28	75.30	87.35	100.45	
y/y gwth		14%	-11%	21%	33%	27%	19%	19%	16%	15%	
ASP (\$)	\$ 28.41	\$ 28.27	\$ 27.64	\$ 27.46	\$ 28.22	\$ 27.37	\$ 26.83	\$ 26.29	\$ 25.76	\$ 25.25	
y/y gwth		-1%	-2%	-1%	3%	-3%	-2%	-2%	-2%	-2%	
Revenue (\$M)	\$ 727.45	\$ 827.24	\$ 717.96	\$ 864.50	\$ 1,181.56	\$ 1,455.56	\$ 1,697.47	\$ 1,979.59	\$ 2,250.40	\$ 2,536.20	16.5%
y/y gwth		14%	-13%	20%	37%	23%	17%	17%	14%	13%	
<i>Line Card Revenue Share</i>											
Interconnect Bus Front-End Switches, Retimers & Redrivers (PCIe, etc) in AI Applications											
Switches/Retimers/Redrivers per GPU/ASIC	4	4	4	4	4	4	4	4	4	4	
Units (M)	7.63	6.07	15.41	31.47	38.49	48.56	51.38	56.86	63.22	68.28	
y/y gwth		-21%	154%	104%	22%	26%	6%	11%	11%	8%	
ASP (\$)	\$ 50.00	\$ 50.00	\$ 50.00	\$ 42.00	\$ 41.16	\$ 41.16	\$ 40.34	\$ 39.53	\$ 38.74	\$ 37.96	
y/y gwth		0%	0%	-16%	-2%	0%	-2%	-2%	-2%	-2%	
Revenue (\$M)	\$ 381.64	\$ 303.34	\$ 770.63	\$ 1,321.66	\$ 1,584.21	\$ 1,998.77	\$ 2,072.34	\$ 2,247.76	\$ 2,449.11	\$ 2,592.37	10.4%
y/y gwth		-21%	154%	72%	20%	26%	4%	8%	9%	6%	

Source: Lightcounting, 650 Group, TD Cowen estimates

Figure 96 Section C: Layer 2 & 3 Data & Network Layer Processing (Switch ASICs & NICs, including Scale Up)

Ethernet Scale-Out										
Ethernet Switch ASICs	2021	2022	2023	2024	2025E	2026E	2027E	2028E	2029E	2030E
Switch ASIC SerDes Lane by Speed (K)										
1 Gbps	9,592	8,655	7,718	7,555	7,811	5,871	5,924	5,977	6,029	6,082
10 Gbps	27,739	25,478	20,857	20,975	17,309	-	-	-	-	-
25 Gbps	133,615	160,454	170,967	168,074	64,346	71,238	59,756	28,516	16,613	9,679
50 Gbps	60,620	108,142	123,980	213,239	220,052	198,047	178,242	169,330	143,931	115,144
100 Gbps	-	386	4,637	21,872	284,333	497,582	651,833	684,425	574,502	482,233
200 Gbps	-	-	-	-	8,745	48,228	139,861	292,309	495,463	629,734
400 Gbps	-	-	-	-	-	-	5,120	25,088	76,769	152,771
Total	231,565	303,115	328,159	431,715	602,596	820,967	1,040,735	1,205,644	1,313,308	1,395,643
	3.5	3.7	3.8	4.2	4.9	5.7	6.6	7.3	7.5	7.4
Switch ASIC SerDes Lane by Speed (% y/y)										
1 Gbps	-3%	-10%	-11%	-2%	3%	-25%	1%	1%	1%	1%
10 Gbps	-11%	-8%	-18%	1%	-17%	-100%				
25 Gbps	8%	20%	7%	-2%	-62%	11%	-16%	-52%	-42%	-42%
50 Gbps	82%	78%	15%	72%	3%	-10%	-10%	-5%	-15%	-20%
100 Gbps			1101%	372%	1200%	75%	31%	5%	-16%	-16%
200 Gbps						452%	190%	109%	70%	27%
400 Gbps								390%	206%	99%
Total	17%	31%	8%	32%	40%	36%	27%	16%	9%	6%

Source: Lightcounting, 650 Group, TD Cowen estimates

Figure 97 Section C (cont): Layer 2 & 3 Data & Network Layer Processing (Switch ASICs & NICs, including Scale Up)

Switch ASIC ASP per SerDes Lane	\$	19	\$	17	\$	17	\$	14	\$	14	\$	14	\$	13	\$	13	\$	13
y/y gwth (%)		-9%		-8%		-3%		-17%		-1%		-1%		-1%		-1%		-1%
Switch Revenue (\$M)	\$	4,341	\$	5,251	\$	5,542	\$	6,031	\$	8,334	\$	11,240	\$	14,107	\$	16,179	\$	17,447
y/y gwth (%)		7%		21%		6%		9%		38%		35%		26%		15%		8%
Switch ASIC Units		2,384,816		2,853,744		2,619,253		2,935,954		3,912,151		4,858,508		5,679,999		6,051,868		6,125,420
SerDes Lanes per ASIC		97.10		106.22		125.29		147.04		154.03		168.98		183.23		199.22		214.40
y/y gwth (%)		7%		20%		-8%		12%		33%		24%		17%		7%		1%
Switch ASIC ASP (\$)	\$	1,820	\$	1,840	\$	2,116	\$	2,054	\$	2,130	\$	2,314	\$	2,484	\$	2,673	\$	2,848
y/y gwth (%)		0%		1%		15%		-3%		4%		9%		7%		8%		7%
Switch Revenue Share (%)																		
Broadcom		50%		49%		49%		54%		54%		54%		54%		54%		54%
Cisco		29%		29%		30%		22%		22%		22%		22%		22%		22%
HPE		0%		0%		1%		1%		1%		1%		1%		1%		1%
Huawei		1%		1%		1%		1%		1%		1%		1%		1%		1%
Intel		0%		1%		0%		0%		0%		0%		0%		0%		0%
Juniper		3%		3%		2%		2%		2%		2%		2%		2%		2%
Marvell (Innovium)		2%		3%		3%		4%		4%		3%		3%		2%		2%
Nokia		9%		7%		6%		5%		5%		5%		5%		5%		5%
NVIDIA (Mellanox)		2%		3%		4%		7%		10%		10%		10%		10%		10%
Other		6%		6%		6%		6%		2%		2%		3%		3%		3%
Ethernet NICs (Including DPUs & SmartNICs)		2021		2022		2023		2024		2025E		2026E		2027E		2028E		2029E
From Section A: Total Ports																		
Ethernet NIC Ports--Total (M)		28.7		27.9		27.9		28.7		35.6		46.8		45.4		51.8		58.5
y/y gwth (%)		13%		-3%		0%		3%		24%		32%		-3%		14%		13%
Ethernet NIC ASP (\$ per Port)	\$	73	\$	94	\$	134	\$	163	\$	199	\$	217	\$	234	\$	253	\$	271
y/y gwth (%)		6%		28%		43%		22%		22%		9%		8%		8%		7%
Ethernet NIC Revenues (\$M)	\$	2,105	\$	2,614	\$	3,734	\$	4,685	\$	7,087	\$	10,170	\$	10,654	\$	13,117	\$	15,860
y/y gwth (%)		20%		24%		43%		25%		51%		43%		5%		23%		21%
Ethernet NIC Revenue Share (%)																		
Amazon (Annapurna)		15%		13%		11%		15%		14%		14%		13%		12%		11%
AMD (Xilinx)		1%		5%		8%		10%		10%		10%		9%		9%		8%
Broadcom		14%		14%		10%		8%		9%		9%		12%		14%		15%
Cisco		2%		2%		1%		1%		1%		1%		2%		3%		3%
Intel		16%		14%		10%		10%		9%		7%		7%		6%		5%
Marvell		10%		9%		8%		7%		7%		5%		5%		5%		4%
NVIDIA (Mellanox)		22%		30%		42%		38%		42%		42%		39%		40%		40%
Other		20%		13%		11%		12%		9%		13%		13%		12%		14%

Source: Lightcounting, 650 Group, TD Cowen estimates

Figure 98 Section C (cont): Layer 2 & 3 Data & Network Layer Processing (Switch ASICs & NICs, including Scale Up)

Infiniband Switches & NICs	2021	2022	2023	2024	2025E	2026E	2027E	2028E	2029E	2030E
Infiniband Switch ASIC Units (K)	62	115	447	634	711	639	602	566	531	515
y/y growth (%)	26%	84%	290%	42%	12%	-10%	-6%	-6%	-6%	-3%
Infiniband Switch ASP (\$ per ASIC)	\$ 2,858	\$ 2,938	\$ 3,095	\$ 3,199	\$ 3,258	\$ 3,319	\$ 3,380	\$ 3,443	\$ 3,506	\$ 3,571
y/y growth (%)	3%	3%	5%	3%	2%	2%	2%	2%	2%	2%
Infiniband Switch ASIC Revenue (\$M)	\$ 178	\$ 337	\$ 1,383	\$ 2,028	\$ 2,316	\$ 2,122	\$ 2,036	\$ 1,949	\$ 1,863	\$ 1,840
y/y growth (%)	30%	89%	311%	47%	14%	-8%	-4%	-4%	-4%	-1%
Infiniband NIC Units (K)	193	226	787	1,112	1,220	1,085	1,053	939	834	709
y/y growth (%)	57%	17%	248%	41%	10%	-11%	-3%	-11%	-11%	-15%
Infiniband NIC ASP (\$ per NIC)	\$ 704	\$ 893	\$ 986	\$ 1,052	\$ 1,072	\$ 1,091	\$ 1,110	\$ 1,129	\$ 1,148	\$ 1,171
y/y growth (%)	1%	27%	10%	7%	2%	2%	2%	2%	2%	2%
Infiniband NIC Revenue (\$M)	\$ 136	\$ 202	\$ 776	\$ 1,170	\$ 1,307	\$ 1,184	\$ 1,168	\$ 1,060	\$ 958	\$ 830
y/y growth (%)	58%	49%	284%	51%	12%	-9%	-1%	-9%	-10%	-13%
Total Infiniband Revenue	\$ 314	\$ 538	\$ 2,159	\$ 3,198	\$ 3,624	\$ 3,306	\$ 3,204	\$ 3,009	\$ 2,820	\$ 2,671
y/y growth (%)	41%	71%	301%	48%	13%	-9%	-3%	-6%	-6%	-5%

Source: 650 Group, TD Cowen estimates.

Figure 99 Section C (cont): Layer 2 & 3 Data & Network Layer Processing (Switch ASICs & NICs, including Scale Up)

Scale Up Switches	2021	2022	2023	2024	2025E	2026E	2027E	2028E	2029E	2030E
Scale-Up Switch ASIC Units (M)										
NV Link Switch Units				0.4	2.1	2.7	3.4	4.0	4.5	5.1
y/y growth (%)					430%	25%	30%	15%	14%	13%
NVIDIA Accelerator Units				3.5	3.9	4.6	4.7	5.0	5.3	5.5
Scale Up Domain per Switch				2.0	1.9	1.8	1.7	1.6	1.6	1.5
Implied Switch Units				1.7	2.1	2.6	2.8	3.1	3.3	3.7
UA Link (Native) [Infinity Fabric]				0.1	0.1	0.1	0.3	0.3	0.5	0.6
y/y growth (%)					100%	21%	108%	32%	47%	15%
PCIe/CXL						0.5	0.8	0.5	0.7	0.8
y/y growth (%)							78%	-44%	50%	15%
Ethernet (Either SUE or UAL with Ethernet PHY)						0.6	0.7	1.7	2.4	2.7
y/y growth (%)							14%	130%	43%	11%
Accelerator Units ex NVIDIA & TPU				2.5	3.2	4.8	5.3	6.2	7.1	8.0
Scale Up Domain per Switch				8.0	5.0	4.0	3.0	2.5	2.0	2.0
Implied Switch Units				0.3	0.6	1.2	1.8	2.5	3.6	4.0
UA Link (Native) % Share				100%	100%	10%	14%	13%	14%	14%
PCIe % Share					0%	37%	45%	18%	19%	19%
Ethernet (SUE or UAL with Ethernet PHY) % Share					0%	53%	41%	68%	67%	67%
Other (Google's ICI, Others)					0.3	0.3	0.4	0.4	0.5	0.5
y/y growth (%)						15%	13%	12%	9%	8%
Total				0.5	2.5	4.2	5.6	6.9	8.6	9.6
Scale-Up Switch ASIC ASP (\$)										
NV Link				\$ 925.73	\$ 1,962.55	\$ 2,089.42	\$ 2,195.94	\$ 2,386.66	\$ 2,725.27	\$ 3,111.93
y/y growth (%)					112%	6%	5%	9%	14%	14%
UA Link				\$ 301.12	\$ 321.44	\$ 369.66	\$ 412.32	\$ 512.69	\$ 557.78	\$ 606.82
y/y growth (%)					7%	15%	12%	24%	9%	9%
PCIe/CXL						\$ 650.00	\$ 812.50	\$ 942.50	\$ 961.35	\$ 980.58
y/y growth (%)							25%	16%	2%	2%
Ethernet						\$ 1,730.65	\$ 1,903.71	\$ 2,189.27	\$ 2,381.79	\$ 2,591.24
y/y growth (%)							10%	15%	9%	9%
Other					\$ 283.52	\$ 290.78	\$ 303.75	\$ 311.28	\$ 319.84	\$ 328.64
y/y growth (%)						3%	4%	2%	3%	3%

Source: 650 Group, TD Cowen estimates.

Figure 100 Section C (cont): Layer 2 & 3 Data & Network Layer Processing (Switch ASICs & NICs, including Scale Up)

Scale-Up Switch ASIC Revenue (\$M)														
NV Link	\$	370.29	\$	4,160.60	\$	5,536.98	\$	7,565.01	\$	9,455.34	\$	12,308.41	\$	15,881.82
y/y growth (%)				1024%		33%		37%		25%		30%		29%
UA Link	\$	15.06	\$	32.14	\$	44.72	\$	103.64	\$	170.11	\$	271.83	\$	340.09
y/y growth (%)				114%		39%		132%		64%		60%		25%
PCIe/CXL	\$	-	\$	292.50	\$	650.00	\$	424.13	\$	648.91	\$	761.17		
y/y growth (%)						122%		-35%		53%		17%		
Ethernet	\$	-	\$	1,105.63	\$	1,390.95	\$	3,682.65	\$	5,721.91	\$	6,893.96		
y/y growth (%)						26%		165%		55%		20%		
Other (Google ICI, etc)	\$	85.06	\$	100.32	\$	118.42	\$	135.92	\$	152.22	\$	168.92		
y/y growth (%)						18%		18%		15%		12%		11%
Total	\$	385	\$	4,278	\$	7,080	\$	9,828	\$	13,868	\$	19,103	\$	24,046
y/y growth (%)				1010%		66%		39%		41%		38%		26%
Scale-Up Switch ASIC Share (%)														
NVIDIA (NVLink, Ethernet)				97%		78%		77%		68%		64%		66%
Broadcom (Ethernet, PCIe)				0%		13%		13%		22%		25%		24%
Astera Labs (UA Link, PCIe)				3%		4%		5%		3%		3%		3%
Marvell (UA Link, PCIe, Ethernet)				0%		2%		2%		2%		2%		2%
Other				0%		3%		3%		5%		5%		5%
NVIDIA (NVLink, Ethernet)	\$	4,161	\$	5,537	\$	7,565	\$	9,455	\$	12,308	\$	15,882		
Broadcom (Ethernet, PCIe)	\$	-	\$	943	\$	1,243	\$	3,031	\$	4,707	\$	5,667		
Astera Labs (UA Link, PCIe)	\$	107	\$	301	\$	483	\$	408	\$	634	\$	763		
Marvell (UA Link, PCIe, Ethernet)	\$	3	\$	118	\$	210	\$	286	\$	443	\$	531		

Source: TD Cowen.

Company-Specific Revenue Builds

Figure 101 Broadcom Estimated Datacenter Revenue (Buchalter)

Broadcom	2021	2022	2023	2024	2025E	2026E	2027E	2028E	2029E	2030E	CAGR (25-30)
Section B											
AECs											
Optics	\$ 35.35	\$ 68.22	\$ 91.53	\$ 185.65	\$ 321.60	\$ 441.17	\$ 522.88	\$ 581.11	\$ 657.60	\$ 731.96	
Line Cards	\$ 293.02	\$ 335.96	\$ 344.66	\$ 402.58	\$ 550.22	\$ 677.81	\$ 790.47	\$ 921.84	\$ 1,047.95	\$ 1,181.04	
PCIe Retimers	\$ 189.31	\$ 148.52	\$ 374.68	\$ 713.43	\$ 871.31	\$ 1,099.32	\$ 1,139.79	\$ 1,236.27	\$ 1,347.01	\$ 1,425.80	
Section C											
Ethernet Switches	\$ 2,153.13	\$ 2,571.11	\$ 2,694.36	\$ 3,255.42	\$ 4,498.55	\$ 6,067.46	\$ 7,614.77	\$ 8,733.14	\$ 9,417.88	\$ 9,908.24	
Ethernet NICs & DPUs	\$ 286.37	\$ 353.88	\$ 372.31	\$ 364.94	\$ 637.86	\$ 915.30	\$ 1,278.48	\$ 1,836.42	\$ 2,379.01	\$ 3,012.65	
Scale Up Switches	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 943.00	\$ 1,242.76	\$ 3,030.95	\$ 4,707.31	\$ 5,667.40	
Total Broadcom Networking Infrastructure Silicon	\$ 2,957.18	\$ 3,477.69	\$ 3,877.54	\$ 4,922.01	\$ 6,879.54	\$ 10,144.07	\$ 12,589.15	\$ 16,339.73	\$ 19,556.76	\$ 21,927.10	26.1%
Other Broadcom Networking Revenue	\$ 4,425.10	\$ 5,822.31	\$ 7,947.96	\$ 12,179.57	\$ 18,279.40	\$ 35,021.94	\$ 42,712.01	\$ 55,020.93	\$ 68,990.02	\$ 82,726.64	
Service Provider & Custom Routing	\$ 1,478	\$ 2,264	\$ 2,390	\$ 1,624	\$ 2,396	\$ 2,704	\$ 2,784	\$ 2,922	\$ 3,153	\$ 3,277	
Routing SerDes Lanes (M)	2.63	4.36	4.72	3.88	5.77	6.58	6.85	7.26	7.91	8.31	
\$ per SerDes Lane	\$ 562	\$ 520	\$ 507	\$ 419	\$ 415	\$ 411	\$ 407	\$ 403	\$ 399	\$ 395	
AI ASICs (from Acceleration section)	\$ 1,500.00	\$ 1,893.59	\$ 3,144.28	\$ 9,133.33	\$ 14,840.14	\$ 30,307.43	\$ 38,287.37	\$ 50,221.00	\$ 64,077.86	\$ 77,630.51	
Other	\$ 1,447.27	\$ 1,664.82	\$ 2,413.80	\$ 1,421.96	\$ 1,043.29	\$ 2,010.42	\$ 1,640.50	\$ 1,878.25	\$ 1,759.38	\$ 1,818.81	
Total Broadcom Networking Revenue (Reported/Est.)	\$ 7,382.28	\$ 9,300.00	\$ 11,825.50	\$ 17,101.59	\$ 25,158.94	\$ 45,166.01	\$ 55,301.16	\$ 71,360.66	\$ 88,546.78	\$104,653.74	33.0%

Source: TD Cowen, Company reports

Figure 102 Marvell Estimated Datacenter Revenue (Buchalter)

Marvell	2021	2022	2023	2024	2025E	2026E	2027E	2028E	2029E	2030E	CAGR (25-30)
Section B											
AECs	\$ -	\$ -	\$ -	\$ -	\$ 54.28	\$ 88.51	\$ 126.13	\$ 151.27	\$ 171.85	\$ 196.85	
Optics	\$ 114.34	\$ 138.71	\$ 216.65	\$ 522.55	\$ 930.95	\$ 1,160.99	\$ 1,176.48	\$ 1,079.21	\$ 1,127.31	\$ 1,254.79	6.2%
Line Cards	\$ 33.68	\$ 48.68	\$ 39.98	\$ 77.89	\$ 106.45	\$ 131.14	\$ 152.93	\$ 178.35	\$ 202.75	\$ 228.50	
PCIe Retimers	\$ 3.82	\$ 3.03	\$ 7.71	\$ 13.22	\$ 15.84	\$ 19.99	\$ 20.72	\$ 22.48	\$ 24.49	\$ 25.92	
Section C											
Ethernet Switches	\$ 66.85	\$ 148.57	\$ 153.90	\$ 231.86	\$ 291.68	\$ 337.21	\$ 352.67	\$ 323.57	\$ 348.94	\$ 367.11	
Ethernet NICs & DPUs	\$ 215.56	\$ 247.75	\$ 281.40	\$ 308.45	\$ 496.11	\$ 508.50	\$ 532.70	\$ 655.86	\$ 634.40	\$ 753.16	
Scale Up Switches	\$ -	\$ -	\$ -	\$ -	\$ 3.21	\$ 118.25	\$ 209.91	\$ 285.97	\$ 443.06	\$ 530.94	
Total Marvell Networking Infrastructure Silicon	\$ 434.24	\$ 586.74	\$ 699.64	\$ 1,153.96	\$ 1,898.53	\$ 2,364.58	\$ 2,571.55	\$ 2,696.71	\$ 2,952.80	\$ 3,357.28	12.1%
Other Marvell Datacenter Revenue	\$ 1,350.62	\$ 1,822.06	\$ 1,517.06	\$ 2,087.70	\$ 2,655.00	\$ 3,000.00	\$ 3,054.00	\$ 3,111.24	\$ 3,171.91	\$ 3,236.23	
Storage Controllers	\$ 654	\$ 994	\$ 508	\$ 650	\$ 900	\$ 900	\$ 954	\$ 1,011	\$ 1,072	\$ 1,136	
Fibre Channel HBAs	\$ 205	\$ 188	\$ 130	\$ 176	\$ 250	\$ 250	\$ 250	\$ 250	\$ 250	\$ 250	
DCI Optics (Coherent)	\$ 492	\$ 640	\$ 879	\$ 1,262	\$ 1,505	\$ 1,850	\$ 1,850	\$ 1,850	\$ 1,850	\$ 1,850	
AI ASICs (from Acceleration Section)	\$ -	\$ -	\$ -	\$ 909	\$ 1,455	\$ 1,489	\$ 2,560	\$ 3,689	\$ 5,019	\$ 7,160	
Total Marvell Datacenter Revenue (Reported/Est.)	\$ 1,785	\$ 2,409	\$ 2,217	\$ 4,151	\$ 6,009	\$ 6,853	\$ 8,185	\$ 9,497	\$ 11,144	\$ 13,754	18.0%

Source: TD Cowen, Company reports

Figure 103 NVIDIA Estimated Networking Revenue (Buchalter)

NVIDIA	2021	2022	2023	2024	2025E	2026E	2027E	2028E	2029E	2030E	CAGR (25-30)
Section B											
AECs											
Optics											
Line Cards											
PCIe Retimers											
Section C											
Ethernet Switches	\$ 69.15	\$ 142.49	\$ 224.67	\$ 424.06	\$ 833.38	\$ 1,124.02	\$ 1,410.67	\$ 1,617.86	\$ 1,744.71	\$ 1,835.55	
Ethernet NICs & DPUs	\$ 470.60	\$ 772.68	\$ 1,570.63	\$ 1,760.93	\$ 2,976.66	\$ 4,271.38	\$ 4,155.08	\$ 5,246.90	\$ 6,344.03	\$ 7,531.63	
Infiniband NICs and Switches	\$ 314.07	\$ 538.47	\$ 2,158.99	\$ 3,198.40	\$ 3,623.75	\$ 3,305.88	\$ 3,203.96	\$ 3,008.85	\$ 2,820.49	\$ 2,670.59	
Scale Up Switches	\$ -	\$ -	\$ -	\$ -	\$ 4,160.60	\$ 5,536.98	\$ 7,565.01	\$ 9,455.34	\$ 12,308.41	\$ 15,881.82	
Total NVIDIA Networking Infrastructure Silicon	\$ 853.81	\$ 1,453.64	\$ 3,954.29	\$ 5,383.39	\$ 11,594.38	\$ 14,238.26	\$ 16,334.72	\$ 19,328.95	\$ 23,217.63	\$ 27,919.59	19.2%
Other NVIDIA Networking Revenue	\$ 711.69	\$ 1,264.63	\$ 4,620.23	\$ 7,606.21	\$ 8,078.54	\$ 7,542.82	\$ 8,964.77	\$ 9,326.08	\$ 9,402.63	\$ 9,858.19	
System-Level Revenue for Ethernet NICs & Switches	\$ 198	\$ 407	\$ 642	\$ 1,210	\$ 6,965	\$ 8,776	\$ 10,880	\$ 13,249	\$ 15,385	\$ 17,557	
System-Level Revenue for Infiniband Switches	\$ 897	\$ 1,538	\$ 6,362	\$ 10,018	\$ 8,547	\$ 7,468	\$ 6,854	\$ 5,951	\$ 4,927	\$ 4,339	
System-Level Revenue for NVLink (Scale Up) Switches	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	
Total NVIDIA Networking Revenue	\$ 1,566	\$ 2,718	\$ 8,575	\$ 12,990	\$ 19,673	\$ 21,781	\$ 25,299	\$ 28,655	\$ 32,620	\$ 37,778	13.9%

Source: TD Cowen, Company reports

Figure 104 Credo Estimated Datacenter Revenue (O'Loughlin)

Credo	2021	2022	2023	2024	2025E	2026E	2027E	2028E	2029E	2030E	CAGR (25-30)
Section B											
AECs	\$ 28	\$ 92	\$ 68	\$ 128	\$ 495	\$ 718	\$ 814	\$ 882	\$ 912	\$ 956	14.1%
Optics	\$ 10	\$ 31	\$ 36	\$ 86	\$ 169	\$ 348	\$ 523	\$ 719	\$ 939	\$ 1,115	45.8%
Line Cards	\$ 7	\$ 17	\$ 14	\$ 26	\$ 35	\$ 44	\$ 51	\$ 59	\$ 68	\$ 76	
PCIe Retimers	\$ -	\$ -	\$ 8	\$ 13	\$ 32	\$ 40	\$ 41	\$ 45	\$ 49	\$ 52	
Section C											
Ethernet Switches											
Ethernet NICs & DPUs											
Scale Up Switches											
Total Credo Networking Infrastructure Silicon	\$ 45	\$ 139	\$ 126	\$ 254	\$ 732	\$ 1,150	\$ 1,429	\$ 1,706	\$ 1,968	\$ 2,199	24.6%
Other Credo Datacenter Revenue	\$ 12	\$ 4	\$ 2	\$ 34	\$ 107	\$ 114	\$ 148	\$ 166	\$ 179	\$ 195	
System-Level Revenue for AECs (Cables in addition to chi	\$ 39	\$ 96	\$ 70	\$ 163	\$ 603	\$ 832	\$ 962	\$ 1,049	\$ 1,091	\$ 1,151	
Units (Full Cables)	0.21	0.71	0.53	1.22	4.52	6.57	7.67	8.45	8.88	9.47	
ASP (Full Cables)	\$ 185	\$ 135	\$ 133	\$ 133	\$ 133	\$ 127	\$ 125	\$ 124	\$ 123	\$ 122	
Total Credo Datacenter Revenue	\$ 57	\$ 143	\$ 128	\$ 288	\$ 839	\$ 1,263	\$ 1,577	\$ 1,873	\$ 2,147	\$ 2,394	23.3%

Source: TD Cowen, Company reports

Figure 105 Astera Labs Estimated Datacenter Revenue (O'Loughlin)

Astera	2021	2022	2023	2024	2025E	2026E	2027E	2028E	2029E	2030E	CAGR (25-30)
Section B											
AECs	\$ -	\$ 1	\$ 1	\$ 30	\$ 129	\$ 177	\$ 206	\$ 227	\$ 238	\$ 253	
Optics											
Line Cards											
PCIe Retimers	\$ -	\$ 80	\$ 116	\$ 366	\$ 532	\$ 779	\$ 746	\$ 809	\$ 882	\$ 933	
Section C											
Ethernet Switches											
Ethernet NICs & DPUs											
Scale Up Switches	\$ -	\$ -	\$ -	\$ -	\$ 107	\$ 301	\$ 483	\$ 408	\$ 634	\$ 763	
Total Astera Networking Infrastructure Silicon	\$ -	\$ 81	\$ 117	\$ 396	\$ 767	\$ 1,256	\$ 1,436	\$ 1,444	\$ 1,754	\$ 1,949	
Other Astera Datacenter Revenue	\$ -	\$ -	\$ -	\$ -	\$ 8	\$ 30	\$ 36	\$ 43	\$ 52	\$ 63	
CXL Controllers & Switches (Leo)					\$ 8	\$ 30	\$ 36	\$ 43	\$ 52	\$ 63	
Total Astera Datacenter Revenue	\$ 81	\$ 117	\$ 396	\$ 396	\$ 775	\$ 1,287	\$ 1,472	\$ 1,487	\$ 1,806	\$ 2,012	21.0%

Source: TD Cowen, Company reports

Figure 106 MACOM Estimated Datacenter Revenue (O'Loughlin)

MACOM	2021	2022	2023	2024	2025E	2026E	2027E	2028E	2029E	2030E	CAGR (25-30)
Section B											
ACCs	\$ -	\$ -	\$ 1.11	\$ 1.76	\$ 19.97	\$ 15.30	\$ 22.85	\$ 34.37	\$ 40.51	\$ 47.36	18.9%
Optics (Electronics Only)	\$ 45.70	\$ 59.64	\$ 74.79	\$ 141.13	\$ 249.38	\$ 377.83	\$ 462.80	\$ 527.76	\$ 620.70	\$ 719.87	23.6%
Interconnect Bus Redrivers											
Total MACOM Networking Infrastructure Silicon	\$ 45.70	\$ 59.64	\$ 75.89	\$ 142.89	\$ 269.35	\$ 393.13	\$ 485.65	\$ 562.13	\$ 661.20	\$ 767.24	
Other MACOM Datacenter Revenue (incl. Lasers)	\$ 88.00	\$ 89.36	\$ 81.17	\$ 70.71	\$ 43.80	\$ 43.80	\$ 43.80	\$ 43.80	\$ 43.80	\$ 43.80	
Total MACOM Datacenter Revenue	\$ 133.70	\$ 149.00	\$ 157.06	\$ 213.60	\$ 313.15	\$ 436.92	\$ 529.45	\$ 605.93	\$ 705.00	\$ 811.03	21.0%

Source: TD Cowen, Company reports

Figure 107 Semtech Potential Datacenter Revenue (O'Loughlin)

Semtech	2021	2022	2023	2024	2025E	2026E	2027E	2028E	2029E	2030E	CAGR (25-30)
Section B											
ACCs	\$ -	\$ -	\$ 1.11	\$ 3.02	\$ 51.92	\$ 114.74	\$ 166.21	\$ 229.16	\$ 249.28	\$ 270.64	39.1%
Optics (incl. Analog)	\$ 44.94	\$ 58.70	\$ 73.67	\$ 139.11	\$ 226.71	\$ 328.55	\$ 424.23	\$ 506.65	\$ 620.70	\$ 746.53	26.9%
Interconnect Bus Redrivers											
Total Semtech Networking Infrastructure Silicon	\$ 44.94	\$ 58.70	\$ 74.77	\$ 142.13	\$ 278.63	\$ 443.28	\$ 590.44	\$ 735.81	\$ 869.97	\$ 1,017.18	
Other Semtech Datacenter Revenue											
Total Potential Semtech Datacenter Revenue	\$ 44.94	\$ 58.70	\$ 74.77	\$ 142.13	\$ 278.63	\$ 443.28	\$ 590.44	\$ 735.81	\$ 869.97	\$ 1,017.18	29.6%

Source: TD Cowen, Company reports

VALUATION METHODOLOGY AND RISKS

Valuation Methodology

Semiconductors:

Our valuation methodology is primarily based on forward P/E multiples plus cash followed by EV/EBITDA. In many cases, we use EV/sales as a third methodology.

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Stocks Mentioned In Important Disclosures

Ticker	Company Name
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AVGO	Broadcom Inc.
CRDO	Credo Technology Group Holding Ltd
MTSI	MACOM Technology Solutions Holdings
MVVL	Marvell Technology Group
SMTC	Semtech Corp.

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- Assumption:** The expected total return calculation includes annual return and forecasted dividend yield

TD Cowen Equity Research Rating Distribution

Distribution of Ratings/Investment Banking Services (IB) as of 06/30/25

Rating	Count	Ratings Distribution	Count	IB Services/Past 12 Months
Buy	823	65.84%	322	39.13%
Hold	410	32.80%	103	25.12%
Sell	17	1.36%	4	23.53%

TD Cowen Equity Research Rating Distribution Table does not include any subject company for which the equity research rating is currently suspended.

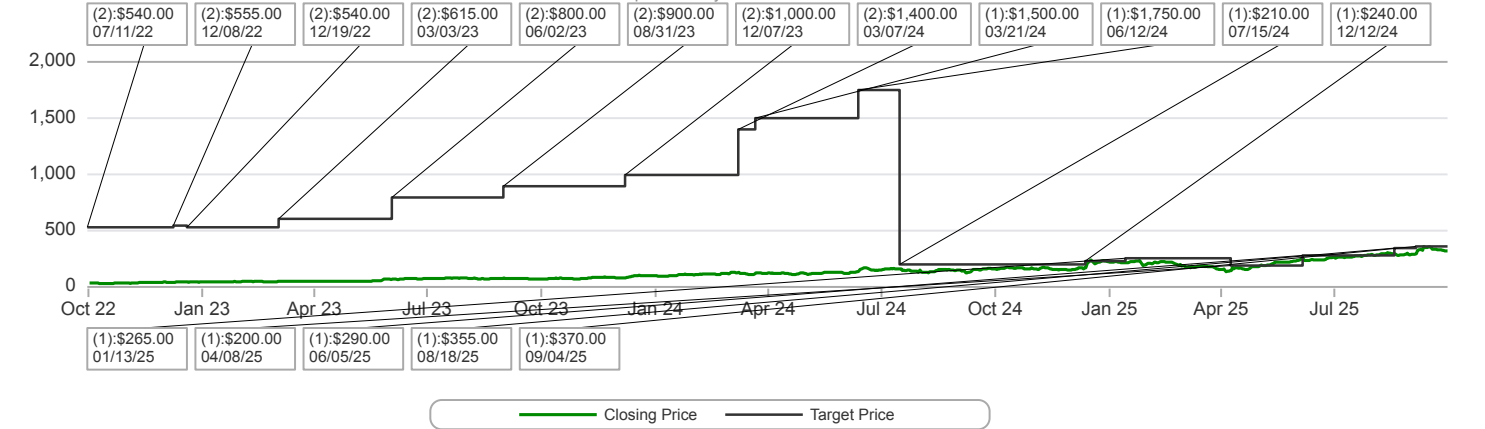
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